

Source-First Exhaustion of the Physical Carrier Universe

A Candidate-Completeness Theorem for the Neutrino Propagation Role Complex

Amos Jay Maley
Independent Researcher

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Abstract

Earlier work isolates a nonempty, nested physical-role complex for the realized neutrino source–propagation–detection target,

$$\mathcal{R}_\nu^{\text{osc}} = (r_{\triangleright\Delta}, r_\Delta; r_\Delta \succ r_{\triangleright\Delta}), \quad r_\Delta \Rightarrow r_{\triangleright\Delta}, \quad r_{\triangleright\Delta} \not\Rightarrow r_\Delta,$$

and constructs a complete carrier-neutral physical interface for that target. Those results determine what any realization must accomplish, but they do not identify what physical architecture carries the required roles. The present paper addresses the logically prior candidate-completeness problem: construct the full class of target-relative carrier architectures from an independently specified physical ontology, without using the later success, failure, ranking, or survival of familiar neutrino models.

The analysis begins with a semantic class $\text{CarrierReal}_{\text{sem}}(\mathbb{T}_\nu^{(3)})$ of same-target physical carrier realizations. Each realization is reduced to its currently participating carrier core, thereby separating actual target-relative carrier work from merely latent capability under a hypothetical future coupling. A carrier-facing trace

$$\chi_{\text{car}} : \text{CarrierReal}_{\text{sem}}(\mathbb{T}_\nu^{(3)}) \longrightarrow \mathcal{T}_{\text{car}}(\mathbb{T}_\nu^{(3)})$$

records support, persistence mode, derivation status, constituent kind, intrinsic typed data, primitive parameter domains, target attachment, and the complete physical incidence architecture of that core. Source activity is evaluated pointwise over parameter space: source-null incidences are removed before carrier-path closure, whereas every source-active incidence is retained without a numerical weak-coupling threshold. The resulting trace is proved to be a complete invariant of carrier-semantic equivalence.

Independently of candidate outcomes, the target carrier signature $\mathfrak{D}_{\text{car}}(\mathbb{T}_\nu^{(3)})$ is derived from the realized physical ontology. It generates a source carrier theory $\mathbb{S}_{\text{car}}(\mathbb{T}_\nu^{(3)})$ containing the relevant source sorts, lawful operations, constitutive equations, and type-indexed activity/null structure. Local typing is separated from source-constitutive coherence, and both are kept distinct from later physical admissibility. The principal carrier strata—existing-substrate and law-only structures, new local carriers, composites and emergent sectors, topological and defect-supported sectors, solitonic and boundary structures, environmental co-carriers, hidden or enlarged-state-space sectors, and finite hybrids—arise as normal-form regions of this source theory rather than as a list of preferred mechanisms. Symbolic representation labels, multiplicities, couplings, global data, and other source parameters remain open.

The generated candidate universe is

$$\mathfrak{U}_\nu = \left(\bigsqcup_{u \in \text{FamPres}_\nu} \Theta_u^{\text{part}} \right) / \simeq_{\text{inst}},$$

where u ranges over source-coherent normal-form templates, Θ_u^{part} is the source-defined locus with nonempty active target participation, and instance equivalence is equality of the complete activity-reduced carrier trace up to lawful source reparameterization. Distinct microscopic architectures remain distinct even when they induce the same target interface, while target-inert spectators and source-null degenerate sectors do not create false multiplicity.

The main result is the canonical candidate-completeness isomorphism

$$\boxed{\text{CarrierReal}_{\text{sem}}(\mathbb{T}_\nu^{(3)}) / \equiv_{\text{car}} \cong \mathfrak{U}_\nu.}$$

Thus every finitely presented same-target carrier architecture in the declared comparison class has a source-generated normal form, and every generated instance has a coherent physical carrier interpretation. A

separate independence theorem shows that the universe factors only through the fixed source data: deleting or altering downstream interface values, simultaneous-survival data, rankings, and expected-occupant labels leaves the scientific candidate universe unchanged.

The theorem is deliberately prior to model selection. Membership in $\mathfrak{U}_\nu$ implies neither interface adequacy, anomaly or vacuum compatibility, simultaneous survival, role eligibility, realization, nor experimental preference. The result supplies the complete candidate domain on which those later questions may be posed without survivor backfitting.

Keywords: neutrino oscillation; physical carrier; candidate completeness; source-first generation; carrier normal form; typed hypergraph; composite sector; topological sector; open system; AASC; role occupancy.

Main result

Fix the previously established neutrino-propagation role packet, the complete target-relative interface, the known physical substrate, and the comparison class of finitely presented same-target carrier architectures. Every semantic carrier realization then admits a source-generated, source-coherent carrier-incidence normal form, unique up to carrier-semantic equivalence and lawful parameter reparameterization. Pointwise source activity fixes candidate identity before downstream evaluation: source-null degenerate incidences and disconnected latent sectors are removed, while every nonnull active incidence is retained without a magnitude threshold. The resulting parametric universe is canonically isomorphic to the semantic carrier class modulo carrier equivalence and is independent of all later candidate outcomes.

Scope and reading guide

The theorem is relative to a fixed target, a fixed role-level physical ontology, and the finite occupancy-bearing presentation condition stated in Section 2. It is not a claim of unrestricted ultraviolet completeness. The central proof proceeds in four stages: semantic carrier reduction and trace construction; derivation of the source carrier theory and its constitutive coherence; normal-form reconstruction of every finite carrier architecture; and exact quotienting by carrier-semantic equivalence. The paper then proves generation independence, examines representative counterexample classes, and states the downstream simultaneous-constraint problem without using it in candidate formation.

Contents

1	Introduction and theorem strategy	1
1.1	Why candidate completeness is a theorem burden	1
1.2	Source-first generation	1
1.3	Four distinct completeness problems	1
1.4	Comparison class	2
1.5	Semantic and source constructions	2
1.6	Main theorem	3
1.7	Proof architecture	3
1.8	Organization	4
2	Fixed target and imported results	4
2.1	Target fiber	4
2.2	Imported theorem package	5
2.3	Dependency restriction	5
2.4	Permitted interface information	6
2.5	Excluded microscopic premises	6
2.6	Source objecthood versus downstream viability	6

3	Semantic physical carrier realizations	7
3.1	Raw worlds and the carrier question	7
3.2	Semantic carrier uses	8
3.3	Semantic candidate class	9
3.4	Carrier-use factorization	9
3.5	The physical meaning of “carrier”	10
3.6	Semantic falsifier	10
4	Carrier-facing trace and carrier-semantic identity	10
4.1	Independent target carrier signature	10
4.2	Primitive carrier projections	10
4.3	Trace extensionality	11
4.4	Candidate identity is finer than interface identity	11
4.5	Parameters, activity changes, and family identity	11
4.6	Trace falsifier	12
5	Primitive carrier dimensions and source ontology	12
5.1	From physical roles to a carrier signature	12
5.2	Primitive decomposition of a carrier node	13
5.3	Principal source strata	14
5.4	Why no family is privileged	14
5.5	Open parameter spaces	15
5.6	Ontology falsification burden	15
6	Source-first constructor algebra and constitutive coherence	15
6.1	The three source-level burdens	15
6.2	The independently fixed source carrier theory	15
6.3	Pointwise source activity and null incidence	16
6.4	Raw typed term algebra	18
6.5	Source coherence and the scientific candidate algebra	18
6.6	Finite coherent-hypergraph presentation	20
6.7	Saturation and the unlisted-constructor objection	20
6.8	Constructive versus scientific completeness	21
7	Existing-substrate and no-new-carrier branches	21
7.1	Why the null-newness branch comes first	21
7.2	Law-only and constitutive realizations	21
7.3	Law-only versus integrated-out hidden carrier	22
7.4	Collective reorganization of existing carriers	22
7.5	Open parameter domain	22
8	Local primitive carrier families	22
8.1	Generic local carrier type	22
8.2	No representation cutoff	23
8.3	Fermionic, bosonic, and general source types	23
8.4	Chiral, vectorlike, real, and pseudoreal structures	23
8.5	Multiplicity without cardinality backfitting	23
8.6	Local interactions and attachment	23
8.7	Existing versus new local structure	23
8.8	Named mechanisms as representative constructions	24

9	Composite, collective, and emergent carrier families	24
9.1	Derived carriers are not primitive local fields	24
9.2	Constituent retention	24
9.3	Emergent local behavior does not imply a new primitive node	25
9.4	Collective and condensate carriers	25
9.5	Dual descriptions and effective equivalence	25
10	Topological, extended, and defect-supported carriers	25
10.1	Why point-local inventories are insufficient	25
10.2	Topological carrier family	26
10.3	Extended and defect-supported carriers	26
10.4	Non-invertible structures	26
10.5	Topological sectors versus spectators	26
10.6	Global-form dependence	27
11	Solitonic, localized nonperturbative, boundary, and edge carriers	27
11.1	Localized nonperturbative carriers	27
11.2	Boundary and edge carriers	27
11.3	Source/detector localized structures	28
11.4	Boundary-topological hybrids	28
12	Environmental, medium, hidden, and enlarged-state-space carriers	28
12.1	Environmental co-carriage	28
12.2	Open-system carriers	29
12.3	Minimal dilations and candidate equivalence	29
12.4	Memory and non-Markovian structure	29
12.5	Medium as carrier versus target input	29
13	Hybrid closure and carrier-incidence reconstruction	30
13.1	Why mixed architectures are not an escape hatch	30
13.2	Non-factorizing incidence	31
13.3	Incidence reconstruction theorem	31
13.4	Hybrid family templates	31
13.5	Hybrid source independence	31
14	Spectator stripping and the occupancy-bearing candidate core	32
14.1	The spectator multiplication problem	32
14.2	Pointwise parameter activity and degenerate incidence	33
14.3	World identity versus candidate identity	34
14.4	Spectator topological sectors	34
14.5	Spectator stripping commutes with lawful redescription	34
15	Lawful candidate equivalence and quotient discipline	34
15.1	Candidate equivalence	34
15.2	Family presentations, activity strata, and scientific family classes	35
15.3	Candidate universe as pointwise activity-reduced instances	37
15.4	Candidate automorphisms	37
15.5	Interface equivalence is a coarser relation	37
15.6	Canonical normal form without a privileged representative	37

16 Source-coherent normal form and reconstruction theorem	38
16.1 Normal-form object	38
16.2 Reduction invariance and oriented confluence	38
16.3 Semantic reduction theorem	38
16.4 Primitive component decomposition	39
16.5 Source normal-form existence	39
16.6 Source normal-form soundness	39
16.7 Normal-form exactness	40
16.8 Family templates and symbolic parameters	40
16.9 What the normal form does not determine	40
17 Semantic candidate completeness	40
17.1 Coverage and soundness maps	40
17.2 The two completeness directions	41
17.3 Parametric family closure	41
17.4 Omission disposition	42
17.5 No downstream inference	43
17.6 Comparison-class boundary	43
17.7 Strongest falsifier	43
18 Independence of candidate generation	43
18.1 Scientific factorization	43
18.2 Operational check of independence	44
18.3 Outcome information excluded from source formation	44
18.4 Post-generation representative constructions	45
18.5 Source provenance and falsifiability	45
19 Representative stress tests and counterexample classes	45
19.1 Purpose	45
19.2 Same interface, distinct carriers	46
19.3 Spectator and inflow TQFTs	46
19.4 Latent carrier copy and actual activation	47
19.5 One family across an active/null incidence boundary	47
19.6 Malformed fusion data	47
19.7 Boundary physics	47
19.8 Unbounded local families	47
19.9 New incidence relations or constitutive equations	47
19.10 Law-only versus hidden-sector ambiguity	47
19.11 Disposition of representative cases	48
20 Discussion, scope, and conclusion	48
20.1 Data supplied to subsequent constraint analysis	48
20.2 The subsequent simultaneous-constraint problem	49
20.3 Exact nonclaims	49
20.4 Main theorem	49
20.5 Interpretation	50

A	Formal target carrier signature	50
A.1	Typed source objects	50
A.2	Essentially algebraic source theory	51
A.3	Pointwise activity over source parameter families	51
A.4	Incidence typing	51
A.5	Finite presentations	52
A.6	Target attachment type	52
A.7	Source versus constraint typing	52
A.8	Universe convention	53
B	Detailed proofs of the completeness bridge	53
B.1	Proof of trace recovery	53
B.2	Carrier-bearing reduction without unique set-theoretic minimality	53
B.3	Pointwise activity and degenerate source strata	54
B.4	Graph-induction proof of constructor completeness	54
B.5	Typing, source coherence, and viability	54
B.6	Proof of the completeness isomorphism	54
B.7	Alternative set-theoretic formulation	55
B.8	Proof of omission disposition	55
B.9	No inference from family density	55
B.10	Latent-capability boundary	55
C	Principal family classes and representative constructions	55
C.1	Status of the table	55
C.2	Named representative constructions	56
C.3	Representative construction failure protocol	57
C.4	Source-coherence qualification	57
C.5	Activity loci are source data, not selection cuts	57
D	Detailed stress-test analyses	57
D.1	Unlisted local representation	57
D.2	Composite with elementary-like interface	57
D.3	Target-bearing TQFT	58
D.4	Decoupled TQFT	58
D.5	Decoupled latent-capable copy	58
D.6	Activation by actual incidence	58
D.7	Non-invertible defect	58
D.8	Typed but incoherent fusion datum	58
D.9	Environment-mediated realization	58
D.10	Hidden dilation	58
D.11	Law-only modification	58
D.12	New incidence relation	59
D.13	Essentially infinitary architecture	59
D.14	Source-null point inside one symbolic family	59
D.15	Downstream selection of a degenerate point	59
E	Glossary	59

F	Principal results and falsification conditions	61
F.1	Purpose	61
F.2	Logical dependence	62
F.3	Exact scope of completeness	63
F.4	Direct falsifier	63
F.5	Near misses that do not refute the theorem	63
F.6	Scientifically permitted endpoint branches	64
F.7	Nonclaims adjacent to the main theorem	64
G	Source-formation grammar	64
G.1	Role of the grammar	64
G.2	Node constructors	64
G.3	Incidence constructors	65
G.4	Type-indexed source-coherence profiles	66
G.5	Type-indexed source activity profiles	67
G.6	Parameter-domain constructors	67
G.7	Family normal-form schema	68
G.8	Formation rules by principal stratum	68
G.9	Constructor completeness burden	69
G.10	Why the formation table is not the completeness proof	69
	References	70

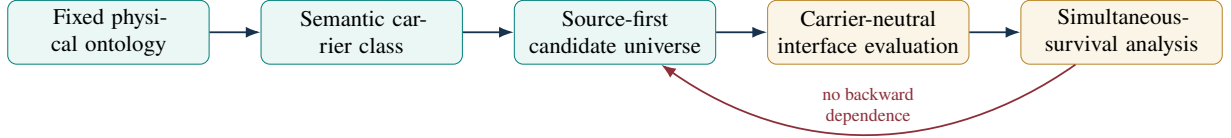


Figure 1: Dependency order. Candidate generation is completed before interface or survival outcomes are introduced.

1 Introduction and theorem strategy

1.1 Why candidate completeness is a theorem burden

A unique survivor has physical force only relative to a candidate domain known to be complete. If the domain omits a lawful carrier architecture, uniqueness may be an artifact of the search space rather than a consequence of the physical constraints. The problem is especially acute for neutrino propagation because the same target-facing behavior may, in principle, arise from very different microscopic architectures: deformations of the existing substrate, new local degrees of freedom, composites, environmental or hidden sectors, topological data, defects, boundary structures, or finite hybrids of these possibilities.

The present paper therefore asks a question that is logically prior to model selection:

Which physical carrier architectures exhaust the fixed neutrino-propagation target?

The answer cannot be a finite list of familiar mechanisms. It must be a source-generated structural universe whose coverage is proved independently of the later success or failure of its members.

Two previous results provide the fixed inputs. The first proves that the realized oscillation target requires the nested role complex

$$\mathcal{R}_\nu^{\text{osc}} = (r_{\triangleright\Delta}, r_\Delta; r_\Delta \succ r_{\triangleright\Delta}), \quad r_\Delta \Rightarrow r_{\triangleright\Delta}, \quad r_{\triangleright\Delta} \not\Rightarrow r_\Delta.$$

The second constructs the complete target-relative interface and the exact semantic quotient of its physical distinctions [14, 22]. Neither result identifies a carrier. The task here is to construct the domain of possible carriers without using candidate-specific values of that interface.

1.2 Source-first generation

The term *source-first* denotes a semantic dependency condition, not merely bibliographic provenance. Carrier families are generated from upstream physical roles, support and persistence categories, constituent kinds, lawful incidence relations, constitutive equations, and source parameter domains. Their existence and identity must be recoverable while all downstream interface values, compatibility results, rankings, and expected occupants are hidden.

The AASC kernel supplies the fixed-domain discipline for this order: reference preserves the target, standing blocks unsupported use, and irreversibility prevents an incomplete candidate domain from being repaired retrospectively as though the original search had been complete [18, 23]. The realized physical interior supplies the role-level ontology of locality, bounded transport, fields, particles, gauge structure, deformation response, vacuum, time, and causality [24]. The Standard-Model carrier results supply the already-realized substrate and the structure/parameter distinction [15, 17, 19, 25]. The symmetry results supply the controlled vocabulary for topological, extended, higher-form, higher-group, and non-invertible structures without promoting any of them into a preferred occupant [16, 26].

1.3 Four distinct completeness problems

Four notions of completeness must remain separate.

C1. Interface completeness. Every physical distinction admitted by the fixed target appears in the

target-relative interface.

- C2. Candidate completeness.** Every same-target occupancy-bearing carrier architecture is represented in the source-generated universe. This is the theorem proved here.
- C3. Survivor completeness.** Every generated candidate is evaluated under one simultaneous shared-witness constraint system. This is a subsequent problem governed by the pullback principle of [21].
- C4. Role-realization completeness.** Candidate, quotient, role-set, eligibility, compatibility, live-residue, and failure completeness jointly license realization through Role-Occupancy Closure [20].

No one of these statements implies the next. In particular,

$$X \in \mathcal{U}_v \not\Rightarrow \Pi_v(X) \in \mathfrak{S}_v^{\text{adm}}, \quad X \in \mathcal{U}_v \not\Rightarrow \text{RealizationLicensed}(X).$$

1.4 Comparison class

The result is target-relative rather than universal over arbitrary ontology. The comparison class $\mathfrak{C}_v^{(3)}$ contains same-target architectures whose occupancy-bearing part is typed by the fixed physical carrier ontology, participates through currently instantiated physical incidence and target attachment, and admits a finite carrier-incidence presentation. Finiteness concerns the number of source nodes and incidence relations in a presentation; it does not impose finite parameter domains, bounded multiplicity, a finite representation list, or a finite set of family instances.

This class includes local fields, composites, emergent sectors, TQFT-like structures, defects, solitons, boundary modes, physical media, hidden dilations, constitutive-law deformations, and arbitrary lawful finite hybrids. A proposed architecture requiring a new primitive support sort, persistence mode, incidence type, operator category, constitutive equation, or essentially infinitary presentation is not silently discarded. It is an explicit counterexample or a comparison-class extension.

1.5 Semantic and source constructions

The proof has two independently defined sides.

On the semantic side, a same-target physical world is reduced to its currently participating carrier core. This removes target-inert spectators while preserving every component lying on, or constitutively required by, an instantiated target-relative carrier path. The carrier-facing trace

$$\chi_{\text{car}} : \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) \rightarrow \mathcal{T}_{\text{car}}(\mathbb{T}_v^{(3)})$$

retains the full support, persistence, constituent, derivation, parameter, attachment, and incidence structure of that core. Carrier-semantic equivalence is defined independently through all lawful carrier uses, and the trace is proved to be a complete invariant of that equivalence.

On the source side, the carrier signature

$$\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)}) = (\text{Supp}_{\mathbb{T}_v^{(3)}}, \text{Pers}_{\mathbb{T}_v^{(3)}}, \text{Der}_{\mathbb{T}_v^{(3)}}, \text{Kind}_{\mathbb{T}_v^{(3)}}, \text{Attr}_{\mathbb{T}_v^{(3)}}, \text{Inc}_{\mathbb{T}_v^{(3)}}, \text{Att}_{\mathbb{T}_v^{(3)}})$$

is derived from the fixed physical ontology. The source carrier theory augments this signature with lawful operations, constitutive equations, and type-indexed activity/null structure. Its raw term algebra generates finite typed carrier-incidence diagrams. Only source-coherent instances enter the scientific family class, and candidate identity is assigned pointwise after source activity reduction. The resulting source theory therefore distinguishes local typing, constitutive objecthood, and later physical viability.

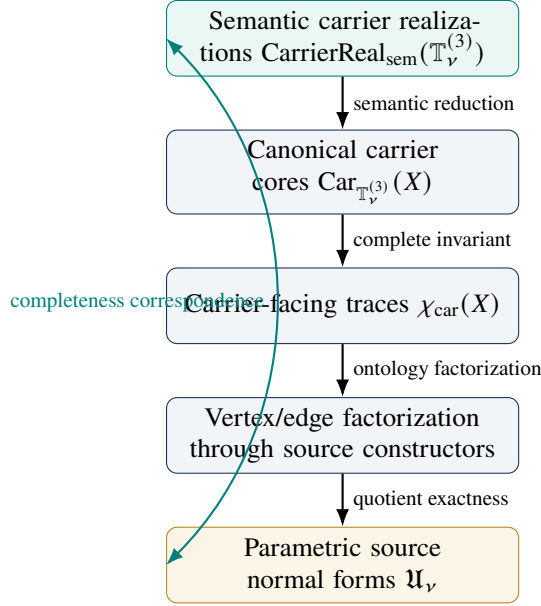


Figure 2: The semantic-to-source bridge. The source theory represents an independently defined semantic carrier class rather than defining physical possibility by enumeration.

1.6 Main theorem

The parametric candidate universe is

$$\mathfrak{U}_v := \left(\bigsqcup_{u \in \text{FamPres}_v} \Theta_u^{\text{part}} \right) / \simeq_{\text{inst}}, \quad (1)$$

where $[u, \theta]$ denotes the carrier-semantic class of the activity-reduced instance $X_u^{\text{act}}(\theta)$. The complete coherent locus Θ_u^{coh} remains part of the family data, while only points with nonempty active target participation enter the candidate universe. Different activity strata of one symbolic family may describe distinct carrier architectures or collapse to classes represented elsewhere.

Theorem 1.1 (Source-first physical carrier completeness, informal). *Within $\mathbb{T}_v^{(3)}$ and $\mathfrak{C}_v^{(3)}$, the carrier-trace map and source-normal-form reconstruction induce a canonical bijection*

$$\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) / \equiv_{\text{car}} \cong \mathfrak{U}_v.$$

The bijection is independent of candidate-specific interface values and every later survivor or occupancy datum. It preserves distinct carrier architectures even when the preceding interface map sends them to the same interface class.

The precise result is Theorem 17.5.

1.7 Proof architecture

The proof uses nine recurring methods.

Semantic reduction. Carrier-bearing components are identified by current target-relative participation rather than latent counterfactual capability.

Trace recovery. Primitive physical projections recover support, persistence, constituent identity, intrinsic attributes, parameter domains, incidence, and target attachment.

Coherent hypergraph representation. Every finite carrier core is represented by a typed incidence hyper-

graph together with the constitutive equations of the source objects it instantiates.

Source-coherence separation. Endpoint typing is distinguished from constitutive objecthood, and constitutive objecthood from later target compatibility.

Pointwise source activity. Source-null incidences are removed at each parameter point before carrier-path closure; nonnull incidences remain active without a numerical threshold.

Constructor induction. Source nodes, incidence hyperedges, attachments, coherence cells, and finite hybrids are reconstructed by finite induction.

Spectator reduction. Disconnected target-inert components are removed without deleting actually participating hidden, environmental, topological, boundary, or inflow structures.

Quotient exactness. Candidate identity is equality of the complete active carrier architecture up to lawful ontology-preserving reparameterization, not equality of the target interface.

Generation independence. The scientific universe is invariant under deletion or alteration of every downstream candidate outcome.

1.8 Organization

Section 2 states the target, comparison class, and inherited results. Sections 3–5 construct the semantic carrier class, carrier trace, and source ontology. Section 6 develops the source carrier theory and constructor algebra. Sections 7–12 analyze the principal carrier strata. Sections 13–15 establish hybrid closure, spectator reduction, pointwise activity, and the candidate quotient. Sections 16–17 prove the reconstruction and completeness theorems. The final sections establish generation independence, examine representative limiting cases, state the exact scope of the result, and formulate the subsequent simultaneous-constraint problem.

2 Fixed target and imported results

2.1 Target fiber

The target is

$$\mathbb{T}_v^{(3)} = (T_v^{\text{osc}}, \mathcal{R}_v^{\text{osc}}, \mathfrak{I}_v^{\text{adm}}, \mathfrak{B}_{\text{fixed}}, \mathfrak{O}_{\text{phys}}, \text{FP}). \quad (2)$$

Here T_v^{osc} is the established oscillation target, $\mathcal{R}_v^{\text{osc}}$ is the previously proved nested role complex, $\mathfrak{I}_v^{\text{adm}}$ is the complete target-relative interface type and semantic quotient, $\mathfrak{B}_{\text{fixed}}$ is the realized electroweak/lepton and physical-interior substrate, $\mathfrak{O}_{\text{phys}}$ is the imported role-level physical ontology, and FP is the finite-presentation condition stated below. No microscopic carrier coordinate is included in the target.

Definition 2.1 (Carrier comparison class). *The class $\mathfrak{C}_v^{(3)}$ consists of same-target physical carrier proposals X satisfying:*

- (i) *the source/detection boundary and target-facing physical semantics are those of T_v^{osc} ;*
- (ii) *the occupancy-bearing portion of X is typed by the imported physical carrier ontology and is selected by actual target-relative participation in the current architecture, not by latent capability under a hypothetical future coupling;*
- (iii) *the occupancy-bearing architecture admits a finite presentation by carrier nodes, instantiated target attachments, and typed physical incidence relations, with source activity evaluated pointwise at the instantiated source parameter value;*
- (iv) *primitive continuous and discrete parameter domains may be arbitrary source-defined sets or spaces;*

(v) *any new operator category, spacetime regime, source/detection semantics, or primitive support ontology is declared as a target/comparison-class extension rather than a same-scope repair.*

Remark 2.2 (Finite presentation is not a finite scan). *A family may contain arbitrary representation labels, multiplicities, couplings, topological data, or continuous parameters. The requirement is that each candidate architecture have a finite carrier-incidence description at the target-relative level. No symbolic family is replaced by a finite sample.*

2.2 Imported theorem package

Only the theorem-level roles listed in table 1 are imported.

Table 1: Imported results and their use in this paper.

Source	Imported result	Use here
Kernel and AMetric boundary	Fixed-domain identity, standing, no same-act repair, and exclusion of hidden boundary selectors.	Preserve target fidelity and the logical order between source formation and downstream evaluation.
Realized Physical Interior	Role-level locality, transport, field, particle, gauge, mass/deformation, vacuum, time, and causality architecture.	Derive a general physical carrier ontology without assuming a neutrino model.
Standard-Model carrier results	Existing matter/gauge carrier inventory, structure/parameter separation, target-update discipline, and carrier closure.	Fix the realized substrate and leave the completion-carrier variable open.
Symmetry and generalized-symmetry results	Global form, extended operators, higher-form/higher-group and non-invertible structures, with target-indexed relevance.	Admit nonlocal and defect-supported carrier kinds without presuming their necessity or viability.
Preceding role theorem	Carrier-neutral nested role complex and open-completion target family.	Fix the physical work to be occupied without importing microscopic content.
Preceding interface theorem	Complete target-relative interface and exact semantic interface quotient.	Supply target and attachment typing only; candidate-level interface values do not enter generation.
Simultaneous-constraint theorem	Pairwise compatibility insufficiency and shared-record pullback.	Reserve simultaneous compatibility for downstream analysis and forbid componentwise pre-gluing here.
Role-Occupancy Closure	Candidate, quotient, eligibility, compatibility, live-residue, and failure-completeness requirements.	Identify the later need for candidate completeness; no occupancy calculation is performed here.

2.3 Dependency restriction

Partition candidate-related information into an upstream source packet S_{src} and a downstream outcome packet O_{out} :

$$S_{\text{src}} = (\mathbb{T}_{\nu}^{(3)}, \mathfrak{D}_{\text{phys}}, \mathfrak{B}_{\text{fixed}}, \mathbb{S}_{\text{car}}(\mathbb{T}_{\nu}^{(3)}), \text{source parameter and activity domains}), \quad (3)$$

$$O_{\text{out}} = (\text{interface values, failures, survivor labels, rankings, expected occupants}). \quad (4)$$

Definition 2.3 (Source-first admissible datum). *A datum is source-first admissible iff it is fixed by S_{src} before any candidate-specific interface projection or simultaneous-constraint computation.*

Definition 2.4 (Endpoint-contaminated datum). *A datum is endpoint-contaminated iff its formation depends on a candidate's interface adequacy, constraint failure, survivor status, expected role assignment, named-model success, or downstream identity encoding those quantities.*

Theorem 2.5 (No-backflow). *The scientific generation map factors through the source packet alone:*

$$\text{Gen}_{\text{car}} = \widehat{\text{Gen}_{\text{car}}} \circ \pi_{S_{\text{src}}}.$$

Consequently, for every two downstream packets $O_{\text{out}}, O'_{\text{out}}$,

$$\text{Gen}_{\text{car}}(S_{\text{src}}, O_{\text{out}}) \simeq_{\text{cand}} \text{Gen}_{\text{car}}(S_{\text{src}}, O'_{\text{out}}).$$

Proof. Every primitive carrier sort is derived from $\mathfrak{D}_{\text{phys}}$, every parameter domain from a source type, every constructor from a source-licensed physical relation, and every pointwise activity/null decision from the type-indexed source activity structure in $\mathbb{S}_{\text{car}}(\mathbb{T}_{\nu}^{(3)})$. The coherent locus and its source-defined participating locus are therefore fixed before downstream evaluation. No formation rule has an argument slot for an interface value, failure, or survivor status. Any formal representation of the generator that receives such data must be independent of them. $\square$

Corollary 2.6 (Interface-value blindness). *Deleting every candidate-specific interface value leaves $\mathbb{S}_{\text{car}}(\mathbb{T}_{\nu}^{(3)})$, the constructor algebra, the full coherent family loci, pointwise activity profiles, participating loci, family normal forms, and $\simeq_{\text{cand}}$ unchanged.*

2.4 Permitted interface information

The preceding interface theorem contributes only the identity of the target fiber, the type of attachment required for later evaluation, and the fact that physically distinct carrier architectures may have the same interface image. This paper does not decide whether an attachment realizes $\mathfrak{S}_{\nu}^{\text{adm}}$. A candidate is generated because its carrier architecture is source-typed and source-coherent, not because it succeeds.

2.5 Excluded microscopic premises

Dirac and Majorana masses, the Weinberg operator, seesaw mechanisms, sterile states, decay, decoherence, nonstandard interactions, Lorentz-violating propagation, TQFT sectors, higher-form completions, and non-invertible defects are not primitive inputs to the source grammar. They enter only as representative constructions mapped into already-derived source families.

Nonclaim 2.7 (No known-solution exclusion). *The nonoccupancy result for the minimal massless-neutrino Standard Model does not exclude every existing-substrate or no-new-carrier deformation from $\mathfrak{U}_{\nu}$. Those branches are generated independently and may later fail under simultaneous physical constraints.*

2.6 Source objecthood versus downstream viability

Source typing requires physically interpretable carrier nodes, parameter domains, and source-constitutive coherence. Source activity then determines, at each coherent parameter point, which instantiated incidences and target attachments are physically nonnull and therefore enter the current carrier core. This decision is neither a numerical weak-coupling cutoff nor a downstream survivor result. Anomaly cancellation, complete gauge compatibility, vacuum stability, response multiplicity, interface adequacy, and a shared simultaneous witness record remain separate later constraints. Using them during formation would collapse candidate completeness into survivor completeness.

Logical boundary

Finite tables and computational consistency checks may test a particular representation of the source construction. They do not establish semantic completeness of the carrier signature or constructor algebra. The completeness proof is the representation theorem developed in Sections 16–17.

3 Semantic physical carrier realizations**3.1 Raw worlds and the carrier question**

Let $\mathfrak{W}_v^{\text{raw}}$ denote same-target physical presentations containing the fixed substrate together with any proposed completion structure. A raw presentation may contain laboratory apparatus, distant decoupled sectors, gauge representatives, coordinate choices, regularization data, and other structure that is irrelevant to the carrier question. The analysis does not classify entire possible worlds. It classifies the physical subconstruction that is *currently attached* to the open role complex through an instantiated carrier-incidence architecture.

Definition 3.1 (Currently instantiated target attachment). *For $W \in \mathfrak{W}_v^{\text{raw}}$, let $\text{Att}_{\mathbb{T}_v^{(3)}}(W)$ be the family of source-level physical attachment maps that are actually present in W , are physically non-null at the instantiated source point, and whose target is one of the open attachment slots of $\mathcal{R}_v^{\text{osc}}$ in the fixed target fiber. An attachment is recorded at source level and need not yet satisfy target-interface adequacy. A merely possible future coupling, an attachment available only after extending W , a source-null attachment at the instantiated parameter point, or a map introduced in a different target fiber is not an element of $\text{Att}_{\mathbb{T}_v^{(3)}}(W)$. The source-level null/non-null criterion is fixed independently in definition 6.4 and is not a downstream outcome.*

Definition 3.2 (Current target-relative carrier path). *A current target-relative carrier path in W is a finite source-coherent physical incidence hyperdiagram whose terminal morphism lies in $\text{Att}_{\mathbb{T}_v^{(3)}}(W)$, and whose remaining nodes, hyperedges, and maps are all currently instantiated and physically non-null in W . The permitted relations include support, persistence, derivation, co-carriage, composition, projection or dilation, binding, inflow, fusion or junction, environmental mediation, hidden-sector realization, and the source-constitutive maps required to interpret them. A relation that is present only as a symbolic slot but evaluates to the source-null relation at the instantiated parameter point is not an active hyperedge. No counterfactual extension, newly added coupling, replacement attachment, or downstream survivor condition may be used to form a current path.*

Definition 3.3 (Current carrier-support closure). *For a family $\mathcal{P}$ of current target-relative carrier paths, $\text{cl}_{\text{cur}}^W(\mathcal{P})$ is the smallest full physical subdiagram of W containing every node, hyperedge, and attachment in $\mathcal{P}$, together with every source-constitutive object or map required to interpret those paths as physical carrier structure.*

Definition 3.4 (Occupancy-bearing carrier reduction). *The occupancy-bearing reduction of W is*

$$\text{Car}_{\mathbb{T}_v^{(3)}}(W) := \text{cl}_{\text{cur}}^W \left(\bigcup_{P \in \text{Path}_{\mathbb{T}_v^{(3)}}(W)} P \right),$$

where $\text{Path}_{\mathbb{T}_v^{(3)}}(W)$ is the family of all current target-relative carrier paths in W . If no currently instantiated target attachment exists, the reduction is empty. If no new primitive carrier is introduced, $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$ may lie entirely in $\mathfrak{B}_{\text{fixed}}$ together with a law, constitutive, collective, or environmental incidence structure.

Definition 3.5 (Carrier-bearing component). *A component c of W is carrier-bearing precisely when it is a node, hyperedge, attachment, or source-constitutive map of $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$. Equivalently, c either lies on a current target-relative carrier path or is required to interpret such a path as a source-coherent physical*

object. Carrier bearing is therefore a property of the currently instantiated architecture. It may be witnessed by a lawful variation or deletion within that architecture, but the witness may not add a new carrier-incidence map, introduce a new coupling, or replace the target attachment.

Definition 3.6 (Latent carrier capability). *A component $c \subseteq W$ has latent carrier capability when there exists a source-coherent same-target extension $W \hookrightarrow W^+$ that adds at least one physical carrier-incidence relation or target attachment absent from W , such that c is carrier-bearing in W^+ . Latent capability by itself does not make c carrier-bearing in W .*

Lemma 3.7 (Latent capability is not present carrier bearing). *Let $c \subseteq W$ have latent carrier capability but lie on no current target-relative carrier path in W . Then*

$$c \notin \text{Car}_{\mathbb{T}_v^{(3)}}(W).$$

In particular, a source-coherent but completely decoupled copy of an admitted carrier architecture is absent from the occupancy-bearing reduction until an actual target-relative incidence is instantiated.

Proof. By hypothesis, every route by which c could reach the target requires at least one incidence or attachment absent from W . Current paths are formed only from the relations already instantiated in W . Hence no current path contains c , and source closure adds only objects required to interpret existing paths. Therefore c is not in $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$. $\square$

Proposition 3.8 (Activation by actual incidence). *Let Z be disjoint from $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$, and let W_e be a source-coherent same-target architecture obtained by adding an actually instantiated, physically nontrivial incidence or attachment e such that a current target-relative carrier path in W_e passes through Z and e . Then Z and e belong to $\text{Car}_{\mathbb{T}_v^{(3)}}(W_e)$. This is a new carrier architecture, not a reclassification produced by latent capability alone.*

Proof. The added relation creates a current path ending at an attachment in $\text{Att}_{\mathbb{T}_v^{(3)}}(W_e)$. By definition 3.4, the full source-constitutive closure of that path belongs to $\text{Car}_{\mathbb{T}_v^{(3)}}(W_e)$, including the participating part of Z and the relation e . Before e was instantiated, no such path existed. Thus the change is architectural. $\square$

Gauge labels, bases, mere bookkeeping witnesses, and target-inert spectators are not carrier-bearing. A field, defect, medium, hidden sector, or topological system may be capable of becoming a carrier in another architecture without belonging to the carrier core of the present one.

Lemma 3.9 (Canonicity of the carrier reduction). *The isomorphism class of $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$ is invariant under lawful same-target redescription of W .*

Proof. Lawful redescription preserves the target, currently instantiated physical attachments, source-coherent incidence, and the distinction between actual relations and counterfactual extensions. It therefore maps current carrier paths to current carrier paths and required source-constitutive closure to required closure. The induced map restricts to an isomorphism of the corresponding occupancy-bearing reductions. $\square$

3.2 Semantic carrier uses

The semantic prior must not depend on grammar membership.

Definition 3.10 (Carrier-semantic use). *A carrier-semantic use is a lawful same-target physical functional $U : \mathfrak{B}_v^{\text{raw}} \rightarrow \mathcal{Y}_U^\perp$ satisfying:*

- (i) **target typing**: U is meaningful without changing $\mathbb{T}_v^{(3)}$;
- (ii) **redescription invariance**: lawful same-target redescriptions preserve its value;

(iii) **carrier dependence**: any change of value is witnessed by a change in the current occupancy-bearing reduction $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$;

(iv) **spectator extensionality**: changing structure outside $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$ does not change the value.

The class of all such uses is $\text{CarUse}_{\text{sem}}(\mathbb{T}_v^{(3)})$.

Examples include asking for the support locus of a bearer, whether it is primitive or composite, which physical node carries a target attachment, which components share a hidden dilation, whether a boundary mode is attached to a bulk field, or what incidence relation makes a medium a co-carrier. These are carrier questions. Asking for a particular oscillation probability is an interface question and is not used to define candidate identity.

Definition 3.11 (Carrier-semantic equivalence). For $W, W' \in \mathfrak{B}_v^{\text{raw}}$,

$$W \equiv_{\text{car}} W' \iff U(W) = U(W') \quad \forall U \in \text{CarUse}_{\text{sem}}(\mathbb{T}_v^{(3)}).$$

Proposition 3.12. The relation $\equiv_{\text{car}}$ is an equivalence relation on the same-target comparison class.

Proof. Reflexivity and symmetry are immediate. Transitivity follows from equality of every carrier-semantic functional. $\square$

3.3 Semantic candidate class

Definition 3.13 (Semantic carrier realization). A semantic carrier realization is an occupancy-bearing reduction $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$ of some $W \in \mathfrak{C}_v^{(3)}$, before quotienting by carrier-semantic equivalence. Thus

$$\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) := \{\text{Car}_{\mathbb{T}_v^{(3)}}(W) : W \in \mathfrak{C}_v^{(3)}\}.$$

The quotient $\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})/\equiv_{\text{car}}$ is formed only in the candidate-completeness theorem. The term realization here means a physically interpretable carrier proposal at source level, not an interface-adequate or licensed occupant.

Nonclaim 3.14 (No survivor semantics). Membership in $\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$ does not imply gauge consistency, anomaly cancellation, vacuum compatibility, interface adequacy, or role eligibility. It means only that the current architecture has a physical carrier interpretation within the source comparison class.

3.4 Carrier-use factorization

Theorem 3.15 (Occupancy-bearing reduction). Every carrier-semantic use factors uniquely through the occupancy-bearing reduction up to lawful output equivalence. For each $U \in \text{CarUse}_{\text{sem}}(\mathbb{T}_v^{(3)})$ there exists $\widehat{U}$ such that

$$U(W) = \widehat{U}(\text{Car}_{\mathbb{T}_v^{(3)}}(W)).$$

Proof. By spectator extensionality, U is constant on all changes outside $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$. By carrier dependence, every change of U is detected inside that subconstruction. Hence U is constant on each fiber of the reduction map $W \mapsto \text{Car}_{\mathbb{T}_v^{(3)}}(W)$. The standard quotient factorization defines $\widehat{U}$. Uniqueness holds on the image of the reduction. $\square$

Corollary 3.16 (World structure and latent capability do not multiply candidates). Two raw worlds with carrier-equivalent occupancy-bearing reductions define the same candidate even if they differ in target-inert sectors, external apparatus, or decoupled sectors possessing latent carrier capability.

3.5 The physical meaning of “carrier”

The word carrier has two distinct uses in the upstream corpus. The foundational no-carrier theorem excludes a number, probability, invariant, certificate, or other constructed predicate from becoming a primitive authorizer of standing. The term *physical carrier* is used downstream: a field, sector, defect, environment, law-bearing process, or other physical subconstruction that participates in the current target-relative architecture occupying a required physical role once the admissibility regime is already fixed. No conflict follows.

3.6 Semantic falsifier

Semantic-carrier falsifier

A direct defeat of the semantic prior is a same-target physical question whose answer changes while $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$ remains fixed, or a component outside the proposed reduction whose *currently instantiated* incidence changes occupancy-bearing carrier work. A merely hypothetical future attachment is not such a defeat. A genuine case requires enlarging the reduction before any constructor completeness claim can stand.

4 Carrier-facing trace and carrier-semantic identity

4.1 Independent target carrier signature

The carrier trace is defined semantically rather than extracted from a finite candidate table. It is the full target-relative restriction of the occupancy-bearing reduction to the physical sorts and relations supplied by the fixed carrier ontology. Define

$$\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)}) = (\text{Supp}_{\mathbb{T}_v^{(3)}}, \text{Pers}_{\mathbb{T}_v^{(3)}}, \text{Der}_{\mathbb{T}_v^{(3)}}, \text{Kind}_{\mathbb{T}_v^{(3)}}, \text{Attr}_{\mathbb{T}_v^{(3)}}, \text{Inc}_{\mathbb{T}_v^{(3)}}, \text{Att}_{\mathbb{T}_v^{(3)}}). \quad (5)$$

The entries are source categories or groupoids, not fixed finite lists. They record support loci, persistence/dynamical modes, derivation/newness status, constituent carrier kinds, intrinsic typed data, lawful physical incidence, and target attachments. Their principal strata are developed in Sections 5–12.

Definition 4.1 (Carrier-facing trace). *For $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$, the carrier trace $\chi_{\text{car}}(X)$ is the isomorphism class of the finite $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$ -typed incidence diagram underlying $\text{Car}_{\mathbb{T}_v^{(3)}}(X)$, together with:*

- (a) *the type and physical identity class of every carrier-bearing node;*
- (b) *support, persistence, derivation, and constituent-kind labels;*
- (c) *all intrinsic typed attributes meaningful at source level;*
- (d) *the complete source-active, currently instantiated target-bearing incidence diagram;*
- (e) *the source-constitutive structure and commuting diagrams required for the instantiated source objects and maps;*
- (f) *primitive parameter domains and instantiated parameter values whenever they carry carrier-semantic content, modulo their lawful reparameterization groupoids;*
- (g) *source-active currently instantiated target attachment maps, recorded at source level but not evaluated for target-interface adequacy.*

The trace omits presentation, auxiliary enumeration labels and presentation metadata, target-inert spectator components, and candidate outcome data.

4.2 Primitive carrier projections

For every component of χ_{car} there is a target-typed projection evaluator:

$$\pi_{\text{supp}}, \pi_{\text{pers}}, \pi_{\text{der}}, \pi_{\text{kind}}, \pi_{\text{attr}}, \pi_{\text{inc}}, \pi_{\text{coh}}, \pi_{\Theta}, \pi_{\text{att}}.$$

Each is a carrier-semantic use: it is invariant under lawful redescription, depends only on the occupancy-bearing reduction, and is insensitive to target-inert world structure.

Lemma 4.2 (Trace separation). *If $\chi_{\text{car}}(X) \neq \chi_{\text{car}}(Y)$, then at least one primitive carrier projection or one finite incidence query has different values on X and Y .*

Proof. The trace is an isomorphism class of a finite typed diagram. Non-isomorphic finite source-coherent diagrams differ in node count/type, an intrinsic node attribute, an incidence hyperedge or its attachment, a constitutive coherence structure, a parameter domain, or a target attachment. Each difference is returned by the corresponding primitive projection or finite incidence query. $\square$

4.3 Trace extensionality

Lemma 4.3 (Semantic descent through the carrier trace). *Every $U \in \text{CarUse}_{\text{sem}}(\mathbb{T}_v^{(3)})$ descends through χ_{car} : there exists a lawful functional h_U such that*

$$U = h_U \circ \chi_{\text{car}}.$$

Proof. By theorem 3.15, U depends only on the occupancy-bearing reduction. By lawful-redescription invariance, it is constant on isomorphic $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$ -typed incidence diagrams with lawfully equivalent parameterization. These are exactly the fibers of χ_{car} . $\square$

Theorem 4.4 (Carrier-trace complete invariant). *For $X, Y \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$,*

$$\boxed{\chi_{\text{car}}(X) = \chi_{\text{car}}(Y) \iff X \equiv_{\text{car}} Y.}$$

Proof. If the traces are equal, every carrier-semantic use agrees by lemma 4.3; hence $X \equiv_{\text{car}} Y$. Conversely, if the traces differ, lemma 4.2 supplies a primitive carrier-semantic use distinguishing them, so $X \not\equiv_{\text{car}} Y$. $\square$

Corollary 4.5 (Architecture-sensitive, spectator-blind identity). *Carrier identity preserves every occupancy-bearing node, attribute, source-constitutive structure, parameter domain, incidence relation, and target attachment, while ignoring target-inert spectator structure and presentation.*

4.4 Candidate identity is finer than interface identity

The preceding interface theorem identifies two completions when every target-admitted interface evaluator agrees. They are identified here only when their carrier traces agree. The two relations therefore satisfy, in general,

$$X \equiv_{\text{car}} Y \implies \Pi_v(X) = \Pi_v(Y),$$

whenever both projections are defined, but not conversely.

Example 4.6 (Same interface, distinct dilation). *Two inequivalent hidden-sector architectures can induce the same record-complete reduced channel. If the hidden nodes, coupling incidence, or primitive parameter domains differ, the carrier traces differ even though the target-relative interface images agree. Stinespring uniqueness only identifies minimal dilations under the relevant isometry, not arbitrary enlarged architectures [2, 12, 28].*

Theorem 4.7 (No interface-quotient backflow). *Equality of the target-relative interface packets cannot by itself justify a candidate merger. A merge is lawful only when the complete carrier traces are isomorphic up to source-authorized reparameterization.*

4.5 Parameters, activity changes, and family identity

A parameter is primitive at the present scope when it is part of the independently defined source object or relation, whether or not varying it changes the active carrier trace at every point. Couplings, representation

labels, multiplicities, support topology, fusion data, boundary attachment maps, environmental parameters, and hidden-sector dimensions may therefore remain symbolic. Source semantics can make an incidence active on one parameter locus and null on another. The change is recorded pointwise rather than freezing the generic graph topology or waiting for subsequent simultaneous evaluation to discover the degenerate locus.

Definition 4.8 (Activity-aware carrier-trace parameterization). *A parametric source family consists of a source-coherent template H_u over its full coherent locus Θ_u^{coh} , together with the source-fixed pointwise activity reduction $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))$. On the target-participating locus, the carrier instance is*

$$X_u^{\text{act}}(\theta) = \text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))).$$

The raw source template and constitutive equations are fixed across the family, but the active carrier-trace type is permitted to change across source-defined activity strata. Within a fixed activity stratum the structural trace type is constant up to lawful source isomorphism. No subsequent simultaneous-constraint analysis survivor condition enters this pointwise assignment.

4.6 Trace falsifier

Carrier-trace falsifier

If two same-target carrier architectures have equal current traces but differ under a lawful carrier-semantic use, the trace is incomplete and all downstream normal-form and quotient theorems reopen. The correct response is to add the missing physical trace coordinate, not to create a special-case family label.

5 Primitive carrier dimensions and source ontology

5.1 From physical roles to a carrier signature

The source ontology is derived from the physical work required for a carrier to be meaningful, not from the names of successful neutrino models. Every actually target-participating, carrier-bearing component must answer a finite collection of structural questions: where is it supported, how does it persist, whether it is already present or newly introduced, what kind of constituent it is, which intrinsic physical attributes belong to it, how it attaches to the fixed target, and how it is related to other carrier-bearing components. These questions are prior to candidate survival and are available from the realized-physical ontology [24].

Definition 5.1 (Support category). $\text{Supp}_{\mathbb{T}_v^{(3)}}$ is the target-relative category of physical supports on which occupancy-bearing work can reside. Objects include, where meaningful in the source ontology, bulk-local or distributed support, localized nonperturbative support, extended support, boundary/edge support, topological/global support, environmental/medium support, hidden/enlarged support, and support inherited from the existing substrate. Morphisms are lawful support embeddings, restrictions, projections, and attachments.

Definition 5.2 (Persistence category). $\text{Pers}_{\mathbb{T}_v^{(3)}}$ records the physical mode by which a carrier-bearing component persists or transmits standing-bearing content: propagating, localized persistent, collective/condensed, topological, boundary-localized, open-system/co-carried, or law-bearing on existing nodes. Morphisms preserve the corresponding persistence semantics.

Definition 5.3 (Derivation category). $\text{Der}_{\mathbb{T}_v^{(3)}}$ records whether a carrier-bearing node is an existing-substrate object, a new primitive object, a composite or emergent object derived from a lower-level carrier diagram, or a law/constitutive bearer with no new primitive node.

Definition 5.4 (Constituent and attribute categories). $\text{Kind}_{\mathbb{T}_v^{(3)}}$ contains physical constituent sorts admitted by the upstream ontology—field/particle-like, composite/collective, topological, defect/operator, solitonic, boundary, environmental, hidden, or law/process objects—while $\text{Attr}_{\mathbb{T}_v^{(3)}}$ contains the intrinsic typed data

meaningful for those sorts, such as representation, spin/statistics, chirality, multiplicity, winding, fusion data, boundary labels, hidden-space dimension, or constitutive parameters.

Definition 5.5 (Incidence and attachment categories). $\text{Inc}_{\mathbb{T}_v^{(3)}}$ is the category of lawful physical relations among carrier-bearing nodes: coupling, shared support, composition/binding, emergence, projection/dilation, boundary attachment, fusion/junction, inflow, environmental co-carriage, constraint sharing, and any other source-admitted target-relative incidence. $\text{Att}_{\mathbb{T}_v^{(3)}}$ records the typed attachment of the carrier core to the fixed target and known substrate. Attachment is typed but not evaluated for interface adequacy.

The categories above are deliberately open under source-authorized objects and morphisms. Every admitted sort or morphism carries the constitutive identity laws that make it an object or map of that source category; those laws are collected into the source carrier theory in definition 6.1. A new physical support, incidence relation, or constitutive identity is not forced into an “other” box; it enlarges the source theory and reopens the theorem at the stated scope. This is how the theorem remains falsifiable without reducing completeness to a finite vocabulary list.

5.2 Primitive decomposition of a carrier node

For an actually target-participating carrier-bearing node v , define its primitive source type by

$$\text{type}(v) = (N_v, S_v, D_v, K_v, A_v, \Theta_v, T_v), \quad (6)$$

with entries in $\text{Der}_{\mathbb{T}_v^{(3)}}$, $\text{Supp}_{\mathbb{T}_v^{(3)}}$, $\text{Pers}_{\mathbb{T}_v^{(3)}}$, $\text{Kind}_{\mathbb{T}_v^{(3)}}$, $\text{Attr}_{\mathbb{T}_v^{(3)}}$, a primitive parameter domain, and a target-attachment slot.

Lemma 5.6 (Primitive coordinate necessity). *Every carrier-bearing node in $\mathfrak{C}_v^{(3)}$ has a well-defined value in each coordinate of equation (6), up to lawful source equivalence.*

Proof. An actually target-participating carrier-bearing node must have a physical support; otherwise its carrier work has no locus. It must have a persistence mode; otherwise it cannot bear or transmit physical work. It is either already realized, newly primitive, derived/composite, or a law-bearing operation on existing nodes; these cases exhaust derivation status. It has a constituent sort and whatever intrinsic attributes are meaningful for that sort. Its primitive freedom is represented by a source parameter domain, and its relevance to the target requires a typed target attachment. Each statement follows from the definition of carrier-bearing work and the fixed physical ontology, not from the constructor grammar. $\square$

Theorem 5.7 (Primitive carrier signature adequacy). *Within $\mathfrak{C}_v^{(3)}$, every connected actually target-participating carrier-bearing component is completely typed by the node coordinates equation (6), its incidence relations in $\text{Inc}_{\mathbb{T}_v^{(3)}}$, and the constitutive equations attached to the instantiated source sorts. If a same-target component requires a physical distinction or constitutive law not representable in the source theory, it is a direct falsifier of the current source ontology.*

Proof. By lemma 5.6, all node-local carrier information is represented. Carrier information that is not node-local concerns how the component is composed, attached, projected, supported, or co-instantiated with other components, hence belongs to $\text{Inc}_{\mathbb{T}_v^{(3)}}$ or $\text{Att}_{\mathbb{T}_v^{(3)}}$. The carrier trace contains both the local/incidence data and the source-constitutive structure by which those data form an object. There is no remaining target-relative carrier-semantic locus inside the declared physical ontology. $\square$

5.3 Principal source strata

The principal families are strata of $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$, not primitive assumptions. Let F_α denote the full subspace of traces with the corresponding derivation/support/persistence profile:

- F_{base} : existing-substrate carrier architecture,
- F_{local} : new local primitive carrier,
- F_{comp} : composite, collective, or emergent carrier,
- F_{top} : topological or global carrier,
- F_{ext} : extended or defect-supported carrier,
- F_{sol} : localized solitonic or nonperturbative carrier,
- F_{bdy} : boundary or edge carrier,
- F_{env} : environmental or medium co-carrier,
- F_{hid} : hidden or enlarged-state-space carrier,
- F_{law} : no-new-node law or constitutive carrier architecture.

A carrier trace can meet more than one condition. Such an object is not assigned an ad hoc mixed label; it is a hybrid incidence diagram whose vertices lie in the principal strata.

Theorem 5.8 (Carrier-stratum decomposition). *Every connected actually target-participating carrier node or derived component in $\mathfrak{C}_v^{(3)}$ lies in at least one principal source stratum determined by its primitive source type. Every finite carrier architecture is a finite incidence composition of such components.*

Proof. The derivation coordinate places a node in the existing, primitive, derived/composite, or law-bearing branch. The support and persistence coordinates refine it into local, topological, extended, solitonic, boundary, environmental, or hidden strata where applicable. A connected component with several support or persistence modes is already a finite subdiagram and therefore decomposes into its typed vertices and incidence. The second statement follows from the finite-presentation clause of definition 2.1. $\square$

The complete primitive carrier packet is summarized in figure 3.

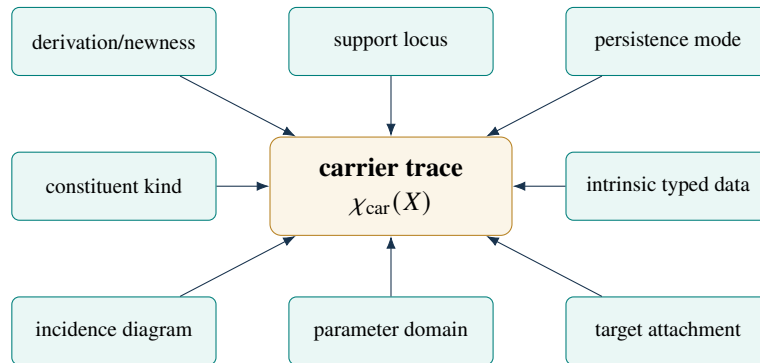


Figure 3: Primitive carrier dimensions. Named model families are strata of this structural packet, not premises.

5.4 Why no family is privileged

The local primitive stratum has no logical priority over topological, defect, environmental, hidden, or law-only strata. Local fields are familiar and often useful, but familiarity is not a source-formation rule. Conversely, an exotic topological or non-invertible description has no privileged status merely because it motivated the analysis. The source theorem admits both only when their physical sorts are present in $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$.

Nonclaim 5.9 (No one-to-one carrier ontology). *The principal strata are not asserted to be mutually exclusive ontological kinds. A composite may be local, a topological sector may support extended defects, a hidden sector may be environmental, and a boundary mode may be emergent. The complete object is the trace and incidence diagram.*

5.5 Open parameter spaces

A source parameter domain Θ_v may be continuous, discrete, categorical, lattice-valued, representation-valued, or a moduli object. No compactness, finite cardinality, or preferred measure is assumed. The only requirement is source typing and lawful reparameterization. Numerical ranges, anomaly-free subloci, and interface-compatible subdomains remain downstream constraints.

Corollary 5.10 (No finite-cutoff requirement). *Candidate completeness can be established over symbolic parameter domains. A finite scan of representation labels, multiplicities, couplings, or topological data is neither necessary nor sufficient.*

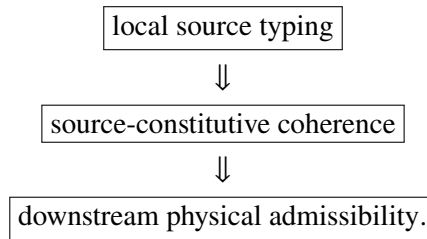
5.6 Ontology falsification burden

A critic can defeat theorem 5.7 by exhibiting a same-target carrier-bearing distinction that is neither a node-local source coordinate nor a lawful incidence/attachment relation. Such a distinction would enlarge $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$. The theorem does not shield itself by classifying unfamiliar physics as inadmissible.

6 Source-first constructor algebra and constitutive coherence

6.1 The three source-level burdens

The semantic side supplies physically interpretable carrier diagrams. The source side must generate them without reading whether any instance later satisfies the target-relative interface. Three logically distinct burdens must therefore be kept separate:



The first says that nodes and maps have declared source types. The second says that the typed data actually constitute the source object they purport to describe. The third asks whether that coherent source object satisfies gauge, anomaly, vacuum, interface, and simultaneous-realization constraints. Only the first two enter the candidate-completeness theorem.

6.2 The independently fixed source carrier theory

Definition 6.1 (Source carrier theory). *The source carrier theory is the essentially algebraic theory*

$$\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)}) = (\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)}), \mathfrak{M}_{\text{car}}, \mathfrak{E}_{\text{src}}, \mathfrak{A}_{\text{src}}),$$

where $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$ supplies the source sorts, $\mathfrak{M}_{\text{car}}$ supplies the lawful source operations and incidence maps, $\mathfrak{E}_{\text{src}}$ is the type-indexed family of constitutive equations and coherence diagrams that define those source objects and operations, and $\mathfrak{A}_{\text{src}}$ is the type-indexed source activity structure assigning physical participation support and null classes to incidence and attachment sorts. The theory is fixed from the source ontology of section 5 before candidate generation and contains no candidate-specific interface value, simultaneous-constraint outcome, survivor label, or expected occupant.

Definition 6.2 (Source-constitutive equations). *The family $\mathfrak{E}_{\text{src}}$ contains exactly the equations required for the declared source sorts and maps to be well defined. It includes, where the relevant sort is instantiated:*

- (i) *internal object laws, such as representation composition and identity laws;*
- (ii) *functoriality and compatibility of support, restriction, projection, and attachment maps;*
- (iii) *constituent/derivation compatibility for composite, emergent, condensed, or effective nodes;*
- (iv) *associativity, unit, module, junction, and higher-coherence diagrams for fusion or defect data;*
- (v) *the defining identities of claimed embeddings, reductions, dilations, quotient maps, and law/process actions;*
- (vi) *overlap and gluing commutativity for finite hybrids.*

These are constitutive equations, not viability conditions. Anomaly cancellation, vacuum compatibility, dynamical stability, target-interface adequacy, common-witness compatibility, and role occupancy are not members of $\mathfrak{E}_{\text{src}}$.

Example 6.3 (Representative constitutive laws). *A representation-labelled node must satisfy*

$$\rho(g_1 g_2) = \rho(g_1) \rho(g_2), \quad \rho(e) = \mathbf{1}.$$

A fusion datum with associator α must satisfy the pentagon identity and the relevant unit triangles. A module or boundary action must satisfy its module-associativity diagrams. A claimed dilation (V, π) must satisfy the identities defining that dilation, rather than merely having maps of the correct source and target types. A derived node must commute with the constituent and derivation maps that define it. These requirements are enforced before the object enters the candidate universe.

6.3 Pointwise source activity and null incidence

Source coherence says that an incidence map is a well-defined source object; it does not by itself say that the map performs nontrivial carrier work at every point of a symbolic parameter family. Activity semantics are therefore fixed at source level before any family is evaluated by subsequent simultaneous-constraint analysis.

Definition 6.4 (Source activity support). *For each source incidence or attachment sort $r \in \text{Inc}_{\mathbb{T}_V^{(3)}} \sqcup \text{Att}_{\mathbb{T}_V^{(3)}}$, the fixed source ontology supplies a lawful activity-support assignment*

$$\text{asupp}_r : m \mapsto \text{asupp}_r(m)$$

from an instantiated source morphism m of sort r to the typed sub-incidence on which that morphism performs constitutive carrier work. The assignment is invariant under lawful source isomorphism and reparameterization. The source-null class is

$$\mathfrak{N}_{\text{src}}(r) := \{m : \text{asupp}_r(m) = \emptyset\}.$$

A morphism is active iff its activity support is nonempty. Nullity is type-indexed source semantics: it is not defined by a universal numerical threshold and is not a the established interface theorem or subsequent simultaneous-constraint analysis failure predicate.

For a scalar interaction written in a source presentation as $m_g = gM$, source semantics can give $\text{asupp}(m_g) = \emptyset$ exactly when the physical interaction morphism is the zero morphism. By contrast, a zero value of an intrinsic parameter such as a mass, angle, or order parameter does not remove a carrier node merely because the number is zero; removal occurs only if the relevant source object or incidence belongs to its type-indexed null class. Rank-changing maps are treated by their actual activity support, which may shrink without becoming empty.

Definition 6.5 (Pointwise source activity reduction). *Let $H \in \mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)})$ be a fully instantiated source-coherent diagram. Its pointwise activity reduction $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H)$ is obtained by:*

- (i) *evaluating every source incidence and target attachment at the instantiated source point;*
- (ii) *replacing each such morphism by its activity support and deleting source-null incidences and attachments;*
- (iii) *taking the union of all remaining finite active hyperpaths terminating in an active target attachment; and*
- (iv) *adjoining the complete source-constitutive closure required to interpret those active paths.*

If no active target attachment remains, $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H) = \emptyset$. The reduction uses only $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$, the instantiated source datum, and the fixed target attachment types.

Proposition 6.6 (Activity is source-first and redescription invariant). *If $H \cong_{\text{src}} H'$ by a lawful source-equational isomorphism or lawful source reparameterization, then*

$$\text{ActRed}_{\mathbb{T}_v^{(3)}}(H) \cong_{\text{src}} \text{ActRed}_{\mathbb{T}_v^{(3)}}(H').$$

Moreover, the activity decision is unchanged by deleting, randomizing, or replacing all candidate-specific interface and later outcome data.

Proof. Lawful source isomorphism preserves each incidence sort, its instantiated source morphism up to the corresponding source isomorphism, and the type-indexed activity-support assignment. Hence it preserves the null/active partition and maps active target-relative hyperpaths to active hyperpaths. Source-constitutive closure is functorial under the same isomorphism. Downstream outcome data are not arguments of asupp or $\text{ActRed}_{\mathbb{T}_v^{(3)}}$. $\square$

Lemma 6.7 (Activity-coherence preservation). *For every $H \in \mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)})$,*

$$\text{ActRed}_{\mathbb{T}_v^{(3)}}(H) \neq \emptyset \implies \text{ActRed}_{\mathbb{T}_v^{(3)}}(H) \text{ is a finite model of the retained part of } \mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)}).$$

Deleting a null incidence cannot leave behind a node or coherence cell whose interpretation requires that incidence: dependent structure is removed unless independently supported by another active path.

Proof. The reduction first restricts source morphisms to their source-defined activity supports, then takes the full source-constitutive closure of the surviving target-relative paths. Every retained sort and operation therefore carries all equations required for its interpretation. Any constitutive datum depending only on a deleted null relation and on no surviving path lies outside the closure and is removed. Thus all equations inherited by the retained diagram continue to hold. $\square$

Theorem 6.8 (Pointwise activity–interpretation congruence). *Let $H \in \mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)})$ be fully instantiated and let W_H be its raw physical source interpretation before occupancy-bearing reduction. Then*

$$\boxed{\text{Car}_{\mathbb{T}_v^{(3)}}(W_H) \cong \text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_v^{(3)}}(H))}$$

whenever $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H) \neq \emptyset$. If $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H) = \emptyset$, then W_H has no currently instantiated target-participating carrier core and contributes no candidate instance.

Proof. By definitions 3.1 and 3.2, the semantic carrier core contains exactly the physically non-null incidence paths that are actually instantiated and terminate in a non-null target attachment, together with their constitutive closure. By definitions 6.4 and 6.5, $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H)$ constructs exactly the same source subdiagram before interpretation. The two constructions therefore retain and discard the same physical relations and required constitutive data. Source coherence of the retained diagram is lemma 6.7. If no active target attachment remains, the semantic path family is empty as well. $\square$

6.4 Raw typed term algebra

The raw constructive object is a free finite typed-hypergraph algebra over $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$, enriched by source-authorized derivation and attachment maps.

Definition 6.9 (Primitive node constructor). *For every source object*

$$\kappa = (N, S, D, K, A, \Theta, T)$$

in the node-type space of $\mathfrak{D}_{\text{car}}(\mathbb{T}_v^{(3)})$, $\text{Atom}(\kappa)$ introduces one carrier node with those types and an open parameter $\theta \in \Theta$.

Definition 6.10 (Derived-node constructor). *For every source-authorized derivation map $d \in \text{Der}_{\mathbb{T}_v^{(3)}}$,*

$$\text{Derive}_d(H_1, \dots, H_m)$$

introduces a composite, emergent, condensed, effective, or otherwise derived carrier node together with its constituent subdiagram and derivation witness. The constructor does not erase the constituent architecture unless source equivalence proves that the derived object alone is the carrier trace.

Definition 6.11 (Incidence constructor). *For every lawful relation type $\iota \in \text{Inc}_{\mathbb{T}_v^{(3)}}$,*

$$\text{Incede}_\iota(v_1, \dots, v_k; w_1, \dots, w_\ell)$$

introduces a typed incidence hyperedge among existing nodes. Its domain and codomain typing are fixed by the source relation.

Definition 6.12 (Attachment constructor). *For every $a \in \text{Att}_{\mathbb{T}_v^{(3)}}$, $\text{Attach}_a(H, \mathbb{T}_v^{(3)})$ records an instantiated source-level target attachment of the carrier diagram. The constructor asserts that the attachment is presently part of the candidate architecture and is correctly typed; it does not assert that the attachment realizes $\mathfrak{S}_v^{\text{adm}}$. A diagram lacking any instantiated route to the target is world structure rather than an occupancy-bearing candidate until such an attachment or carrier-incidence path is actually introduced.*

Definition 6.13 (Finite hybrid constructor). *If $H_1, \dots, H_n$ are source-generated carrier diagrams and E is a finite set of lawful incidence maps joining them, then*

$$\text{Hybrid}(H_1, \dots, H_n; E)$$

is the finite union equipped with those maps. It becomes a source candidate only when the required overlap and coherence diagrams commute.

Definition 6.14 (Raw source carrier algebra). *The raw source carrier algebra $\mathcal{G}_{\text{car}}^{\text{raw}}(\mathbb{T}_v^{(3)})$ is the least class containing all primitive node constructors and closed under derived-node formation, incidence introduction, target attachment, finite hybrid composition, lawful source redescription, and spectator stripping. Membership in $\mathcal{G}_{\text{car}}^{\text{raw}}(\mathbb{T}_v^{(3)})$ asserts formation and typing, not constitutive coherence.*

6.5 Source coherence and the scientific candidate algebra

Definition 6.15 (Source-constitutive coherence). *For a finite raw source diagram $H \in \mathcal{G}_{\text{car}}^{\text{raw}}(\mathbb{T}_v^{(3)})$,*

$$\text{Coh}_{\text{src}}(H)$$

holds iff the interpretation of every instantiated source sort and operation in H satisfies the corresponding equations in $\mathfrak{E}_{\text{src}}$, and every required overlap, attachment, derivation, and higher-incidence diagram commutes. Equivalently, H is a finite model of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$.

Definition 6.16 (Source-coherent carrier algebra). *The source-coherent carrier algebra is*

$$\mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)}) := \{H \in \mathcal{G}_{\text{car}}^{\text{raw}}(\mathbb{T}_v^{(3)}) : \text{WellTyped}(H) \wedge \text{Coh}_{\text{src}}(H)\} / \cong_{\mathfrak{E}_{\text{src}}},$$

where $\cong_{\mathfrak{E}_{\text{src}}}$ identifies only lawful source-equational presentations. In the remainder of the manuscript, $\mathcal{G}_{\text{car}}(\mathbb{T}_v^{(3)})$ denotes $\mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)})$, not the unconstrained raw term algebra.

Definition 6.17 (Source-formed carrier proposal). *A source-formed proposal is an isomorphism class of fully instantiated finite source-coherent terms $H \in \mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)})$ for which $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H) \neq \emptyset$. Its scientific interpretation is the source interpretation of $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H)$, not the unreduced symbolic template. A disconnected source-coherent component, or a component connected only through a source-null incidence at that parameter point, is lawful source structure but is stripped before candidate identity is formed. A raw term that fails Coh_{src} is not a candidate of the asserted source kind. A term requiring a new target coordinate, source sort, source operation, constitutive equation, or activity-support law returns an explicit source-ontology or target update rather than entering $\mathfrak{U}_v$.*

Theorem 6.18 (Constitutive coherence is not downstream viability). *Source-constitutive coherence is necessary for candidate formation and is independent of downstream viability. More precisely:*

- (i) *a raw typed term that violates $\mathfrak{E}_{\text{src}}$ fails to constitute the source object it names and is excluded before candidate formation;*
- (ii) *a source-coherent proposal may still fail anomaly cancellation, vacuum compatibility, dynamical stability, target-interface adequacy, subsequent simultaneous-constraint analysis simultaneous compatibility, or a later role-occupancy analysis occupancy.*

Proof. Item (i) follows from the definition of a model of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$: the constitutive laws are part of the identity conditions of the source object. Item (ii) follows because none of the listed downstream predicates occurs in $\mathfrak{E}_{\text{src}}$. The first layer establishes that an object of the claimed source kind exists; later layers ask whether that object realizes the neutrino target. $\square$

Theorem 6.19 (Source-coherence and activity soundness). *There is a source interpretation map*

$$\text{Interpret}_{\text{src}} : \{H \in \mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)}) : \text{ActRed}_{\mathbb{T}_v^{(3)}}(H) \neq \emptyset\} \longrightarrow \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$$

such that every source-formed carrier proposal has a physically interpretable, actually target-participating carrier trace in the declared comparison class. No raw term failing source typing or Coh_{src} , and no fully instantiated term with empty pointwise active target core, lies in the candidate domain of the present paper.

Proof. Interpret primitive nodes as the physical source objects from which their types were drawn and source morphisms as their instantiated physical relations. Because the term is a model of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$, all identities needed for those local interpretations to assemble into one source object hold. Apply the independently fixed activity reduction before candidate identity is formed. By lemma 6.7, the retained diagram remains source coherent, and by theorem 6.8 its interpretation is exactly the semantic occupancy-bearing carrier core. Source-null degenerate incidences and any components supported only by them are therefore removed pointwise, while every non-null active relation remains regardless of numerical smallness. This proves source interpretation and current carrier participation, not downstream viability. $\square$

Theorem 6.20 (Semantic carrier traces are source coherent). *For every $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$, every finite source presentation of $\chi_{\text{car}}(X)$ obtained from the fixed source ontology satisfies Coh_{src} .*

Proof. A semantic carrier realization already instantiates the source objects and physical incidence it claims to contain. The equations in $\mathfrak{E}_{\text{src}}$ are exactly the constitutive identity laws of those source objects and maps.

Failure of one of those laws would mean that the presentation did not represent the claimed source object. Hence the induced finite presentation is a model of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$. $\square$

6.6 Finite coherent-hypergraph presentation

Let $H(X)$ be the finite typed incidence diagram of $\chi_{\text{car}}(X)$. Vertices are primitive or derived carrier nodes; hyperedges are physical incidence relations; source attributes include any associator, module, dilation, derivation, or gluing data required by the relevant sort; and $\text{Coh}_{\text{src}}(H(X))$ records the constitutive equations they satisfy. The diagram retains non-factorizing many-body and higher incidence and need not reduce to pairwise couplings.

Lemma 6.21 (Vertex generation). *Every vertex of $H(X)$ is generated by either a primitive node constructor or a finite derived-node constructor, with its source-coherence data.*

Proof. By theorem 5.8, each vertex is existing, primitive, derived/composite, or law-bearing. Existing and new primitive nodes use **Atom**; derived/composite nodes use **Derive**; law-bearing structures are represented by source process nodes or incidence on existing nodes. Their constitutive laws are inherited from $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$. $\square$

Lemma 6.22 (Incidence and coherence generation). *Every incidence map and every constitutive coherence cell of $H(X)$ is generated by the corresponding source operation and equation in $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$.*

Proof. Every physical relation in the carrier trace is an object of $\text{Inc}_{\mathbb{T}_v^{(3)}}$, and every identity required to make that relation part of the claimed source object lies in the type-indexed family $\mathfrak{E}_{\text{src}}$. The raw constructor introduces the map; the coherence condition restricts to the models in which the required diagrams commute. $\square$

Theorem 6.23 (Coherent carrier-constructor factorization). *Every $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$ admits a finite source-coherent derivation in $\mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)})$ whose interpreted carrier trace is isomorphic to $\chi_{\text{car}}(X)$.*

Proof. Choose a finite presentation of $H(X)$, permitted by definition 2.1. By theorem 6.20, the presentation is source coherent. Generate all vertices, then the finitely many incidence maps, attachments, and coherence data by the preceding lemmas. The resulting model of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$ has the same finite typed diagram, attributes, parameter point, constitutive structure, and target attachment as $\chi_{\text{car}}(X)$. The chosen construction order is not part of candidate identity. $\square$

Corollary 6.24 (Source-coherent hybrid closure). *Every lawful finite hybrid of source-generated carrier components is generated by the same algebra, provided the source overlap and gluing equations hold. A typed but incoherent gluing is excluded at source formation rather than deferred to later physical evaluation.*

6.7 Saturation and the unlisted-constructor objection

The grammar is not a finite list of named software rules. It is saturated over every source object and lawful operation in $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$. A newly recognized physical relation on existing source types is admitted automatically if it is already an operation of the source theory and its constitutive equations hold. If a new same-target relation requires a genuinely new source sort, operation, or constitutive equation, the source ontology changes and the source-completeness theorem reopens. This is the candidate-side analogue of the target-evaluator saturation used in the established interface theorem.

Theorem 6.25 (No hidden output dependence). *No source sort, constitutive equation, generated node, incidence map, source-null/activity-support law, family boundary, parameter domain, pointwise activity reduction, or candidate merge depends on the established interface theorem pass/fail values, subsequent simultaneous-constraint analysis survivor data, or an expected occupant label.*

Proof. The source theory is fixed from the upstream ontology. Its constructors accept only source objects, previously generated coherent diagrams, source parameters, and target-type attachments. Its coherence equations and type-indexed activity-support laws are source semantics of those objects and relations. There is no output-status argument. The result also follows from theorem 2.5. $\square$

6.8 Constructive versus scientific completeness

Finite enumeration can list selected templates or instantiate parameter samples. That operation is not the completeness theorem. Scientific completeness follows from theorem 6.23: every semantic carrier trace has a source-coherent derivation because the grammar is saturated over the independently defined source theory.

Source-coherence falsifier

The theorem is falsified by either of the following same-target constructions: (i) a coherent physical carrier diagram containing a source sort, operation, or constitutive equation absent from $\mathbb{S}_{\text{car}}(\mathbb{T}_\nu^{(3)})$; or (ii) a source-coherent finite carrier architecture whose trace has no derivation in $\mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_\nu^{(3)})$. A malformed associator, representation, dilation, or gluing map is not a new candidate; it is a failure to instantiate the claimed source object.

7 Existing-substrate and no-new-carrier branches

7.1 Why the null-newness branch comes first

A source-first universe must remain adversarial to the expectation of new matter. The first generated stratum therefore contains every same-target carrier architecture whose primitive carrier nodes are already part of the fixed physical substrate. The preceding role analysis proves that one particular minimal massless-neutrino fiber does not occupy the role complex; it does not prove that every lawful reorganization, collective structure, constitutive law, or environmental embedding of existing nodes is impossible.

Definition 7.1 (Existing-substrate family). F_{base} is the class of carrier traces whose primitive nodes lie in $\mathcal{B}_{\text{fixed}}$, while incidence, effective organization, collective structure, or law-bearing processes may be newly introduced provided they are source-typed and same-target.

Theorem 7.2 (Existing-physics inclusion). Every same-target occupancy-bearing architecture with no new primitive carrier node and with all support inherited from $\mathcal{B}_{\text{fixed}}$ is represented in F_{base} or in its law-only subfamily.

Proof. The primitive nodes are generated from existing-substrate source objects. Any new occupancy-bearing content must therefore appear in the incidence, derivation, collective, environmental, or process-law coordinates. These are generated by the source constructor algebra. No local-new-carrier assumption is required. $\square$

7.2 Law-only and constitutive realizations

Definition 7.3 (Law-only carrier architecture). A law-only architecture has no new primitive carrier node. Its new occupancy-bearing content is a source-typed process, constitutive map, transport law, nonlocal kernel, effective channel, or incidence relation acting on existing carrier nodes.

The word “law-only” does not mean “nonphysical.” A constitutive relation, channel, or nonlocal kernel can carry standing-bearing physical content when it changes target-relevant propagation or co-instantiation. Nor does it mean “free parameterization”: the law object and its parameter domain are explicit parts of the carrier trace and are later subject to subsequent simultaneous constraints.

Proposition 7.4 (Law versus presentation). A change of equation, channel, constitutive map, or transport kernel defines a law-bearing candidate only when it changes the carrier trace. A basis, gauge, parametrization,

or representation of the same law is candidate skin.

Proof. The carrier trace retains law/process objects and their incidence on existing nodes while quotienting lawful redescription. A purely presentational change leaves the trace invariant; a physical law change does not. $\square$

7.3 Law-only versus integrated-out hidden carrier

An effective law on existing nodes and an explicit hidden-sector dilation can induce the same target-relative interface. They remain different candidates unless a source equivalence theorem identifies their complete carrier traces.

Theorem 7.5 (Law/hidden distinction). *Let X_{law} be a law-only architecture and X_{hid} an enlarged-state-space architecture. Then*

$$X_{\text{law}} \equiv_{\text{car}} X_{\text{hid}}$$

only if the hidden nodes and incidence are carrier-semantically eliminable and the induced law object preserves the complete carrier trace. Equality of reduced channels or the target-relative interface images is insufficient.

7.4 Collective reorganization of existing carriers

A collective or condensed phase formed from existing substrate nodes can be occupancy-bearing without enlarging primitive inventory. Such a candidate belongs to $F_{\text{base}} \cap F_{\text{comp}}$, and its collective node, order parameter where physical, constituent incidence, and target attachment remain explicit.

Nonclaim 7.6 (No existence claim for a successful no-new-carrier realization). *The analysis proves that the branch is generated, not that any member realizes $\mathcal{R}_v^{\text{osc}}$ or passes the target-relative interface.*

7.5 Open parameter domain

The law-only branch includes symbolic families of admissible process types and source-defined parameters. The source universe does not impose a preferred derivative order, locality ansatz, operator dimension, or numerical cutoff unless the comparison class itself requires one. A proposal outside the declared process category is a comparison-class extension, not an unlisted instance silently discarded.

8 Local primitive carrier families

8.1 Generic local carrier type

The local branch is generated from source-local carrier sorts, not from a list of neutrino mass mechanisms. Let v be a new primitive local node. Its source type may be written schematically as

$$\kappa_{\text{loc}} = (d, \mathfrak{s}, \rho_G, \epsilon, \mu, \mathcal{L}, \Theta), \quad (7)$$

where d is the spacetime regime fixed by the target, $\mathfrak{s}$ is a source spin/statistics type, ρ_G a representation object of the fixed gauge/global-form category when applicable, ϵ records chirality/reality structure, μ multiplicity data, $\mathcal{L}$ local support and persistence typing, and Θ the primitive parameter domain. This notation is a source schema, not a Lagrangian ansatz.

Definition 8.1 (Local primitive family). *F_{local} is the family of carrier traces containing at least one new primitive bulk-local or localized field/particle-like node whose source type is equation (7), together with arbitrary lawful finite incidence and target attachment.*

8.2 No representation cutoff

The representation coordinate ranges over the source-defined representation category of the fixed gauge/global form. The candidate construction does not list only small representations, anomaly-free representations, or phenomenologically popular charges. If the source category is infinite, the family is symbolic.

Theorem 8.2 (Local carrier generation). *Every same-target architecture whose occupancy-bearing core contains new primitive local carrier nodes has a source normal form in F_{local} , with open representation, multiplicity, chirality/reality, and parameter data.*

Proof. Each primitive local node is an object of the local subcategory of $\text{Supp}_{\mathbb{T}_V^{(3)}} \times \text{Pers}_{\mathbb{T}_V^{(3)}} \times \text{Kind}_{\mathbb{T}_V^{(3)}} \times \text{Attr}_{\mathbb{T}_V^{(3)}}$, hence is introduced by `Atom`. Every local coupling, mixing, source/detector attachment, or shared support relation is an incidence object and is introduced by `Incide`. Finiteness of the carrier presentation yields a finite source term. No anomaly or interface value is inspected. $\square$

8.3 Fermionic, bosonic, and general source types

The familiar fermionic and bosonic branches are retained, but the theorem is formulated at the source category level so that any additional local statistics or generalized local object admitted by the comparison class is included. A spin/statistics relation may constrain the source type; the theorem does not use that constraint to select among candidates beyond source well-typing [6, 7, 10, 31].

8.4 Chiral, vectorlike, real, and pseudoreal structures

Chirality and reality type are intrinsic source attributes where meaningful. The generator keeps chiral, vectorlike, real, pseudoreal, and mixed representation structures as separate carrier families or parameter strata. It does not pre-cancel anomalies by pairing them, since that would use subsequent simultaneous-constraint analysis consistency result as a generation rule.

Nonclaim 8.3 (No anomaly filtering). *A source-typed local carrier may later fail gauge, mixed, global, or gravitational anomaly constraints. The source-generated universe includes it; a later simultaneous-constraint analysis records the failure in the shared witness structure.*

8.5 Multiplicity without cardinality backfitting

Multiplicity is a source parameter. It is not capped because a known model uses one, two, or three copies, and it is not set after inspecting which multiplicity passes. Symbolic multiplicity domains are retained:

$$\mu \in \mathbb{N}_0 \quad \text{or a source-defined submonoid.}$$

A downstream theorem may restrict μ ; the present theorem may not.

Proposition 8.4 (Parametric multiplicity closure). *A local family with symbolic multiplicity parameter represents every finite multiplicity instance without enumerating them separately. Candidate completeness therefore does not require a finite multiplet table.*

8.6 Local interactions and attachment

Candidate generation records only source-licensed incidence types: local couplings, mixing/transition maps, gauge attachment, vacuum coupling, boundary/source/detector attachment, and possible coupling to hidden or environmental nodes. Numerical coupling values and operator coefficients remain open. Whether an interaction is allowed by every gauge, anomaly, vacuum, and EFT condition belongs to subsequent simultaneous-constraint analysis.

8.7 Existing versus new local structure

Theorem 8.5 (Newness classification). *A local node is new at the present scope iff it is not carrier-semantically isomorphic to a node already in $\mathcal{B}_{\text{fixed}}$ and cannot be derived from the existing substrate by a*

source-authorized composite/emergent constructor without adding primitive content.

Proof. The derivation coordinate of the carrier trace distinguishes existing, primitive-new, and derived nodes. A mere relabeling or alternative field representation does not change the trace. A genuinely new primitive node changes it. $\square$

8.8 Named mechanisms as representative constructions

After the grammar is fixed, familiar Dirac, Majorana, Weinberg, and seesaw constructions can be mapped into local or hybrid normal forms. Their success in landing in the grammar is a coverage check only. The family definitions above do not mention them.

Local-branch omission test

A previously unlisted local field or particle-like object defeats the local generator only if its source support, persistence, constituent type, attributes, or incidence cannot be represented by the open local source categories. An unfamiliar representation label alone is already covered by the symbolic representation parameter.

9 Composite, collective, and emergent carrier families

9.1 Derived carriers are not primitive local fields

A physical carrier can be occupancy-bearing even when it is not an elementary node of the microscopic description. Bound states, collective modes, condensates, quasiparticles, and other emergent sectors can possess stable support, transport, and target attachment. Effective field theory and local quantum physics provide familiar examples of descriptions in which the operative low-energy bearer is not identical to a fundamental field coordinate [9, 10, 30]. The analysis therefore treats derivation status as a primitive source coordinate.

Definition 9.1 (Composite carrier datum). *A composite carrier datum is a tuple*

$$C = (H_{\text{const}}, d_{\text{bind}}, S_C, D_C, A_C, \Theta_C),$$

where H_{const} is a finite constituent carrier diagram, d_{bind} is a source-authorized binding, collective, condensation, quotient, or emergent derivation map, and the remaining entries give the support, persistence, intrinsic attributes, and primitive parameter domain of the derived node.

Definition 9.2 (Composite/emergent family). F_{comp} is the class of carrier traces containing at least one node introduced by $\text{Derive}_{d_{\text{bind}}}$ from a nonempty constituent diagram.

9.2 Constituent retention

A composite node does not automatically erase its constituents from candidate identity. Whether the constituents remain occupancy-bearing depends on their physical incidence after the derived node is formed.

Lemma 9.3 (Constituent-retention criterion). *Let c be a derived carrier node with constituent diagram H_{const} . A constituent subdiagram is removed from the carrier core only if every carrier-semantic use factors through c and no target-bearing incidence separately detects the constituent structure.*

Proof. If a carrier-semantic use distinguishes constituent support, statistics, hidden incidence, or target attachment, the constituent diagram is part of $\text{Car}_{\text{TV}(3)}$ by definition. If no such use exists, replacing the constituent realization by another source-equivalent derivation leaves the carrier trace unchanged, and the constituent detail is candidate skin at the present scope. $\square$

Theorem 9.4 (Composite generation). *Every finitely presented same-target composite, collective, condensed, or emergent carrier architecture has a source normal form in F_{comp} , including its constituent diagram, derivation map, residual constituent incidence, and open parameter domain.*

Proof. Generate the finite constituent diagram by theorem 6.23. Apply the source-authorized derived-node constructor, retain every constituent edge required by the preceding lemma, and add the target attachments. The resulting trace is isomorphic to the semantic composite trace. $\square$

9.3 Emergent local behavior does not imply a new primitive node

A derived carrier may admit a local effective field presentation. That fact does not reclassify it as a primitive local carrier. Candidate identity retains its derivation status and constituent incidence unless a source equivalence theorem proves that those differences are carrier-semantically invisible.

Proposition 9.5 (Primitive/composite distinction). *A primitive local node and a composite node with the same target-relative interface are candidate-equivalent only if their carrier traces agree, including derivation status and every carrier-detectable constituent relation.*

9.4 Collective and condensate carriers

A collective mode or condensate can be built from existing or new nodes. Its source trace records the order/collective variable only when it has physical standing; gauge-variant representatives are not promoted. The vacuum and gauge support required to make the condensate meaningful remain explicit attachment/incidence data, but whether the collective state is dynamically stable or consistent is deferred.

Definition 9.6 (Collective co-carriage). *Collective co-carriage is a hyperedge connecting a derived collective node to the constituent and environmental/vacuum nodes whose joint state is required for its persistence.*

Corollary 9.7 (Existing-substrate composite branch). *A composite built entirely from $\mathfrak{B}_{\text{fixed}}$ belongs to $F_{\text{base}} \cap F_{\text{comp}}$. It is not removed merely because no new elementary field is introduced.*

9.5 Dual descriptions and effective equivalence

Microscopically different architectures can yield equivalent low-energy effective carriers. This paper does not identify them merely because a low-energy operator basis agrees. A lawful candidate merge requires carrier-trace equivalence, including the declared EFT matching, hidden constituent incidence, and parameter-domain identification. target-interface equality is even weaker and cannot merge them.

Composite omission burden

A purported new composite family is already generated if it consists of a source-typed constituent diagram, a lawful derivation map, and target-bearing incidence. It falsifies the source-completeness theorem only if the physical derivation or collective incidence requires a new source sort not contained in $\text{Der}_{\mathbb{T}_V^{(3)}}$ or $\text{Inc}_{\mathbb{T}_V^{(3)}}$.

10 Topological, extended, and defect-supported carriers

10.1 Why point-local inventories are insufficient

The local field content and Lie algebra of a theory need not determine its global form, genuine line operators, monopole sectors, higher-form symmetries, defect category, or non-invertible fusion data [1, 3, 4, 8, 29]. The preceding interface theorem correctly refrains from making such structures universal coordinates of the point-local neutrino interface. The present theorem addresses a different question: can an independently generated carrier architecture use topological or extended support? The source ontology answers yes whenever such support and incidence are physically meaningful, even if the subsequent analysis later finds the target attachment inert or inconsistent.

10.2 Topological carrier family

Definition 10.1 (Topological carrier node). *A topological carrier node is an actually target-participating carrier-bearing source object whose persistence and instantiated target incidence depend on global/topological data rather than a local propagating degree of freedom. Its trace may contain global form, topological state space, flux/winding class, sector labels, braiding or linking data, and topological response parameters.*

Definition 10.2 (Topological family). *F_{top} is the class of traces with at least one topological/global carrier node or topological incidence relation that lies on a currently instantiated target-relative carrier path to the target attachment.*

Theorem 10.3 (Topological generation). *Every finitely presented topological carrier architecture admitted by the source support and incidence categories has a source normal form in F_{top} , with symbolic global/topological parameter data.*

Proof. Topological nodes are source objects in $\text{Supp}_{\mathbb{T}_V^{(3)}} \times \text{Pers}_{\mathbb{T}_V^{(3)}} \times \text{Kind}_{\mathbb{T}_V^{(3)}} \times \text{Attr}_{\mathbb{T}_V^{(3)}}$, and their linking, braiding, inflow, or attachment relations are source incidence objects. Their constitutive topological identities are enforced by Coh_{src} . The general coherent-constructor factorization applies without assuming a local field realization. $\square$

10.3 Extended and defect-supported carriers

Definition 10.4 (Extended/defect carrier). *An extended or defect-supported carrier is an occupancy-bearing line, surface, higher-dimensional support, codimension defect, topological interface, or junction object whose physical incidence with bulk, boundary, or other extended nodes is part of the carrier trace.*

Definition 10.5 (Extended family). *F_{ext} is the class of traces containing at least one extended/defect node or one fusion/junction incidence that cannot be eliminated by carrier-semantic equivalence.*

Theorem 10.6 (Extended/defect generation). *Every finitely presented source-coherent same-target carrier architecture supported on extended operators or defects is generated by extended-node introduction, lawful fusion/junction incidence, bulk/boundary attachment, and finite source-coherent hybrid composition.*

10.4 Non-invertible structures

A non-invertible defect is recorded through defect objects, fusion multiplicities, junction spaces, associativity and unit data, action on carrier/target operators, and anomaly/inflow structure. The fusion and module data enter the source universe only when their constitutive pentagon, triangle, and related coherence laws hold. The candidate universe does not assume that non-invertibility is required. It only prevents a source-coherent carrier from being excluded because it is not an ordinary group action or particle state.

Proposition 10.7 (Defect constitutive coherence). *A collection of correctly typed defect objects, fusion channels, junction spaces, and associator maps enters F_{ext} only if it is a model of the corresponding source fusion/module theory. In particular, an associator that violates the pentagon identity is not a downstream-failing defect candidate; it fails to constitute the asserted defect carrier at source level.*

Proposition 10.8 (Non-invertible source neutrality). *The presence of non-invertible terminology in the motivating literature does not alter the generation theorem. Removing all candidate outcomes and all expectations about non-invertible survival leaves F_{ext} and its source parameter domains unchanged.*

10.5 Topological sectors versus spectators

A decoupled TQFT appended to a candidate world is not a new occupancy-bearing candidate if it has no instantiated carrier path to the role attachment and is not constitutively required by the current core. A topological sector that supplies inflow, changes global attachment, supports a relevant defect, or participates in target transport is not a spectator and remains in the trace.

Example 10.9 (Spectator and target-bearing TQFT). *Let Z be a decoupled TQFT. In $W \times Z$, if no instantiated carrier incidence or target attachment leaves Z and every carrier-semantic use factors through W , then Z is stripped. If Z carries an inflow edge cancelling an obstruction of the target attachment, or if a defect in Z mediates the carrier incidence, then Z lies in $\text{Car}_{\mathbb{T}_V^{(3)}}(W \times Z)$.*

10.6 Global-form dependence

Different global forms can share local particle content while differing in line operators and topological sectors. The carrier trace therefore retains global-form information whenever it changes the carrier trace or lawful incidence. It does not identify global-form variants solely because the established interface theorem's point-local core does not distinguish them.

Nonclaim 10.10 (No generalized-symmetry prediction). *Generation of F_{top} and F_{ext} does not imply that any topological, higher-form, higher-group, or non-invertible carrier is interface-adequate or survives later simultaneous physical constraints.*

Extended-sector falsifier

A same-target extended carrier with a source support or fusion/junction relation not representable in the current extended source category directly falsifies the source-completeness theorem. It cannot be dismissed because point-local observables fail to see it.

11 Solitonic, localized nonperturbative, boundary, and edge carriers

11.1 Localized nonperturbative carriers

Solitons, vortices, monopoles, textures, instanton-related persistent sectors, and other nonperturbative configurations can carry localized or extended physical identity without being elementary field quanta [11, 27]. They are represented through source support, topological and boundary data, stability/persistence typing, and their incidence with the fixed substrate.

Definition 11.1 (Solitonic family). F_{sol} consists of carrier traces containing at least one localized nonperturbative node whose identity depends on a source topological, boundary, or nonlinear field configuration and whose target attachment is carrier-bearing.

Theorem 11.2 (Solitonic generation). *Every finitely presented same-target solitonic or localized nonperturbative carrier architecture is generated by a source-typed nonlinear/topological node, its support and stability data, and the lawful incidence connecting it to bulk, boundary, environmental, or target nodes.*

Proof. The configuration type and stability/topological data are intrinsic source attributes; the support is an object of $\text{Supp}_{\mathbb{T}_V^{(3)}}$; and all couplings/attachments are incidence relations. The general factorization theorem applies. $\square$

11.2 Boundary and edge carriers

Gauge theories and local subsystems can possess physical boundary or edge data not reducible to bulk gauge redundancy [5]. The analysis distinguishes a target-bearing boundary carrier from a mere change of boundary condition or from detector hardware outside the target carrier question.

Definition 11.3 (Boundary/edge family). F_{bdy} is the class of carrier traces containing a boundary- or edge-supported node with a currently instantiated target-relative carrier path to the target role, together with its bulk attachment, boundary algebra or response, and primitive boundary parameters.

Theorem 11.4 (Boundary versus target update). *A boundary-supported proposal belongs to the same target iff the target-facing source/detection semantics and admitted boundary type are preserved and the*

new boundary node is a carrier attached within that type. A change of the primitive boundary condition, measurement relation, or target region is a target update.

Proof. Same-target attachment requires a morphism in $\text{Att}_{\mathbb{T}_v^{(3)}}$ to the already-fixed boundary object. A proposal requiring a different boundary object or record semantics changes the target signature rather than adding a carrier node to it. $\square$

11.3 Source/detector localized structures

A structure localized at the production or detection apparatus is a carrier in the present comparison class only if it is part of the physical mechanism bearing the propagation role and not merely an external calibration or fixed measurement convention. The carrier trace records its physical incidence with the source/detection boundary and the propagation carrier. The preceding interface theorem determines whether the resulting interface remains the same target.

Proposition 11.5 (Apparatus/carrier distinction). *A source- or detector-localized component is target-inert world structure when it has no instantiated carrier path and every carrier-semantic use remains unchanged under its replacement by a lawful realization of the same boundary operation. It is carrier-bearing when it participates in the current target-relative architecture and its physical identity or incidence changes that architecture.*

11.4 Boundary-topological hybrids

An edge mode may be supported by a topological bulk sector; a soliton may terminate on a boundary; a defect may carry a boundary junction. These are finite hybrid traces, not separate primitive families. Their physical identity lies in the incidence diagram.

Nonclaim 11.6 (No boundary rescue). *This paper does not allow a boundary or soliton to revision a failed interface after the fact. It generates the architecture; subsequent simultaneous-constraint analysis evaluates the complete shared-witness construction from the beginning.*

12 Environmental, medium, hidden, and enlarged-state-space carriers

12.1 Environmental co-carriage

The propagation target already admits environmental matter response as part of the physical interface. An environment can be a genuine co-carrier when its physical state and incidence are necessary for the occupancy-bearing architecture, rather than an external selector or calibration device.

Definition 12.1 (Environmental co-carrier). *An environmental or medium node m is carrier-bearing when it lies on, or is constitutively required by, an instantiated carrier path and varying or deleting its physical state or current coupling incidence changes the support, persistence, transport, or target attachment of the carrier core. The family of such traces is F_{env} .*

Theorem 12.2 (Environmental co-carriage generation). *Every finitely presented same-target architecture in which a physical medium or environment shares carrier work is generated by environmental-node introduction, system–environment incidence, state/phase attributes, and typed target attachment.*

Proof. The environment is a source support/constituent object, its physical state is an intrinsic attribute, and the system–environment relations are incidence hyperedges. If the environment changes the target’s primitive source/detection semantics, the proposal is a target update under theorem 11.4; otherwise it is a same-target carrier diagram. $\square$

12.2 Open-system carriers

Open-system descriptions can represent loss, decoherence, memory, medium response, or reduced dynamics. The analysis distinguishes three objects:

1. a law-only reduced channel on existing nodes;
2. an explicit environment or hidden dilation that is carrier-bearing;
3. a target-inert dilation or ancillary representation that is candidate skin.

Kraus and Stinespring representations provide standard mathematical examples of nonunique dilations of the same reduced map [2, 12, 28].

Definition 12.3 (Hidden/enlarged-state-space family). *F_{hid} consists of traces with at least one actually target-participating carrier-bearing node outside the visible target substrate, together with instantiated projection/dilation incidence connecting it to visible carrier nodes and target attachment.*

Theorem 12.4 (Hidden-sector generation). *Every finitely presented same-target hidden-sector or enlarged-state-space carrier architecture has a source normal form in F_{hid} , including hidden node types, visible–hidden incidence, projection/dilation maps, primitive hidden parameters, and residual target attachment.*

12.3 Minimal dilations and candidate equivalence

Two minimal dilations related by the source-authorized isometry may be candidate-equivalent when the complete carrier trace is preserved. Nonminimal dilations containing additional actually target-participating carrier-bearing nodes are not merged automatically. Source-coherent ancillary factors with no instantiated carrier path are latent spectators and are stripped.

Proposition 12.5 (Dilation-equivalence criterion). *Let D_1, D_2 be two hidden dilations of the same reduced process. They are candidate-equivalent iff there is a carrier-trace isomorphism preserving hidden support, primitive node identity, visible–hidden incidence, parameter domains, and target attachment. Equality of the reduced process alone is insufficient.*

12.4 Memory and non-Markovian structure

A memory carrier can be represented by hidden or environmental nodes and temporal incidence. The candidate universe does not restrict to Markovian semigroups or Lindblad generators. Such restrictions, if used later, are comparison-class or assumptions of subsequent simultaneous-constraint analysis rather than candidate-generation defaults [2, 13].

12.5 Medium as carrier versus target input

Some environmental data are already target anchors. A fixed background density profile, for example, can be part of the target interface without being a candidate carrier. A new dynamical medium degree of freedom that shares occupancy-bearing work belongs to F_{env} . The distinction is whether the datum is varied as part of the carrier architecture or fixed as target background.

Nonclaim 12.6 (No hidden-sector preference). *The inclusion of hidden and environmental families does not imply that invisible states, decoherence, decay, or medium-induced mechanisms survive the interface and simultaneous constraints.*

Open-system omission test

A new reduced process does not automatically define a new carrier family: it may be law-only. A new hidden/environmental family is required only when a distinct occupancy-bearing node or incidence survives carrier-semantic reduction.

13 Hybrid closure and carrier-incidence reconstruction

13.1 Why mixed architectures are not an escape hatch

A realistic completion may combine a local field, an environmental mode, a boundary degree of freedom, an extended defect, and a hidden dilation. A flat family list would either miss such constructions or proliferate an unbounded collection of “mixed” labels. The carrier trace avoids this problem by representing every architecture through a typed incidence hypergraph. Hybrid closure is therefore not a convenience; it is part of candidate completeness.

Definition 13.1 (Carrier-incidence hypergraph). *A carrier-incidence hypergraph is a finite tuple*

$$H = (V, E, s, t, \lambda_V, \lambda_E, \omega_H, \Theta, a_T),$$

where V is a finite set of carrier nodes, E a finite set of typed hyperedges, s, t record ordered or unordered endpoint lists as required by the edge type, λ_V assigns source node types, λ_E assigns incidence types, ω_H records the instantiated source-constitutive structure and its required commuting diagrams, Θ is the joint primitive parameter domain, and a_T is the typed target-attachment diagram. Each instantiated edge and attachment is additionally governed by the type-indexed source activity law of $\mathbb{S}_{\text{car}}(\mathbb{T}_V^{(3)})$; activity is evaluated pointwise rather than encoded by the mere presence of an edge symbol in the template.

Definition 13.2 (Source-coherent hybrid). *A hybrid is source coherent when each node label lies in $\mathfrak{D}_{\text{car}}(\mathbb{T}_V^{(3)})$, each edge is an object of $\text{Inc}_{\mathbb{T}_V^{(3)}}$ with matching endpoint types, parameter domains are source-defined, target attachments lie in $\text{Att}_{\mathbb{T}_V^{(3)}}$, and the full glued diagram satisfies $\text{Coh}_{\text{src}}(H)$. In particular, every required overlap, associativity, module, projection, and attachment diagram must commute.*

The generic incidence construction is illustrated in figure 4.

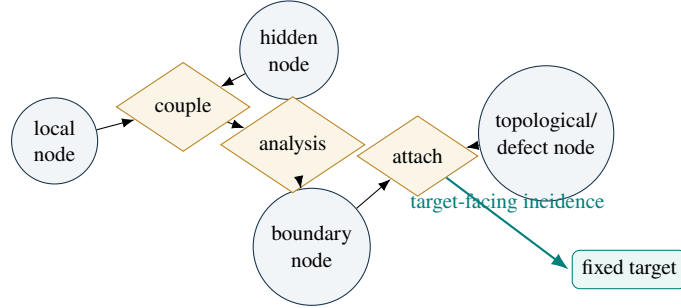


Figure 4: A carrier architecture is a typed incidence hypergraph. Hybrid constructions are generated by the same grammar.

Theorem 13.3 (Finite-hybrid closure). *The source-coherent carrier algebra is closed under every lawful finite source-coherent hybrid of generated carrier diagrams. At each source parameter point, candidate identity is assigned only after the pointwise activity reduction of definition 6.5. Conversely, every finitely presented hybrid semantic carrier architecture is source-generated as a coherent model of the source theory.*

Proof. Raw closure is built into Hybrid, while candidate formation additionally requires the gluing equations of definition 6.15. For the converse, take the finite carrier-incidence hypergraph of the semantic architecture. Generate each connected subdiagram by theorem 6.23, then add the finite set of cross-component hyperedges and the source-coherence data inherited from the semantic architecture. The resulting coherent source term reconstructs the complete trace. $\square$

13.2 Non-factorizing incidence

Some physical relations are genuinely many-body or categorical and cannot be decomposed into independent pairwise couplings without losing carrier-semantic information. Fusion junctions, shared constraints, collective support, common-environment incidence, and anomaly inflow can have this form. Hyperedges are therefore primitive incidence objects where the source ontology requires them. Their endpoint typing is not enough: any constitutive higher-coherence law attached to the relation is enforced by Coh_{src} .

Proposition 13.4 (Pairwise reduction is not assumed). *Candidate completeness does not require every incidence relation to factor into binary edges. A source relation of arity k is represented directly as a k -ary hyperedge.*

13.3 Incidence reconstruction theorem

Theorem 13.5 (Carrier-incidence reconstruction). *Let $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$. Once the primitive/derived carrier nodes of $\text{Car}_{\mathbb{T}_v^{(3)}}(X)$ are generated, the complete carrier trace is recovered by adjoining the finite set of source-typed incidence and attachment relations in $\chi_{\text{car}}(X)$. The reconstructed diagram is carrier-semantically equivalent to X .*

Proof. By theorem 4.4, the trace is a complete invariant. The node generators recover every node and its local attributes. The incidence constructors recover every relation and target attachment, while lemma 6.22 recovers the constitutive coherence structure. Thus the generated trace is isomorphic to the original trace, which implies carrier-semantic equivalence. $\square$

13.4 Hybrid family templates

A hybrid family template contains the full source-coherent symbolic architecture; its active carrier architecture is evaluated pointwise. The template contains:

1. a finite typed vertex template, possibly with symbolic multiplicity;
2. a finite typed edge template;
3. a source-constitutive coherence template ω_u and its required commuting diagrams;
4. a source-defined joint parameter domain together with its full coherent locus, source activity profile, and participating locus;
5. automorphisms and lawful reparameterizations;
6. target attachment slots.

Symbolic multiplicity can encode arbitrary finite repeated nodes without enumerating them. The template remains finite even when its instance space is infinite.

Corollary 13.6 (No eleventh mixed kind). *“Mixed carrier” is not a primitive family. Every lawful mixed architecture is an instance of the common hybrid constructor over the principal source strata.*

13.5 Hybrid source independence

A hybrid edge is generated because it is a source-admitted physical relation, not because the corresponding combination later cancels an anomaly or fits an interface. Whether that instantiated edge is physically active at a particular source parameter point is fixed by its source activity support before subsequent simultaneous-constraint analysis. If an inflow edge, for example, is present as a physical possibility in the source ontology, it is generated for all source-typed endpoints that satisfy the relation’s constitutive equations; subsequent simultaneous-constraint analysis determines whether the resulting coherent source candidate is simultaneously admissible and surviving.

Hybrid falsifier

A same-target finite source-coherent hybrid with a physically meaningful cross-incidence or constitutive gluing equation absent from $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$ is a direct omission counterexample. The revision is to enlarge the source incidence category, not to add a named hybrid exception.

14 Spectator stripping and the occupancy-bearing candidate core

14.1 The spectator multiplication problem

Without spectator control, any candidate can be multiplied by an arbitrary decoupled sector, producing infinitely many false candidate classes. Conversely, an apparently “spectator” sector can carry inflow, global-form data, hidden loss, environmental memory, or other incidence essential to the target. The analysis therefore requires a physical spectator theorem, not a rule deleting all weakly coupled structure.

The controlling distinction is between a sector’s *latent capability* to become a carrier after a future attachment and its *actual participation* in the current target-relative carrier diagram. Candidate identity records the latter.

Definition 14.1 (Target-inert spectator). *A connected subdiagram $Z \subseteq W$ is a target-inert spectator when:*

- (i) $Z \cap \text{Car}_{\mathbb{T}_v^{(3)}}(W) = \emptyset$;
 - (ii) *there is no current target-relative carrier path from Z to an instantiated target attachment or to a component of $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$;*
 - (iii) *replacing Z by any other lawfully decoupled source object leaves every carrier-semantic use unchanged.*
- A spectator may itself be a complete source-coherent field theory, TQFT, defect system, hidden sector, or copy of an admitted carrier type. Such latent capability is irrelevant until a physical incidence or attachment is actually instantiated.*

Theorem 14.2 (Spectator stripping). *If Z is a target-inert spectator of W , then*

$$\chi_{\text{car}}(W) = \chi_{\text{car}}(W \setminus Z), \quad W \equiv_{\text{car}} W \setminus Z.$$

Conversely, any connected subdiagram participating in a current target-relative carrier path remains in $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$.

Proof. By definition 3.4, no node, edge, attachment, or constitutive map of Z belongs to the occupancy-bearing reduction. Removing Z therefore leaves the current path union and its source-constitutive closure unchanged, so the carrier traces agree. Carrier-semantic equivalence follows from trace completeness. Conversely, a subdiagram on a current path is included by the definition of the current support closure. $\square$

Corollary 14.3 (Decoupled products do not multiply candidate classes). *For every target-inert source sector Z ,*

$$[W \times Z]_{\equiv_{\text{car}}} = [W]_{\equiv_{\text{car}}}.$$

This remains true when Z is source-coherent and isomorphic to a carrier architecture admitted elsewhere in the candidate universe.

Theorem 14.4 (Latent-copy invariance and incidence activation). *Let Z_{dec} be a source-coherent sector with latent carrier capability but no current target attachment and no current carrier-incidence path to $\text{Car}_{\mathbb{T}_v^{(3)}}(W)$. Then*

$$\text{Car}_{\mathbb{T}_v^{(3)}}(W \times Z_{\text{dec}}) \cong \text{Car}_{\mathbb{T}_v^{(3)}}(W), \quad \chi_{\text{car}}(W \times Z_{\text{dec}}) = \chi_{\text{car}}(W).$$

Now let e be a source-coherent, physically nontrivial incidence or attachment added to form W_e , and suppose a current target-relative carrier path passes through Z_{dec} and e . Then the participating subdiagram of Z_{dec} and the incidence e belong to $\text{Car}_{\mathbb{T}_V^{(3)}}(W_e)$. If they are not already represented by a lawful isomorphism of the old core, then

$$\chi_{\text{car}}(W_e) \neq \chi_{\text{car}}(W).$$

Proof. The first statement is lemma 3.7 and theorem 14.2. For the second, proposition 3.8 places the new path and its source-constitutive closure in the carrier core. The carrier trace records the participating nodes, incidence, coherence, parameter domains, and attachment. A genuinely new such datum therefore changes the trace. $\square$

14.2 Pointwise parameter activity and degenerate incidence

A sector is not a spectator merely because its coupling is small, its states are invisible, or it is absent from the point-local target interface. Any source-active carrier incidence, topological attachment, inflow relation, environmental memory, or hidden projection path keeps it in the carrier core. The source activity structure of definition 6.4 also fixes the opposite boundary: a symbolic incidence slot that evaluates to a source-null relation at a particular source parameter point does not create a carrier path at that point.

Proposition 14.5 (No weak-coupling threshold; exact null-incidence rule). *Let e_θ be an instantiated source incidence. If $\text{asupp}(e_\theta) \neq \emptyset$, then numerical smallness of any coordinate used to present e_θ is not a spectator criterion and the active support is retained. If $\text{asupp}(e_\theta) = \emptyset$, then e_θ contributes no current carrier edge; any component connected to the target only through e_θ is absent from $\text{Car}_{\mathbb{T}_V^{(3)}}$. The distinction is source-semantic and contains no arbitrary coupling threshold.*

Theorem 14.6 (Degenerate-incidence limiting-pair test). *Let Z be a source-coherent carrier-capable sector and let*

$$H_g = H_{\text{base}} \star_{e_g} Z, \quad g \in \Theta,$$

be one source-coherent symbolic family. Suppose the type-indexed source activity law satisfies

$$\text{asupp}(e_g) \neq \emptyset \quad \text{for } g \in \Theta^\times, \quad \text{asupp}(e_{g_0}) = \emptyset$$

for a degenerate source point g_0 , and suppose no independent active path connects Z to the target at g_0 . Then

$$Z \subseteq \text{Interpret}_{\text{src}}(H_g) \quad (g \in \Theta^\times),$$

while

$$\text{Interpret}_{\text{src}}(H_{g_0}) \cong \text{Interpret}_{\text{src}}(H_{\text{base}}).$$

Consequently the degenerate point has the same carrier trace as the base architecture whenever no other active source datum distinguishes them, whereas every genuinely active non-null point retains the participating sector and incidence.

Proof. For $g \in \Theta^\times$, the source activity support of e_g is nonempty, so an active target-relative path passes through the participating part of Z ; definition 3.4 and proposition 14.5 retain it. At g_0 , the incidence is source-null. Pointwise activity reduction deletes that edge before carrier-path closure. By hypothesis no independent active path reaches Z , so the sector is spectator structure and theorems 6.8 and 14.2 reduce the carrier core to the base architecture. The conclusion is fixed before any subsequent simultaneous evaluation. $\square$

Corollary 14.7 (Downstream selection cannot mutate candidate identity). *If the subsequent analysis later restricts or selects a source parameter value at which an incidence is source-null, it selects the candidate*

class already assigned to that point by $\text{ActRed}_{\mathbb{T}_v^{(3)}}$. A subsequent simultaneous constraint analysis does not retroactively strip a sector, activate a sector, or merge candidate classes.

Proof. The source activity profile is a function only of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$ and the source parameter point. Candidate identity is formed after that activity reduction and before subsequent simultaneous-constraint analysis. Thus any later restriction merely chooses among already classified source instances. $\square$

14.3 World identity versus candidate identity

Candidate identity is intentionally narrower than full-world identity. Two worlds can differ in distant sectors, cosmological history, detector implementation, or other physical facts while presenting the same occupancy-bearing carrier architecture. The candidate quotient removes those differences because the candidate universe is a universe of carrier solutions to one target role, not a inventory of all possible worlds.

14.4 Spectator topological sectors

A decoupled TQFT or global sector is stripped when it has no current carrier-bearing incidence. If it changes line-operator attachment, supplies anomaly inflow, supports a defect, or changes the hidden/environmental realization through an actually instantiated relation, it is not stripped. This distinction is recorded in the hypergraph, not inferred from the label “TQFT.”

14.5 Spectator stripping commutes with lawful redescription

Lemma 14.8. *If $W \simeq W'$ by a lawful ontology-preserving redescription, then the spectator-reduced carrier cores are isomorphic.*

Proof. Lawful redescription preserves every carrier-semantic use, current target attachments, and actual incidence paths. It therefore preserves the spectator/active-carrier partition and induces an isomorphism of the reduced cores. $\square$

False spectator attack

A counterexample to spectator stripping must exhibit a removed sector whose source-active currently instantiated incidence changes a lawful carrier-semantic use while preserving the target, or by exhibiting a source-null parameter point at which the formal template incorrectly retains a disconnected sector. The correct response is to apply the source-defined activity reduction. A counterfactual future attachment, latent carrier capability, external encoding entry, weak-coupling threshold, or downstream survivor result cannot override that current-architecture test.

15 Lawful candidate equivalence and quotient discipline

15.1 Candidate equivalence

The coherent source algebra can generate the same carrier architecture in different orders, labels, bases, gauge representatives, or parameter coordinates. Those redundancies should be removed. Distinct carrier architectures that happen to have the same target interface should not.

Definition 15.1 (Lawful candidate isomorphism). *A lawful candidate isomorphism between source-coherent carrier diagrams H and H' consists of:*

- (a) *a typed hypergraph isomorphism between their spectator-reduced carrier cores;*
- (b) *preservation of support, persistence, derivation status, constituent kinds, and intrinsic attributes;*
- (c) *preservation of every physical incidence, target attachment, and source-constitutive structure ω_H ;*
- (d) *preservation of the equations and commuting diagrams by which each presentation models $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$;*
- (e) *lawful isomorphisms or reparameterizations of primitive parameter domains;*

(f) no use of the target-relative interface equality, failure status, or survivor data.

Definition 15.2 (Candidate equivalence). For source-generated coherent instances X, Y ,

$$X \simeq_{\text{cand}} Y$$

iff there exists a finite zig-zag of lawful candidate isomorphisms.

Theorem 15.3 (Candidate quotient exactness). For source-generated carrier instances,

$$X \simeq_{\text{cand}} Y \iff \chi_{\text{car}}(X) = \chi_{\text{car}}(Y) \iff X \equiv_{\text{car}} Y.$$

Proof. A lawful candidate isomorphism preserves every trace coordinate, including constitutive coherence, hence the left side implies trace equality. Trace equality supplies a typed coherent-hypergraph isomorphism and lawful parameter identification by definition of the trace, so it implies candidate equivalence. The second equivalence is theorem 4.4. $\square$

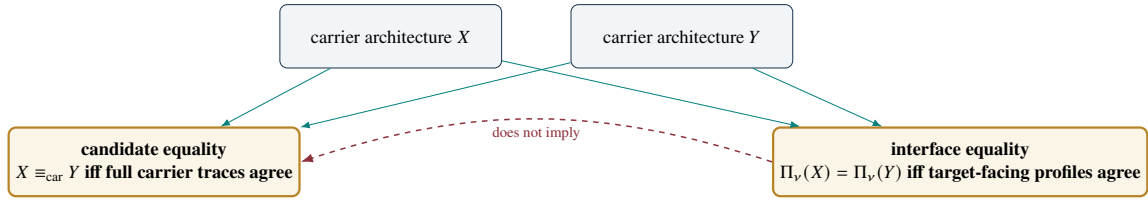


Figure 5: Candidate equivalence preserves the complete carrier architecture and is therefore generally finer than target-interface equivalence.

15.2 Family presentations, activity strata, and scientific family classes

A family presentation begins with a finite source-coherent normal-form template u , a raw product or moduli domain Θ_u^{raw} assembled from the primitive source domains, and a reconstruction map $H_u : \Theta_u^{\text{raw}} \rightarrow \mathcal{G}_{\text{car}}^{\text{raw}}(\mathbb{T}_v^{(3)})$. Constitutive equations can relate several primitive parameters, so the full source domain is first restricted to the coherent locus. Candidate identity is then assigned *pointwise* by the source activity reduction; no assumption is made that one symbolic template has one carrier-core topology over its whole parameter domain.

Definition 15.4 (Full source-coherent family domain). For a parametric source template u , define

$$\Theta_u^{\text{coh}} := \{\theta \in \Theta_u^{\text{raw}} : \text{WellTyped}(H_u(\theta)) \wedge \text{Coh}_{\text{src}}(H_u(\theta))\}.$$

This is the complete source-coherent locus. Restrictions imposed only by anomaly cancellation, vacuum selection, target-interface adequacy, subsequent simultaneous-constraint analysis compatibility, phenomenological fit, or survivor status are not part of Θ_u^{coh} .

Definition 15.5 (Target-participating source locus). The candidate-participating locus of u is

$$\Theta_u^{\text{part}} := \{\theta \in \Theta_u^{\text{coh}} : \text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta)) \neq \emptyset\}.$$

This is not a viability cut. It removes only source-coherent parameter points at which the instantiated source diagram contains no active target attachment and hence no occupancy-bearing carrier proposal for the fixed target. The full coherent source locus Θ_u^{coh} remains part of the family provenance and may not be narrowed by subsequent simultaneous-constraint analysis.

Definition 15.6 (Pointwise activity profile and activity strata). For $\theta \in \Theta_u^{\text{part}}$, define

$$A_u(\theta) := [\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))]_{\cong_{\text{src}}}.$$

For each activity class $[A]$ in the image of A_u , let

$$\Theta_u^{[A]} := \{\theta \in \Theta_u^{\text{part}} : A_u(\theta) = [A]\}.$$

Then

$$\Theta_u^{\text{part}} = \bigsqcup_{[A] \in \text{im } A_u} \Theta_u^{[A]}.$$

The word stratum here is set-theoretic: the construction assumes neither finitely many activity classes nor manifold regularity of their loci. It records only that each source parameter point has a source-fixed active carrier architecture before downstream evaluation.

Theorem 15.7 (Pointwise family interpretation). For every $\theta \in \Theta_u^{\text{part}}$, the candidate instance

$$X_u^{\text{act}}(\theta) := \text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta)))$$

lies in $\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$, and

$$\chi_{\text{car}}(X_u^{\text{act}}(\theta))$$

depends only on the source parameter point modulo lawful source reparameterization. If two points, possibly in different symbolic families, reduce to the same complete active carrier trace, they represent the same candidate instance.

Proof. Source coherence on Θ_u^{coh} , activity-coherence preservation, and theorems 6.8 and 6.19 give the first claim. Redescription invariance is proposition 6.6. The final statement follows from candidate quotient exactness above. $\square$

A scientific family presentation is therefore the source-coherent template u , its complete coherent domain Θ_u^{coh} , its participating locus Θ_u^{part} , its pointwise activity profile, and the active reconstruction map $X_u^{\text{act}} : \Theta_u^{\text{part}} \rightarrow \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$.

Definition 15.8 (Family-presentation class). FamPres_v is the class of all tuples

$$(u, \Theta_u^{\text{coh}}, \Theta_u^{\text{part}}, A_u, X_u^{\text{act}})$$

obtained from source-coherent normal-form templates over $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$. It is defined extensionally from the coherent normal-form theorem; it is not a finite catalogue. Empty-core points in $\Theta_u^{\text{coh}} \setminus \Theta_u^{\text{part}}$ remain recorded as source-coherent degenerations of the template but are not candidate instances.

Definition 15.9 (Family equivalence). Families u and v are family-equivalent if a lawful source reparameterization/groupoid equivalence carries their participating loci onto one another and satisfies

$$X_u^{\text{act}}(\theta) \simeq_{\text{cand}} X_v^{\text{act}}(f(\theta))$$

for every participating point, with every candidate instance of either family occurring in the image of the other. The equivalence may carry one activity stratum to another presentation of the same active carrier class; it may not identify points merely because the established interface theorem gives the same interface value.

Definition 15.10 (Scientific carrier-family class). *The scientific family class is*

$$\text{Fam}_v := \text{FamPres}_v / \simeq_{\text{fam}} .$$

An illustrative family table may display principal strata and selected normal-form coordinates in Fam_v ; it is neither the definition nor a finite enumeration of Fam_v .

Proposition 15.11 (Finite-table non-authority). *Adding, deleting, renaming, or regrouping entries in an illustrative family table does not change Fam_v or the scientific candidate universe, provided the underlying source-coherent templates, pointwise activity reductions, and active normal-form classes are unchanged.*

15.3 Candidate universe as pointwise activity-reduced instances

Definition 15.12 (Instance equivalence). *For family presentations and participating source points,*

$$(u, \theta) \simeq_{\text{inst}} (v, \phi) \iff X_u^{\text{act}}(\theta) \simeq_{\text{cand}} X_v^{\text{act}}(\phi) .$$

The scientific candidate universe is

$$\mathfrak{U}_v := \left(\bigsqcup_{u \in \text{FamPres}_v} \Theta_u^{\text{part}} \right) / \simeq_{\text{inst}} . \quad (8)$$

The quotient relation incorporates both lawful family reparameterization and instance equivalence. The notation $[u, \theta]$ denotes the resulting joint family-and-instance class of the pointwise activity-reduced instance $X_u^{\text{act}}(\theta)$. In particular, a degenerate parameter point in one symbolic family may be identified with a generic point of another family when their active carrier traces coincide; that is a source-first candidate merge, not a downstream viability merge.

Proposition 15.13 (Parametric openness with activity stratification). *Unbounded, noncompact, categorical, or otherwise symbolic source domains do not obstruct the definition of $\mathfrak{U}_v$. Candidate completeness uses every point of every source-coherent participating locus, with candidate identity assigned after pointwise activity reduction. No finite sampling, finite number of activity strata, or generic-parameter assumption occurs in the theorem.*

15.4 Candidate automorphisms

The automorphism group $\text{Aut}_{\text{car}}(X)$ consists of trace-preserving permutations, source-equational isomorphisms, and reparameterizations of indistinguishable nodes, edges, coherence structures, and parameters. Automorphisms remove only internal presentation multiplicity; they do not identify physically distinct support, incidence, or constitutive data.

15.5 Interface equivalence is a coarser relation

If two generated candidates are candidate-equivalent and both admit interface projections, their interface packets are equal. The converse can fail because the interface forgets candidate-internal structure not used by the target. Therefore interface equality cannot appear in definition 15.2.

Theorem 15.14 (No interface-quotient backflow). *There is no lawful candidate quotient whose equivalence classes are defined solely by equality of the target-relative interface images. Any such quotient can merge distinct local/hidden, primitive/composite, law-only/dilation, or topological/non-topological carrier architectures.*

15.6 Canonical normal form without a privileged representative

The scientific normal form is an isomorphism class of source-coherent carrier diagrams, not a selected vertex ordering or canonical label. A software encoder may choose a deterministic representative for reproducibility,

but that representative is verification skin. This prevents canonicalization from acquiring boundary authority.

Quotient falsifier

The theorem fails if $\approx_{\text{cand}}$ merges two traces distinguished by a carrier-semantic use, if it preserves a difference consisting only of lawful graph/source-equational presentation or parameter reparameterization, or if the scientific family class is replaced by the contents of a finite catalogue.

16 Source-coherent normal form and reconstruction theorem

16.1 Normal-form object

For a pointwise activity-reduced, spectator-reduced source-coherent carrier diagram H , define its source normal form by

$$\mathcal{N}_{\text{car}}(H) = [V_H, E_H, \lambda_V, \lambda_E, \omega_H, \Theta_H, a_H]_{\cong_{\text{car}}}, \quad (9)$$

where ω_H denotes the instantiated source-constitutive structure and the bracket denotes lawful coherent-hypergraph isomorphism and parameter reparameterization. The normal form retains the complete carrier trace and forgets only construction order, naming, source-equational presentation, and other lawful skin.

Lemma 16.1 (Normal-form existence). *Every source-formed carrier proposal has a normal form $\mathcal{N}_{\text{car}}$.*

Proof. Pointwise activity reduction followed by spectator stripping yields a finite source-coherent carrier diagram by lemma 6.7 and theorem 14.2. Taking its lawful candidate-isomorphism class gives equation (9). $\square$

16.2 Reduction invariance and oriented confluence

Different generation sequences may introduce components and incidence in different orders. A lawful reduction may remove target-inert spectators, contract a source-authorized presentation redundancy, replace a diagram by a lawfully isomorphic coherent diagram, or rewrite by a constitutive equation in $\mathfrak{E}_{\text{src}}$. Source-equational and isomorphism rewrites can be reversible and need not form a terminating rewrite system. The scientific normal form is therefore the quotient class $\mathcal{N}_{\text{car}}$, not the output of an unrestricted normalization algorithm.

Theorem 16.2 (Normal-form invariance and oriented confluence). *Every lawful source-reduction step preserves $\mathcal{N}_{\text{car}}$; hence every finite chain of lawful reductions from a source-coherent proposal remains in one normal-form class. Moreover, for any explicitly oriented terminating subsystem that removes target-inert spectators and chosen presentation redundancies while preserving source coherence, all terminal representatives lie in that same class. No global termination claim is made for reversible source-equational or isomorphism rewrites, and no unique labeled graph or proof of coherence is selected.*

Proof. Pointwise activity reduction is redescription invariant by proposition 6.6 and spectator stripping preserves the active carrier core and trace by theorem 14.2. Source-authorized presentation contractions, coherent isomorphisms, and constitutive rewrites preserve the modeled source object and the complete carrier trace by definition. Therefore every lawful step preserves $\mathcal{N}_{\text{car}}$, and so does every finite chain. If an oriented subsystem is chosen that strictly decreases a declared finite complexity measure—for example, the number of target-inert connected components together with a finite presentation-redundancy count—then it terminates. Any two terminal representatives reached by that subsystem have the same complete trace and are candidate-equivalent by theorem 15.3. $\square$

16.3 Semantic reduction theorem

Theorem 16.3 (Semantic carrier reduction). *Every $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$ admits a finite spectator-reduced source-coherent carrier-incidence presentation whose normal-form trace equals $\chi_{\text{car}}(X)$.*

Proof. The occupancy-bearing reduction exists by definition 3.4. The comparison-class assumption supplies a finite presentation. The semantic carrier core already contains only source-active current target-relative paths. Source coherence follows from theorem 6.20. The source reconstruction therefore admits a pointwise activity-reduced presentation, and spectator stripping does not change its trace by theorem 14.2. The resulting coherent typed diagram is the desired presentation. $\square$

16.4 Primitive component decomposition

Theorem 16.4 (Primitive carrier decomposition). *Every vertex of the semantic normal form is one of:*

1. *an existing-substrate node;*
2. *a new primitive source node;*
3. *a derived/composite/emergent node with explicit constituent incidence;*
4. *a law/process node acting on existing or generated carriers.*

Its support, persistence, and constituent types place it in one or more principal strata of theorem 5.8; no additional mixed primitive kind is required. Every instantiated sort retains the constitutive equations belonging to that sort.

Proof. This is the derivation-coordinate case split of $\text{Der}_{\mathbb{T}_v^{(3)}}$, followed by the support/persistence stratification of theorem 5.8. Source coherence is inherited from the semantic carrier presentation. $\square$

16.5 Source normal-form existence

Theorem 16.5 (Source normal-form existence). *For every $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$, there exist a scientific source family $[u] \in \text{Fam}_v$ and $\theta \in \Theta_u^{\text{part}}$ such that*

$$\chi_{\text{car}}(X) = \chi_{\text{car}}(X_u(\theta)).$$

Consequently $X \equiv_{\text{car}} X_u(\theta)$.

Proof. Use theorem 16.3 to obtain a finite source-coherent active diagram. Decompose its vertices by theorem 16.4. Generate the vertices, incidence, attachments, and coherence structure using theorem 6.23; include all finite hybrid incidence by theorem 13.3. Replace primitive attribute values by variables over the raw source product Θ_u^{raw} , retain every constitutive equation of the template, and restrict first to the full coherent locus Θ_u^{coh} of definition 15.4. Equip that family with the source-fixed pointwise activity profile of definition 15.6. The original semantic parameter point lies in Θ_u^{part} because its carrier core is nonempty, and its pointwise activity reduction reproduces the semantic active diagram by theorem 6.8. This yields a family presentation $u \in \text{FamPres}_v$; passing to the family-equivalence class gives $[u] \in \text{Fam}_v$. The reconstructed active trace equals the semantic trace. $\square$

16.6 Source normal-form soundness

Theorem 16.6 (Source normal-form soundness). *Every instance $[u, \theta] \in \mathfrak{U}_v$ denotes, through pointwise source activity reduction followed by $\text{Interpret}_{\text{src}}$, a source-coherent physically interpretable carrier proposal in $\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$. Raw typed instances that fail source coherence, parameter points with empty active target core, or instances requiring a source-ontology/target update are excluded before entering $\mathfrak{U}_v$.*

Proof. By definitions 15.10 and 15.12, every universe instance is represented by a source-coherent normal-form template and a point $\theta \in \Theta_u^{\text{part}}$. Apply theorems 6.19 and 15.7: source activity is evaluated before interpretation, source-null incidences are stripped pointwise, and the resulting active diagram has a semantic carrier interpretation. Candidate quotienting preserves that interpretation. Later simultaneous inconsistency does not remove source-level meaning or alter the already assigned active carrier identity. $\square$

16.7 Normal-form exactness

Theorem 16.7 (Normal-form exactness). *For semantic or source-generated carrier instances,*

$$\mathcal{N}_{\text{car}}(X) = \mathcal{N}_{\text{car}}(Y) \iff X \equiv_{\text{car}} Y.$$

For generated instances this is also equivalent to $X \simeq_{\text{cand}} Y$.

Proof. Normal-form equality is equality of the complete carrier trace, including source-constitutive structure, up to lawful isomorphism. Apply theorems 4.4 and 15.3. $\square$

16.8 Family templates and symbolic parameters

The family normal form is obtained by replacing primitive attribute values with variables over raw source domains while retaining the source template, incidence grammar, and source equations, then restricting to the full source-coherent locus Θ_u^{coh} . The active carrier graph is evaluated pointwise by $\text{ActRed}_{\mathbb{T}_v^{(3)}}$, so a single symbolic family may contain several source-defined activity strata, including degenerate points whose carrier trace coincides with another family. Families with different lawful parameterizations are identified by definition 15.9. This yields the extensional scientific family class Fam_v ; the finite family catalogue is only an illustrative projection of that class.

Corollary 16.8 (Open-family closure). *Every semantic architecture is represented by a source-coherent symbolic family and a target-participating parameter point whose pointwise active reduction has the required trace. No finite parameter sampling, generic-point assumption, finite activity stratification, or finite catalogue enumeration is required for normal-form completeness.*

16.9 What the normal form does not determine

The normal form does not decide whether a candidate is anomaly-free, dynamically stable, interface adequate, minimal, unique, or realized. It preserves exactly the source-coherent carrier data subsequent simultaneous-constraint analysis needs to perform those tests without changing candidate identity.

17 Semantic candidate completeness

17.1 Coverage and soundness maps

Define

$$\Phi : \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) / \equiv_{\text{car}} \longrightarrow \mathfrak{U}_v$$

by choosing a semantic representative X , applying theorem 16.5, and setting $\Phi([X]) = [u, \theta]$. Define

$$\Psi : \mathfrak{U}_v \longrightarrow \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) / \equiv_{\text{car}}$$

by

$$\Psi([u, \theta]) := [X_u^{\text{act}}(\theta)] = \left[\text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))) \right].$$

Lemma 17.1 (Well-definedness of Φ). *The map Φ is independent of the semantic representative, the chosen coherent source derivation, and the family presentation.*

Proof. If $X \equiv_{\text{car}} Y$, their traces and normal forms agree by theorems 4.4 and 16.7. Different coherent source derivations reconstructing the same trace yield candidate-equivalent instances by theorems 15.3 and 16.2. Family-equivalent presentations and parameter reparameterizations define the same instance class by definitions 15.9 and 15.12. Therefore the element of $\mathfrak{U}_v$ is unique. $\square$

Lemma 17.2 (Well-definedness of Ψ). *The map Ψ is independent of the family representative, coherent source-equational presentation, and parameter reparameterization.*

Proof. Every universe element has a source-coherent participating representative by definitions 15.10 and 15.12. Family/instance equivalence identifies exactly candidate-equivalent pointwise activity-reduced instances. Candidate equivalence equals carrier-semantic equivalence by theorem 15.3; source activity and interpretation are invariant under lawful source-equational isomorphism by proposition 6.6 and theorem 15.7. $\square$

17.2 The two completeness directions

Theorem 17.3 (Semantic coverage). *For every $X \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$, there exist $[u] \in \text{Fam}_v$ and $\theta \in \Theta_u^{\text{part}}$ such that*

$$X \equiv_{\text{car}} X_u^{\text{act}}(\theta).$$

Thus every semantic carrier class is represented in $\mathfrak{U}_v$.

Proof. This is theorem 16.5 followed by the instance quotient. $\square$

Theorem 17.4 (Source-coherent universe soundness). *Every $[u, \theta] \in \mathfrak{U}_v$ has a source-level physical interpretation*

$$X_u^{\text{act}}(\theta) = \text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))) \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}).$$

In particular, $\mathfrak{U}_v$ contains no merely endpoint-typed but source-incoherent term.

Proof. Apply theorem 16.6. Source-incoherent raw terms never enter FamPres_v . Source-coherent points with empty active target core remain in the recorded coherent source locus but lie outside Θ_u^{part} , and every universe point is interpreted only after the source-fixed activity reduction. Hence each element of $\mathfrak{U}_v$ has a semantic carrier interpretation. $\square$

Theorem 17.5 (Source-first candidate completeness). *Within the fixed target $\mathbb{T}_v^{(3)}$ and comparison class $\mathfrak{C}_v^{(3)}$,*

$$\boxed{\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) / \equiv_{\text{car}} \cong \mathfrak{U}_v.} \quad (10)$$

Equivalently, every semantic carrier architecture has a source-coherent parametric normal form, and every element of the generated universe denotes a source-level physical carrier proposal.

Proof. By theorem 17.3, Φ is defined on every semantic class. By theorem 17.4, Ψ is defined on every universe element. For $[X] \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) / \equiv_{\text{car}}$, the reconstructed coherent instance has the same carrier trace as X , so $\Psi\Phi([X]) = [X]$. For $[u, \theta] \in \mathfrak{U}_v$, interpreting the pointwise active coherent normal form and then applying source-normal-form existence returns the same candidate-normal-form class by theorems 15.7 and 16.7, so $\Phi\Psi([u, \theta]) = [u, \theta]$. Thus the maps are inverse bijections. $\square$

Scientific content of the completeness theorem

The theorem does not say that a finite table lists every model. It says that every same-target carrier architecture in the declared physical comparison class has a finite source-coherent incidence presentation, that the source theory generates every such presentation, and that the candidate quotient is exactly carrier-semantic identity.

17.3 Parametric family closure

Theorem 17.6 (Parametric closure). *The isomorphism of theorem 17.5 remains valid when coherent source domains are infinite, noncompact, representation-valued, categorical, or otherwise symbolic and when pointwise activity changes on arbitrarily complicated source-defined loci. No finite sampling, generic-point assumption, or finite activity-stratum decomposition is part of the theorem.*

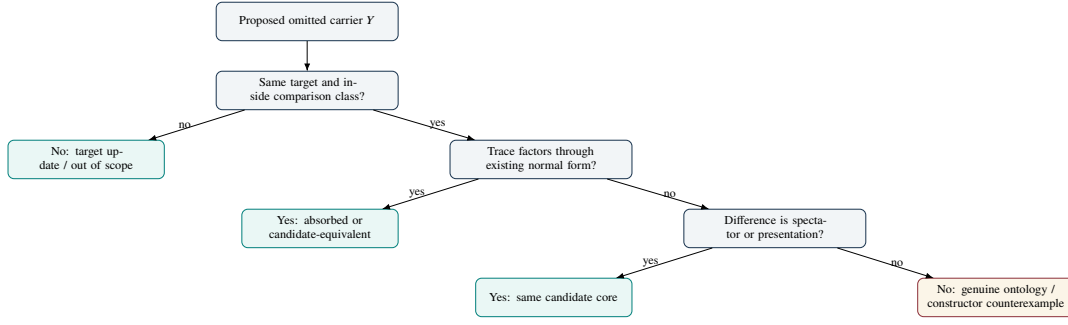


Figure 6: Omitted-carrier disposition. A genuine same-target failure of factorization is an explicit falsifier.

Proof. The source-family layer retains the complete coherent loci, and the candidate universe uses every target-participating point after source-fixed activity reduction. The source-normal-form theorem produces the actual semantic parameter point and its active trace. The proof never invokes cardinality, compactness, genericity, or a finite activity stratification; it uses only source coherence and pointwise source participation. $\square$

17.4 Omission disposition

Theorem 17.7 (Omitted-carrier disposition). *Let X_* be a purported same-target carrier architecture not displayed in an illustrative family table. Exactly one of the following scientific dispositions applies:*

- (i) X_* has a source-coherent normal form already represented by the scientific family class, and the omission is only an presentational omission;
- (ii) X_* is candidate-equivalent to an existing universe instance despite a different presentation;
- (iii) X_* is a lawful finite source-coherent hybrid of existing source families;
- (iv) the additional structure is target-inert, merely latent without a current carrier-incidence path, or candidate-external world structure;
- (v) the purported novelty occurs only on a source-degenerate parameter locus where one or more incidence maps become source-null; pointwise activity reduction maps that instance to its already defined active carrier trace, possibly in another symbolic family;
- (vi) the proposed data are source-well-typed but violate a constitutive source equation, so they do not instantiate the asserted carrier object;
- (vii) X_* changes the target or the declared comparison class;
- (viii) X_* is a genuine same-target coherent counterexample requiring a new source carrier sort, operation, or constitutive equation and reopening [theorem 17.5](#).

There is no disposition “unfamiliar and therefore omitted.”

Proof. If $X_* \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$, candidate completeness supplies a coherent source normal form. If the illustrative table omits it, (i) holds. If the form is present under another presentation, (ii) holds. If its only novelty is finite coherent composition, (iii) holds by hybrid closure. If it is absent from the current target-relative carrier-path closure, including when it is only latently capable under a future attachment, (iv) holds. If the apparent extra architecture disappears because an instantiated source relation lies in its type-indexed null class, (v) holds by [theorems 6.8 and 14.6](#). If a purported source object violates $\mathfrak{E}_{\text{src}}$, (vi) holds. If it is not same-target or violates the comparison class, (vii) holds. If none applies, X_* contradicts the source-coherent normal-form theorem, yielding (viii). $\square$

17.5 No downstream inference

Theorem 17.8 (No survivor inference). *The candidate-completeness isomorphism entails none of the following:*

$$\Pi_v(X_u^{\text{act}}(\theta)) \in \mathfrak{S}_v^{\text{adm}}, \quad \text{Surv}_{\text{sim}}(u, \theta), \quad E_C(u, r), \quad \text{RealizationLicensed}(u).$$

Proof. None of the corresponding predicates occurs in the source carrier theory, carrier trace, coherent constructor algebra, scientific family class, or candidate quotient. They require the independently fixed target-relative interface and subsequent simultaneous-constraint analysis/the later role-occupancy analysis constraint data. $\square$

17.6 Comparison-class boundary

The completeness theorem is relative to the finite-presentation clause. A candidate with an essentially infinitary occupancy-bearing architecture is outside the present class unless it admits a finite symbolic presentation. Such a construction is not refuted; it opens a comparison-class extension. Likewise, a new primitive support, persistence, constituent, incidence sort, or constitutive equation within the same target is an explicit counterexample to $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$, not something the theorem may relabel into an existing family.

Theorem 17.9 (Scope-relative exhaustion). *Within $\mathfrak{C}_v^{(3)}$, every candidate-relevant novelty is either a new value of an existing source parameter, a new finite source-coherent incidence diagram over $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$, a malformed attempted source object, or a source-ontology falsifier. No finite table count or success on a representative suite is needed to establish this disposition.*

17.7 Strongest falsifier

Direct falsifier of this paper

Exhibit a same-target, currently target-participating physical carrier architecture $X_* \in \mathfrak{C}_v^{(3)}$ whose source-coherent carrier trace cannot be reproduced by any generated coherent normal form, while requiring no new target coordinate or comparison-class extension. Such an X_* directly refutes theorem 17.5. No auxiliary encoding, additional test, malformed source term, or post hoc candidate label can cure the scientific failure.

18 Independence of candidate generation

18.1 Scientific factorization

A candidate domain can remain biased even when its individual entries are well formed: source sorts, constitutive equations, or family boundaries might have been chosen after later outcomes were inspected. Independence of generation is therefore a separate theorem. Let S_{src} and O_{out} be the source and outcome packets of Eqs. (3) and (4).

Definition 18.1 (Scientific generator). *The scientific generator is the composite*

$$\text{Gen}_{\text{car}}(S_{\text{src}}) : \mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)}) \longrightarrow \mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_v^{(3)}) \longrightarrow \text{ActRed}_{\mathbb{T}_v^{(3)}} \longrightarrow \text{NF}_{\text{car}} \longrightarrow \mathfrak{U}_v.$$

It has no argument of type O_{out} . In particular, constitutive coherence is fixed by $\mathfrak{C}_{\text{src}}$, pointwise activity and source-null structure are fixed by $\mathfrak{A}_{\text{src}}$, and interface adequacy and simultaneous physical viability are absent from the generator.

Theorem 18.2 (Source-first generation independence). *For any downstream outcome packets $O_{\text{out}}, O'_{\text{out}}$*

compatible with the same source packet,

$$\text{Gen}_{\text{car}}(S_{\text{src}}, O_{\text{out}}) \simeq_{\text{cand}} \text{Gen}_{\text{car}}(S_{\text{src}}, O'_{\text{out}}).$$

Equivalently, the source-coherent family templates, full coherent source domains, target-participating loci, activity profiles, candidate quotient, and universe $\mathfrak{U}_v$ are invariant under deletion, randomization, or replacement of all candidate-specific downstream outcomes.

Proof. The source sorts and carrier semantics are fixed in Sections 3–5. The source operations, constitutive equations, and activity-support/null laws are fixed in definitions 6.1, 6.2 and 6.4 before downstream evaluation. The coherent source algebra and pointwise activity reduction depend only on those data, and the normal-form and instance quotients depend only on the resulting active source-coherent carrier traces. None contains a map from O_{out} . Hence the scientific generator factors through the projection to S_{src} . $\square$

Corollary 18.3 (Named-model independence). *Deleting all names of successful or familiar neutrino mechanisms leaves the generated family universe unchanged. Named constructions can only be mapped into already-generated source-coherent normal forms.*

Corollary 18.4 (Generalized-symmetry narrative independence). *Removing all motivational references to higher-form or non-invertible symmetry leaves the topological and extended source branches unchanged because they are derived from the upstream support/operator ontology. Their constitutive fusion, junction, module, and higher-coherence equations remain fixed by $\mathfrak{E}_{\text{src}}$, not by later phenomenological success.*

18.2 Operational check of independence

The factorization theorem also admits an operational invariance check: starting from one source specification, one may recompute the source-coherent family presentations and pointwise activity profiles after the outcome data are removed or varied. The scientific quotient must remain unchanged up to lawful family and instance equivalence. Such a check illustrates the theorem but does not replace it. The proof is the absence of any map from O_{out} into the source sorts, coherence laws, activity semantics, normal forms, or candidate quotient.

18.3 Outcome information excluded from source formation

No datum consulted in forming candidate identity may encode:

- candidate-specific interface coordinates or interface adequacy;
- simultaneous-constraint failures or survivor status;
- role eligibility or occupancy assignment;
- an expected model or preferred occupant;
- candidate scores, likelihoods, or rankings;
- an encoding from which any such quantity can be reconstructed.

By contrast, support kind, persistence type, representation category, primitive source parameter domains, incidence type, constitutive coherence, source activity laws, and pointwise null/non-null status are legitimate source data because they belong to the carrier object before downstream evaluation.

Theorem 18.5 (Interface-value blindness). *Erasure of every candidate-specific target-relative interface value preserves $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$, Fam_v , the full coherent domains $\{\Theta_u^{\text{coh}}\}$, the participating loci $\{\Theta_u^{\text{part}}\}$, the pointwise activity profiles, $\simeq_{\text{fam}}$, and $\mathfrak{U}_v$.*

18.4 Post-generation representative constructions

Named constructions are introduced only after the source theory, coherence profiles, and family boundaries have been fixed. A map

$$\text{Ex} : \text{NamedConstructions} \rightarrow \mathcal{U}_v$$

then asks where each familiar mechanism lies in the independently derived universe. Failure of a familiar construction to land may expose a missing source sort, operation, incidence relation, or constitutive equation. Successful landing is evidence that the grammar covers that example, not proof of completeness.

Proposition 18.6 (Representative constructions are non-generative). *Adding or removing a named construction after the source theory is fixed cannot change the generated family universe unless that construction supplies a genuine counterexample to the source carrier theory.*

18.5 Source provenance and falsifiability

Each principal source family, constructor, coherence law, and activity rule must be traceable to an imported physical role or a semantic carrier coordinate. Candidate absence is justified only by source ill-typing, source incoherence, absence of current target participation, or explicit scope. Candidate mergers require carrier-trace equivalence and cannot be inferred from interface equality.

Independence falsifier

If changing downstream outcomes alters a source sort, constitutive equation, activity law, coherent family domain, participating locus, candidate merger, or generated normal form, source-first independence has failed. The affected source derivation must then be reconstructed without the downstream data.

19 Representative stress tests and counterexample classes

19.1 Purpose

The cases in this section test the semantic and source grammars after those grammars have been fixed. They do not define the candidate universe. Each case is classified by target identity, occupancy-bearing reduction, carrier trace, source typing, source-constitutive coherence, normal form, candidate equivalence, spectator status, and scope.

Table 2: Representative stress tests.

Challenge	Pressure on the theorem	Required disposition
Same target-relative interface, distinct local and hidden architectures	Tests improper interface-to-candidate quotienting.	Retain separate traces unless a full carrier isomorphism exists.
Decoupled TQFT	Generates false multiplicity from target-inert world structure.	Strip as spectator if no current carrier path exists.
Decoupled copy of an admitted carrier architecture	Latent capability is mistaken for present carrier participation.	Strip while disconnected; retain only after actual incidence is instantiated.
Activated copy after a new physical incidence	Tests whether spectator status is fixed by type rather than architecture.	Add the participating sector and incidence to the carrier trace.
Inflow-supporting TQFT	Appears locally decoupled but bears anomaly or attachment incidence.	Retain in the topological hybrid core.

Challenge	Pressure on the theorem	Required disposition
Law-only reduced channel versus explicit dilation	Same operational map, different carrier ontology.	Separate unless hidden structure is carrier-semantically eliminable.
Composite with elementary-like effective field	Tests primitive/composite collapse.	Preserve derivation and constituent incidence.
Boundary-only carrier	Risks changing the source/detection target.	Same target only under a fixed boundary attachment type.
Medium-induced co-carrier	Risks treating environment as an external selector.	Retain when physical medium incidence is carrier-bearing.
Non-invertible defect	Not an ordinary particle or group action.	Generate through extended nodes and fusion/junction incidence.
Typed but incoherent defect datum	Endpoint types pass while an associator violates pentagon or unit coherence.	Reject before candidate formation as <code>SourceMalformed</code> .
Arbitrary representation/-multiplicity family	Defeats finite scans.	Retain symbolic source domains.
Many-body nonfactorizing incidence	Defeats a pairwise coupling list.	Represent by a primitive hyperedge and the required higher-coherence data.
Weakly coupled active carrier	Small but source-active incidence tempts deletion.	Retain without a numerical threshold.
Source-null degenerate coupling point	One symbolic family contains a point where a carrier incidence becomes physically null.	Apply pointwise activity reduction and merge only when complete active traces agree.
Downstream selection of the degenerate point	A later constraint appears to change candidate identity.	Select the class already determined by pointwise activity; no retrospective merge.
New support notion	Direct pressure on ontology completeness.	Existing source object, target update, or genuine falsifier.
New constitutive equation on existing sorts	Tests whether source objecthood was fully specified.	Existing equation, malformed object, or source-theory reopening.
Essentially infinitary carrier network	Pressures the finite-presentation boundary.	Finite symbolic normal form or explicit comparison-class extension.

19.2 Same interface, distinct carriers

Let X be a law-only effective channel and Y an explicit hidden dilation inducing the same target interface. The target-relative interface may identify their target-visible packets. Candidate identity nevertheless keeps them distinct whenever $\chi_{\text{car}}(Y)$ contains hidden nodes or projection incidence absent from $\chi_{\text{car}}(X)$. This tests theorem 4.7.

19.3 Spectator and inflow TQFTs

Let Z be a topological sector. If $W \times Z$ has no incidence path from Z to the carrier core or target attachment, theorem 14.2 removes it. If Z supplies inflow, a defect action, global-form data, or a target-bearing junction, it remains. The label “TQFT” does not determine the result.

19.4 Latent carrier copy and actual activation

Let Z_{dec} be a source-coherent copy of an admitted carrier architecture, but suppose $W \times Z_{\text{dec}}$ contains no current attachment or incidence path from the copy to the target-bearing core. By lemma 3.7 and theorem 14.4, the copy is a spectator despite its intrinsic capability and does not multiply candidate identity. If a source-coherent physical incidence e is then actually added and a current target-relative path passes through Z_{dec} , the participating copy and e enter the core and alter the carrier trace unless lawfully isomorphic to existing structure.

19.5 One family across an active/null incidence boundary

Let

$$X_g = X_{\text{base}} \star_{e_g} Z$$

be a source-coherent symbolic family whose source activity law makes e_g nonnull for $g \neq g_0$ and null at g_0 . No numerical cutoff is introduced. The activity reduction $\text{ActRed}_{\mathbb{T}_V^{(3)}}(H_g)$ is evaluated pointwise: active points retain Z ; the null point strips Z when no independent active path remains and is instance-equivalent to the base architecture only when their complete active traces agree. A later physical analysis may select g_0 , but it selects a candidate class already fixed here. This is the limiting-pair test treated by theorem 14.6 and corollary 14.7.

19.6 Malformed fusion data

Suppose a proposed defect architecture contains legally typed objects, fusion multiplicities, junction spaces, and an associator of the declared type, but the associator violates the pentagon identity. The proposal belongs to $\mathcal{G}_{\text{car}}^{\text{raw}}(\mathbb{T}_V^{(3)})$ but not $\mathcal{G}_{\text{car}}^{\text{coh}}(\mathbb{T}_V^{(3)})$. It is not a coherent carrier that later fails a physical test; it fails to instantiate the asserted fusion object and receives status `SourceMalformed`. This tests theorem 6.18 and proposition 10.7.

19.7 Boundary physics

A physical edge mode attached to the fixed charged-current boundary is a boundary-family candidate. Replacing the charged-current measurement relation itself changes the target. The distinction is therefore between a boundary-supported carrier and a target update, not between a successful and unsuccessful model.

19.8 Unbounded local families

A source representation category may contain infinitely many objects and multiplicities. The family is represented symbolically. A finite list of low-dimensional representations would not establish completeness even if it contained every familiar model.

19.9 New incidence relations or constitutive equations

A same-target carrier may require a physical relation absent from the current incidence category or a constitutive law absent from $\mathfrak{E}_{\text{src}}$. If that datum is source-authorized and target-relative, it is a genuine counterexample to $\mathbb{S}_{\text{car}}(\mathbb{T}_V^{(3)})$. Adding a named example without enlarging and reproving the source theory would not restore the theorem.

19.10 Law-only versus hidden-sector ambiguity

The distinction is target-relative and semantic. If no carrier-semantic use detects the hidden architecture, spectator reduction and lawful dilation equivalence may collapse it to the law-only trace. If hidden support or incidence is detectable, the candidates remain distinct. In either case, the claimed law or dilation must satisfy its source identities before the comparison is made.

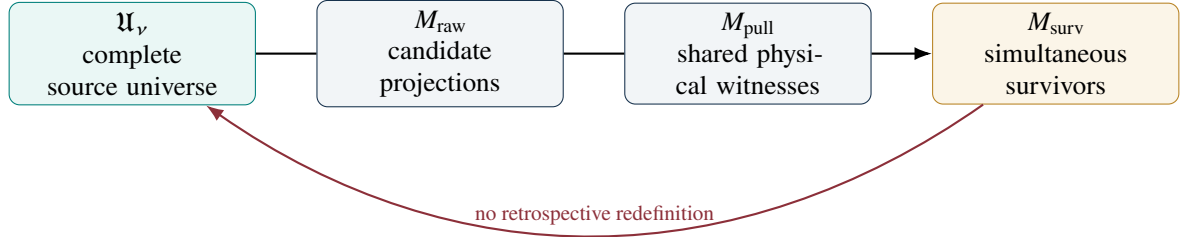


Figure 7: Downstream use of the fixed candidate universe. Physical failure and survival are evaluated only after source-first candidate identity has been established.

19.11 Disposition of representative cases

Each case receives one of the statuses

Generated, Equivalent, Hybrid, Spectator, SourceMalformed, TargetUpdate,
Counterexample, Open.

A completeness theorem cannot use an untyped “not considered” category.

Theorem 19.1 (Representative-case coverage). *Every case in table 2 has a typed disposition under semantic reduction, the source carrier theory, the coherent constructor algebra, spectator reduction, or the scope boundary. None requires candidate-specific interface selection.*

Proof. The dispositions follow from the cited results and are developed in Appendix D. These examples are not used to establish completeness; they test the independently proved grammar and therefore do not make the proof circular. $\square$

20 Discussion, scope, and conclusion

20.1 Data supplied to subsequent constraint analysis

A subsequent simultaneous-constraint analysis receives

$$\mathcal{H}_{3 \rightarrow 4} = (\mathbb{T}_v^{(3)}, \mathcal{R}_v^{\text{osc}}, \mathfrak{S}_v^{\text{adm}}, \mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)}), \mathcal{U}_v, \{(\Theta_u^{\text{coh}}, \Theta_u^{\text{part}}, \mathbf{A}_u)\}_{[u] \in \text{Fam}_v}, \simeq_{\text{cand}}, \text{Prov}_{\text{src}}). \quad (11)$$

This packet contains the source sorts, constitutive equations, type-indexed activity/null laws, complete coherent family domains, source-defined participating loci, pointwise activity profiles, and candidate quotient. It contains no candidate-specific interface values, failure diagnoses, survivor labels, role-eligibility edges, assignment orbits, or preferred occupants.

Theorem 20.1 (Stability under subsequent evaluation). *Subsequent simultaneous evaluation may adjoin candidate projections, shared-witness structures, constraint values, failure diagnoses, and survivor status to $\mathcal{H}_{3 \rightarrow 4}$. It may not retroactively alter a source sort, constitutive equation, activity-support/null law, coherent family identity, coherent source domain, participating locus, pointwise activity profile, candidate-equivalence merge, current-participation carrier-core rule, or latent-capability spectator disposition without reopening the corresponding source theorem.*

Proof. Family identity, constitutive coherence, complete source domains, participating loci, and the active/spectator boundary are functions of the source theory and source parameter point before physical evaluation. Later survivor data are excluded by theorems 2.5 and 18.2. Any retrospective alteration induced by those data would reverse the proved dependency order. $\square$

20.2 The subsequent simultaneous-constraint problem

For every generated candidate instance, the next analysis forms a raw simultaneous physical object, its shared-witness structure, and the coupled constraint bundle:

$$\mathfrak{U}_\nu \longrightarrow M_{\text{raw}}^{\text{sim}} \longrightarrow M_{\text{pull}}^{\text{sim}} \longrightarrow M_{\text{surv}}^{\text{sim}}.$$

The resulting survivor set may be empty, may contain several inequivalent classes, or may collapse to a smaller family. The theorem proved here does not anticipate that outcome.

20.3 Exact nonclaims

Nonclaim 20.2. *The present theorem does not establish that any generated candidate satisfies the complete physical interface.*

Nonclaim 20.3. *The present theorem does not imply that new local matter, a mass term, a mixing matrix, a right-handed neutrino, a Weinberg operator, a hidden sector, a defect, or any other named mechanism is required.*

Nonclaim 20.4. *The present theorem does not establish uniqueness, minimality, live-residue status, role eligibility, compatibility, or realization.*

Nonclaim 20.5. *The candidate-completeness theorem is relative to $\mathbb{T}_\nu^{(3)}$, $\mathfrak{C}_\nu^{(3)}$, the imported physical carrier ontology, source-constitutive coherence, and finite occupancy-bearing presentation. It is not an unrestricted ultraviolet-completion theorem.*

Nonclaim 20.6. *Finite enumeration or computational consistency does not establish semantic candidate completeness.*

20.4 Main theorem

Theorem 20.7 (Source-first exhaustion of the physical carrier universe). *Fix the target fiber $\mathbb{T}_\nu^{(3)}$, the comparison class $\mathfrak{C}_\nu^{(3)}$, the independently defined occupancy-bearing reduction and carrier trace, the independently fixed source carrier theory $\mathbb{S}_{\text{car}}(\mathbb{T}_\nu^{(3)})$, and its saturated source-coherent finite constructor algebra. Then:*

- (i) χ_{car} is a complete invariant of carrier-semantic equivalence;
- (ii) every semantic carrier architecture admits a source-generated parametric normal form;
- (iii) every source-coherent participating normal-form instance has a physical carrier interpretation after source-fixed pointwise activity reduction, while merely typed but incoherent terms and empty-core parameter points are excluded from candidate formation;
- (iv) candidate equivalence is exactly carrier-semantic equivalence and is strictly finer than the target-relative interface equivalence;
- (v) target-inert and merely latent disconnected sectors do not multiply candidates; source-null degenerate incidences are stripped pointwise, every source-active non-null incidence is retained without a weak-coupling threshold, and every finite lawful hybrid is generated;
- (vi) the generated universe is independent of all downstream candidate outcomes;
- (vii) there is a canonical isomorphism

$$\text{CarrierReal}_{\text{sem}}(\mathbb{T}_\nu^{(3)})/\equiv_{\text{car}} \cong \mathfrak{U}_\nu.$$

Proof. Items (i)–(vi) are theorems 4.4, 13.3, 14.2, 14.6, 15.3, 16.5, 16.6 and 18.2. Source soundness uses theorems 6.8, 6.18 and 6.19. Item (vii) is theorem 17.5. $\square$

20.5 Interpretation

The theorem closes the candidate-generation problem at the level required for a later physical exhaustion argument. The resulting universe is not a list of favored neutrino models. It is the parametric quotient of all finitely presented, source-coherent, pointwise activity-reduced, currently target-participating carrier architectures generated by the fixed physical ontology: existing-substrate and law-only structures, new local carriers, composites, topological sectors, defects, solitons, boundary modes, environmental and hidden co-carriers, and every lawful finite hybrid. The proof is independent of which candidates later satisfy the physical constraints.

This distinction matters scientifically. A later conclusion that only one carrier family survives is persuasive only if non-particle and nonlocal alternatives entered the universe independently rather than being introduced retrospectively. The present result establishes that prerequisite. It also preserves candidate identity more finely than the target interface: distinct microscopic carrier architectures remain available for simultaneous testing even when they induce the same carrier-neutral packet.

The remaining question is therefore precise: which elements of the source-first universe satisfy the complete physical interface and all simultaneous physical constraints through one shared witness structure?

Conclusion

Within the fixed target, imported carrier ontology, source-coherent finite-presentation class, and carrier-semantic quotient, the physical carrier universe is exactly the parametric source-generated universe $\mathcal{U}_\nu$. Candidate generation is complete before physical filtering, pointwise activity determines identity before subsequent selection, and neither model names nor survivor results contribute to the definition of the universe.

A Formal target carrier signature

A.1 Typed source objects

The carrier signature used in the main text can be formalized as a small collection of source groupoids and incidence profunctors. The groupoid formulation keeps lawful redescription inside the type system without selecting a privileged representative.

Definition A.1 (Source carrier signature). *A source carrier signature is the tuple*

$$\mathfrak{D}_{\text{car}}(\mathbb{T}_\nu^{(3)}) = (\mathbf{N}, \mathbf{S}, \mathbf{D}, \mathbf{K}, \mathbf{A}, \mathbf{P}, \mathbf{R}, \mathbf{T}),$$

where:

- $\mathbf{N}$ is the derivation/newness groupoid;
- $\mathbf{S}$ is the support groupoid;
- $\mathbf{D}$ is the persistence/dynamical groupoid;
- $\mathbf{K}$ is the constituent-kind groupoid;
- $\mathbf{A}$ is a fibred category of intrinsic attributes over $\mathbf{K}$;
- $\mathbf{P}$ is a fibred category of primitive parameter domains;
- $\mathbf{R}$ is the typed collection of physical incidence relations;
- $\mathbf{T}$ is the category of target attachments to $\mathbb{T}_\nu^{(3)}$.

A primitive node type is an object of the Grothendieck construction

$$\int_{(n,s,d,k) \in \mathbb{N} \times \mathbb{S} \times \mathbb{D} \times \mathbb{K}} \mathbf{A}_k \times \mathbf{P}_k \times \mathbf{T}_k.$$

This notation is not used as a premise in the physical proof; it records the dependency of attributes and parameters on constituent kind.

A.2 Essentially algebraic source theory

Definition A.2 (Formal source carrier theory). *The formal source carrier theory is*

$$\mathbb{S}_{\text{car}}(\mathbb{T}_{\nu}^{(3)}) = (\mathbb{D}_{\text{car}}(\mathbb{T}_{\nu}^{(3)}), \mathbf{M}_{\text{car}}, \mathfrak{E}_{\text{src}}, \mathfrak{A}_{\text{src}}),$$

where $\mathbf{M}_{\text{car}}$ contains the typed formation, incidence, restriction, derivation, attachment, and gluing operations, $\mathfrak{E}_{\text{src}}$ contains the constitutive equations of the source objects and maps, and $\mathfrak{A}_{\text{src}}$ contains the type-indexed source activity-support/null laws for instantiated incidence and attachment morphisms. A source presentation must first be a model of this theory; a coherent parameter point becomes a candidate instance only when its pointwise activity reduction has a nonempty target-participating core.

The equation family is type-indexed. For representation-bearing data it contains identity and composition laws; for fusion or defect data it contains pentagon, triangle, module, and junction coherence; for projection/dilation data it contains the equations defining the claimed embedding, projection, or dilation; for derived nodes it contains constituent/derivation compatibility; and for hybrids it contains the required overlap and gluing commutativity. These equations contain no interface adequacy or downstream survivor status predicate.

A.3 Pointwise activity over source parameter families

For each incidence or attachment sort r , the source theory contains an activity-support map asupp_r and a source-null class $\mathfrak{N}_{\text{src}}(r)$. For a coherent family template H_u , this paper retains the full coherent locus

$$\Theta_u^{\text{coh}} = \{\theta : \text{WellTyped}(H_u(\theta)) \wedge \text{Coh}_{\text{src}}(H_u(\theta))\}.$$

The participating locus is determined source-first by

$$\Theta_u^{\text{part}} = \{\theta \in \Theta_u^{\text{coh}} : \text{ActRed}_{\mathbb{T}_{\nu}^{(3)}}(H_u(\theta)) \neq \emptyset\}.$$

No numerical magnitude threshold is used: every source-nonnull relation is retained, while source-null incidence contributes no current carrier edge. Distinct parameter points, including points of different symbolic families, represent the same candidate exactly when their activity-reduced carrier traces are carrier-equivalent. These assignments are fixed before subsequent simultaneous-constraint analysis.

A.4 Incidence typing

An incidence type $r \in \mathbb{R}$ has a finite ordered list of input node types, a finite ordered list of output node types, and optional physical attributes. Hyperedges can be symmetric when the physical relation requires no order. The source signature includes the following principal incidence classes:

Incidence class	Physical content
Coupling/interaction	A physical relation through which carrier nodes exchange or jointly bear target-relevant structure.
Shared support	Nodes occupy or require a common support object.
Composition/binding	Constituent nodes generate a derived composite or collective node.

Projection/dilation	Visible and hidden nodes are related by embedding, projection, reduction, or dilation.
Boundary attachment	A bulk, extended, or boundary node attaches to the fixed source/detection boundary type.
Fusion/junction	Extended or defect nodes compose through fusion or junction spaces.
Inflow/obstruction support	A topological or boundary sector supplies a physical obstruction/inflow relation.
Environmental co-carriage	A medium or environment shares occupancy-bearing work with another carrier.
Law/process action	A constitutive or dynamical object acts on existing carrier nodes without a new primitive bearer.
Target attachment	The carrier core is typed for later projection to $\mathfrak{S}_v^{\text{adm}}$.

The table is descriptive; $\mathbf{R}$ is source-open. A new same-target relation enlarges the signature and reopens the theorem rather than being assigned to a catch-all edge by fiat.

A.5 Finite presentations

Definition A.3 (Finite carrier presentation). *A finite carrier presentation over $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$ is a finite typed hypergraph with a finite set of node/edge/coherence templates that satisfies the instantiated equations in $\mathfrak{E}_{\text{src}}$, even if node multiplicities and parameter domains are symbolic or infinite.*

A template with multiplicity variable $n \in \mathbb{N}$ is finite as a family presentation. An individual instance at finite n is a finite graph. An essentially infinitary architecture can enter only if it admits a finite symbolic presentation in the source signature.

A.6 Target attachment type

A currently instantiated target attachment has four source-level components:

1. attachment to the fixed physical substrate;
2. attachment to the charged-current source/detection boundary type;
3. an actual source-level attachment map to the propagation-role complex, recorded without evaluating its interface adequacy;
4. declaration of any activated operator or extended-sector category.

A target attachment is not an interface adequacy certificate. It is a physical map presently included in the candidate architecture and later evaluated by subsequent simultaneous-constraint analysis. A merely possible future attachment is not part of the current carrier core. Neither is an instantiated attachment whose source activity support is empty at the parameter point. The core is the source-constitutive closure of the source-active currently instantiated incidence hyperpaths ending at source-active target attachments.

A.7 Source versus constraint typing

The following are source-level type errors:

- an edge whose endpoint types are undefined;
- a node with no physical support or persistence semantics in the comparison class;
- a target attachment requiring a different source/detection boundary without a target update;
- a purported candidate instance with no source-active target attachment after pointwise activity and spectator reduction;

- a parameter with no source domain;
- a representation, fusion, module, dilation, derivation, or gluing datum that violates its constitutive source equations;
- a proposal whose candidate identity is changed by subsequently selected parameter value instead of by the source-fixed pointwise activity law;
- an object generated solely by a survivor label or expected endpoint.

The following are not source type errors and are deferred:

- anomaly noncancellation after a valid source representation has been formed;
- failure to reproduce the visible response packet;
- vacuum or deformation incompatibility;
- failure of a common witness record;
- instability or lack of a viable parameter point;
- role ineligibility or excess live residue.

A.8 Universe convention

All source groupoids are taken inside one fixed Grothendieck universe sufficient for the comparison class. Parameter families and evaluator classes may be large relative to ordinary sets but are small in that universe. This convention prevents set-theoretic size questions from obscuring the physical theorem.

B Detailed proofs of the completeness bridge

B.1 Proof of trace recovery

Let $X, Y \in \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})$. Suppose $\chi_{\text{car}}(X) \neq \chi_{\text{car}}(Y)$. Because the traces are finite typed diagrams up to isomorphism, there exists a finite distinguishing query. It can be chosen from:

1. number and types of carrier-bearing nodes;
2. a support, persistence, derivation, constituent, or intrinsic-attribute projection;
3. existence or type of an incidence hyperedge among a finite tuple of nodes;
4. a source-constitutive structure or commuting-diagram value;
5. a parameter-domain or reparameterization invariant;
6. a target attachment or activated operator-category value.

Each query is a carrier-semantic use. This proves the separating direction of [theorem 4.4](#). The descent direction follows from the definition of carrier-semantic uses as lawful functionals of the occupancy-bearing reduction invariant under ontology-preserving isomorphism.

B.2 Carrier-bearing reduction without unique set-theoretic minimality

The main text defines the core as the source-constitutive closure of all source-active currently instantiated target-relative carrier paths. This avoids an arbitrary choice among several minimal supports and prevents counterfactual carrier capability from entering present candidate identity. If two active subdiagrams each contribute to the current attachment architecture, both are retained unless every carrier use and the complete carrier trace identify them lawfully. A disconnected source-coherent copy with no current path is absent from the core, while adding an actual incidence creates a new path and therefore a new architecture. No choice principle, coupling threshold, or preferred representative is needed.

B.3 Pointwise activity and degenerate source strata

Let $H_u(\theta)$ be a source-coherent family instance. Source coherence establishes that every instantiated relation is a well-defined source object; the type-indexed activity support then establishes whether that relation performs current carrier work at θ . The operations occur in this order:

$$H_u(\theta) \mapsto \text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta)) \mapsto \text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))) \mapsto \chi_{\text{car}}.$$

If a coupling relation e_g becomes source-null at $g = g_0$, the edge is deleted before carrier-path closure. Any component supported only by that edge is then outside the active core. If e_g is source-nonnull, it remains regardless of numerical magnitude. Thus a single coherent symbolic family may meet several source-defined activity strata without any downstream reclassification. When the active trace at a degenerate point equals the active trace of another family instance, candidate equivalence performs the merge already in this paper. A subsequent simultaneous constraint analysis may later select the point but cannot create or remove that identity.

B.4 Graph-induction proof of constructor completeness

Let

$$H = (V, E, \lambda_V, \lambda_E, \omega_H, \Theta, a_T)$$

be a finite typed carrier hypergraph. Choose an enumeration $V = \{v_1, \dots, v_n\}$ and $E = \{e_1, \dots, e_m\}$ for the proof.

For $j = 0, \dots, n$, define H_j as the diagram containing the first j vertices and no edges. H_0 is the empty source diagram. If v_{j+1} is primitive/existing, apply **Atom**; if derived, first generate its finite constituent subdiagram and apply **Derive**. By induction, H_n is generated.

For $k = 0, \dots, m$, let G_k contain all vertices and the first k hyperedges. Since each e_{k+1} is an object of $\text{Inc}_{\mathbb{T}_v^{(3)}}$ with already generated endpoints, $\text{Incide}_{\lambda_E(e_{k+1})}$ generates the next raw incidence diagram. Adjoin the finitely many instantiated constitutive maps and coherence cells in ω_H ; source coherence holds because H is a model of $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$. At $k = m$, add the finite target-attachment diagram. The result is a coherent source term isomorphic to H .

The enumerations are proof devices only. Different enumerations generate candidate-equivalent terms because they have the same trace. This is the substantive content of confluence.

B.5 Typing, source coherence, and viability

A raw generated diagram is locally typed when its nodes and edges have declared source types. It is source sound only when it also satisfies the constitutive equations of the asserted source objects. Thus an associator violating the pentagon identity, a map violating the representation law, or a malformed dilation is excluded before candidate formation. A source-coherent diagram need not be an anomaly-free quantum theory or an interface-adequate completion. The subsequent simultaneous analysis owns those viability questions.

B.6 Proof of the completeness isomorphism

The maps Φ and Ψ of Section 17 are inverses because both semantic and source sides are classified by the same complete carrier trace. Formally, the diagram

$$\begin{array}{ccc} \text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)}) / \equiv_{\text{car}} & \xrightarrow[\cong]{\overline{\chi_{\text{car}}}} & \text{im}(\chi_{\text{car}}) \\ & \searrow \Phi & \uparrow \overline{\mathcal{N}_{\text{car}}} \\ & & \mathfrak{U}_v \end{array}$$

commutes, where $\overline{\chi_{\text{car}}}$ is injective by trace completeness and surjective onto its image, and $\overline{\mathcal{N}_{\text{car}}}$ is bijective by source-normal-form existence, soundness, and exactness. The commutative diagram is a compact

presentation of the two inverse classifications; the following set-theoretic paragraph states the same argument without categorical notation.

B.7 Alternative set-theoretic formulation

Avoiding categorical notation, let Q_{sem} be the set of semantic equivalence classes and Q_{src} the set of generated instance classes. Define $f : Q_{\text{sem}} \rightarrow Q_{\text{src}}$ by source reconstruction and $g : Q_{\text{src}} \rightarrow Q_{\text{sem}}$ by interpretation. Trace equality proves $g \circ f = \mathbf{1}$ and $f \circ g = \mathbf{1}$. This is the formulation actually used in the main theorem.

B.8 Proof of omission disposition

For an unlisted proposal X_* , first test same-target status and comparison-class membership. If either fails, the target-update disposition holds. Otherwise apply candidate completeness to obtain $[u, \theta]$. If the source family is present in the finite catalogue but the instance or alias is missing, the omission is administrative. If another family presentation represents the same class, candidate equivalence applies. If the architecture is explicitly built as a finite coherent composition, hybrid closure applies. If the proposed novelty has no current target attachment or instantiated carrier-incidence path, spectator stripping applies; latent capability under a future coupling does not alter this disposition. If the typed source data violate a constitutive equation, they do not instantiate the asserted carrier object. Failure of all these conditions contradicts the completeness theorem and is a genuine counterexample.

B.9 No inference from family density

The distribution of instances across families is irrelevant to the proof. A local family may contain infinitely many representations while a topological family has a finite set of labels; neither cardinality confers preference. No measure on $\mathcal{U}_v$ is defined.

B.10 Latent-capability boundary

Let Z_{dec} be a complete source-coherent copy of an admitted carrier architecture. In $W \times Z_{\text{dec}}$, no current carrier path traverses the copy when no physical incidence or target attachment is instantiated. Therefore the path-union definition of $\text{Car}_{T_v^{(3)}}$ excludes it, even though a larger architecture could activate it. Adding an actual source-coherent incidence e produces a new path and hence a new carrier core. This establishes the spectator/activation boundary without invoking a coupling threshold or any a downstream outcome.

C Principal family classes and representative constructions

C.1 Status of the table

The table below is a human-readable projection of the principal source strata. It is not the completeness proof, it is not the extensional scientific family class, and it is not required to be finite at the instance or template level. The scientific class is defined in Definitions 15.8 and 15.10 as all source-coherent normal-form templates modulo lawful family equivalence.

Table 4: Principal source families.

Family	Primitive trace profile	Open source data	Illustrative representative construction types
F_{base}	Existing substrate nodes; new incidence or collective organization permitted.	Existing-sector process, collective, environment, attachment parameters.	Existing-substrate deformation; collective reorganization.

Family	Primitive trace profile	Open source data	Illustrative representative construction types
F_{law}	No new primitive node; new law/process object on existing nodes.	Kernel, channel, constitutive, and operator parameters.	Nonstandard propagation law; effective decoherence law.
F_{local}	New primitive bulk-local or localized field/particle node.	Representation, spin/statistics, chirality, multiplicity, couplings, masses/scales.	Dirac/Majorana-like local fields; generic singlet or charged local sectors.
F_{comp}	Derived node with constituent diagram and binding/emergence map.	Constituent types, binding data, collective parameters.	Bound state; quasiparticle; condensate mode.
F_{top}	Topological/global support or incidence.	Global form, topological labels, response, linking data.	TQFT-like co-carrier; flux/winding sector.
F_{ext}	Line, surface, defect, or junction node.	Fusion, junction, action, support, anomaly/inflow data.	Non-invertible defect; higher-form support.
F_{sol}	Localized nonperturbative configuration.	Topological charge, profile, stability, boundary data.	Soliton, vortex, monopole-like sector.
F_{bdy}	Boundary/edge support attached to fixed boundary type.	Boundary algebra, edge labels, attachment maps.	Edge mode; source/detector localized carrier.
F_{env}	Physical medium/environment shares carrier work.	Environment state, coupling, memory, phase parameters.	Medium-induced co-carriage; memory environment.
F_{hid}	Hidden/enlarged state-space node with projection/dilation incidence.	Hidden dimension, representations, couplings, projection maps.	Sterile/hidden sector; explicit open-system dilation.

C.2 Named representative constructions

The following mappings are illustrative and non-generative.

Named construction	Source-normal-form location	Data retained
Dirac-type neutrino mass	F_{local} or local hybrid	New local singlet/right-handed nodes if present, chiral incidence, mass/Yukawa parameters, target attachment.
Majorana-type local mass	F_{local} or local/composite hybrid	Local carrier node, reality structure, lepton-number-changing incidence where sourced.
Weinberg-operator description	F_{law} or F_{comp} , depending on whether UV carriers are explicit	Effective operator/law object; explicit UV carrier nodes retained only when part of the candidate.

Seesaw-type completion	Local/hidden hybrid	Heavy local or hidden nodes, mixing/projection incidence, open parameter domains.
Sterile-neutrino model	F_{hid} or local hidden hybrid	Hidden/sterile nodes, visible attachment, multiplicity and coupling.
Decay or invisible-loss model	F_{law} or hidden/environmental hybrid	Reduced process law or explicit decay products/hidden incidence.
Decoherence model	Law-only or environment/hidden hybrid	Channel generator or explicit environment/memory carrier.
Topological/defect completion	F_{top} , F_{ext} , or hybrid	Defect objects, fusion/junction, topological response, attachment.
Boundary-mediated construction	F_{bdy} or boundary hybrid	Boundary node, bulk attachment, source/detector semantics.
Composite neutrino-like excitation	F_{comp}	Constituent diagram, derivation/binding, effective local behavior.

C.3 Representative construction failure protocol

If a representative construction does not appear in the illustrative table, one must first determine whether its source-coherent normal form already belongs to the extensional scientific family class. If it does not, the next question is whether it requires a new source carrier sort, operation, incidence relation, or constitutive equation. Such a case is a scientific counterexample. New parameter values or a new finite hybrid, by contrast, already lie within the source universe and require only a revised representative map.

C.4 Source-coherence qualification

Every family instance must be a model of the source carrier theory. Local endpoint typing alone is insufficient. A typed parameter value for which a representation law, derivation identity, fusion pentagon, module equation, dilation identity, or hybrid gluing condition fails is outside the scientific family parameter domain and is classified as source-malformed, not as a candidate awaiting subsequent simultaneous evaluation. This restriction is source-first and does not use anomaly, vacuum, interface, or survivor results.

C.5 Activity loci are source data, not selection cuts

Every scientific family retains its complete source-coherent locus Θ_u^{coh} . The source theory then assigns a pointwise activity profile and the participating locus Θ_u^{part} . The rows in this appendix do not assume a generic nonzero coupling, a finite set of strata, or a downstream viable subdomain. A source-null degenerate point may reduce to the same active candidate trace as an instance listed under another principal row; the scientific quotient performs that identification before subsequent simultaneous-constraint analysis.

D Detailed stress-test analyses

D.1 Unlisted local representation

A local field in an arbitrarily large representation is generated because the representation coordinate ranges over the full source representation category. It is not necessary to list the representation. If the representation belongs to a different global-form category, the proposal is a controlled target/global-form fiber rather than an omitted same-fiber instance.

D.2 Composite with elementary-like interface

A composite excitation and a primitive elementary field may project to the same target-relative interface. Their derivation coordinates differ, and the composite trace retains constituents and binding incidence when carrier-detectable. They remain separate candidates.

D.3 Target-bearing TQFT

A TQFT-like sector with an inflow edge, topological attachment, or defect incidence is a topological/extended hybrid. Its local observables may be trivial while its carrier trace is not. It remains in the universe.

D.4 Decoupled TQFT

A TQFT factor with no currently instantiated target-relative carrier path is stripped by theorem 14.2. This prevents arbitrary spectator multiplication without declaring all topological sectors irrelevant.

D.5 Decoupled latent-capable copy

Let Z_{dec} be a source-coherent copy of a carrier architecture admitted elsewhere in the universe, but give it no current target attachment and no instantiated incidence to the carrier core. Although it could become a carrier under a different architecture, lemma 3.7 and theorem 14.4 imply

$$\chi_{\text{car}}(W \times Z_{\text{dec}}) = \chi_{\text{car}}(W).$$

The copy is a spectator, not a new candidate.

D.6 Activation by actual incidence

Now add a source-coherent physical incidence or attachment e that places Z_{dec} on a current target-relative carrier path. The participating subdiagram and e enter the carrier core by proposition 3.8. Unless the added active structure is already trace-isomorphic to the old core, the carrier trace changes. This mandatory limiting-pair test distinguishes actual participation from latent capability without using a coupling threshold.

D.7 Non-invertible defect

A non-invertible defect is represented by an extended node plus fusion/junction incidence, associator/unit/module data, and source attributes. Invertibility is not a source requirement. Constitutive fusion and module coherence is enforced before candidate formation; failure to reproduce the target interface is a result of subsequent simultaneous-constraint analysis.

D.8 Typed but incoherent fusion datum

Take legally typed defect objects, fusion multiplicities, junction spaces, and an associator of the declared formal type, but choose the associator so that the pentagon identity fails. The datum belongs to the raw typed term algebra but not to the coherent source algebra. It receives the disposition `SourceMalformed`, not `Generated` and not a downstream viability failure. This hard case distinguishes constitutive objecthood from viability.

D.9 Environment-mediated realization

A fixed background environment is a target anchor; a dynamical environment sharing carrier work is an environmental node. The trace records the distinction. Treating a fit parameter or external calibration as an environment carrier is source ill-typing.

D.10 Hidden dilation

An explicit hidden dilation and a law-only reduced channel are distinct unless hidden structure is carrier-semantically invisible. Minimal dilation isometries can produce candidate equivalence; nonminimal carrier-bearing extensions remain separate.

D.11 Law-only modification

A new effective process on existing nodes belongs to F_{law} if it changes the carrier trace. If it merely rewrites the same law, it is presentation. If it requires a new operator category outside the comparison class, it is a scope update.

D.12 New incidence relation

A many-body or categorical relation not contained in the current incidence category, or a new constitutive coherence law not contained in the source theory, is the cleanest possible scientific falsifier. The source carrier theory must be enlarged and the completeness proof rerun. A special-case row is insufficient.

D.13 Essentially infinitary architecture

An infinite network with a finite symbolic presentation is represented by a family template. An architecture with no finite occupancy-bearing presentation lies outside $\mathfrak{C}_v^{(3)}$. The theorem does not claim to eliminate it; it states the scope boundary.

D.14 Source-null point inside one symbolic family

Consider $X_g = X_{\text{base}} \star_{e_g} Z$ with a coherent source domain containing g_0 , and suppose the type-indexed source activity law gives nonempty support for e_g away from g_0 but empty support at g_0 . For non-null points, Z lies on an active target-relative path and remains in the carrier core, however small the coupling may be. At g_0 , the null relation is removed before carrier-path closure. If no independent active path reaches Z , then

$$\chi_{\text{car}}(X_{g_0}) = \chi_{\text{car}}(X_{\text{base}}).$$

This is a source-level identity assignment, not a downstream survivor status judgment.

D.15 Downstream selection of a degenerate point

If the subsequent analysis later selects g_0 , it selects the candidate class already assigned to X_{g_0} . A subsequent simultaneous constraint analysis may not retain the inactive Z because it prefers the unreduced symbolic template, nor may it strip a source-nonnull Z because the coupling is numerically small. Either operation would violate the fixed source activity semantics.

E Glossary

Term	Meaning in this paper
Carrier-bearing component	A node, source-active hyperedge, source-active attachment, or source-constitutive dependency lying on a currently instantiated target-relative carrier path. Counterfactual future coupling does not establish present carrier bearing.
Occupancy-bearing reduction	The source-constitutive closure of all source-active currently instantiated target-relative carrier paths in a raw same-target world.
Carrier trace	Complete target-relative typed incidence and constitutive diagram of the occupancy-bearing carrier architecture, excluding presentation and spectators.
Carrier-semantic use	Lawful same-target physical question depending only on occupancy-bearing architecture.
Carrier-semantic equivalence	Equality under every carrier-semantic use; equivalent to source-coherent carrier-trace equality.
Source carrier signature	Categories of support, persistence, derivation, constituent kind, attributes, incidence, parameters, and target attachment.
Source carrier theory	The source signature, lawful operations, type-indexed constitutive equations, and type-indexed activity/null laws that define source carrier objects and their current participation.

Term	Meaning in this paper
Local source typing	Correct assignment of source sorts to nodes, maps, parameters, and endpoints; necessary but not sufficient for source objecthood.
Source-constitutive coherence	Satisfaction of all equations and commuting diagrams constitutive of the claimed source objects and their composition.
Raw source algebra	Free finite typed-hypergraph formation algebra before constitutive coherence is imposed.
Coherent source algebra	Finite models of the source carrier theory. Their full coherent parameter loci are retained; candidate instances are formed pointwise only on the nonempty active-core locus.
Source activity support	Type-indexed source-semantic support on which an instantiated incidence or attachment performs current carrier work; fixed before downstream evaluation.
Source-null incidence	An instantiated relation with empty source activity support; it contributes no current carrier edge at that parameter point.
Pointwise activity reduction	Source-first removal of null incidences/attachments followed by active target-path and constitutive closure before candidate identity is assigned.
Full coherent locus	All source parameter points satisfying local typing and constitutive coherence, without downstream viability restriction.
Participating locus	Subset of the full coherent locus on which pointwise activity reduction leaves a nonempty target-participating carrier core.
Activity stratum	Inverse image of one activity-reduced source-equational isomorphism class; no finiteness or manifold assumption is made.
Source normal form	Lawful source-equational isomorphism class of a spectator-reduced, source-coherent generated carrier diagram.
Candidate equivalence	Ontology-preserving source-coherent carrier-trace equivalence, finer than target-relative interface equivalence.
Family-presentation class	Extensional class of all source-coherent parametric normal-form templates, independent of a finite catalogue.
Scientific carrier-family class	Family-presentation class modulo lawful family equivalence.
Family template	Source-coherent parametric normal-form pattern with a full coherent locus, pointwise activity profile, and source-defined participating locus.
Candidate universe	Quotient of the disjoint union of all scientific family participating loci by activity-reduced instance equivalence.
Principal stratum	Human-readable region of the carrier signature such as local, topological, hidden, or law-only; not a primitive exhaustive list.
Hybrid	Finite source-coherent incidence composition of generated carrier components.
Target-inert spectator	Connected source-coherent structure with no current target-relative carrier path and no effect on carrier-semantic uses; latent capability under a future attachment is irrelevant.
Latent carrier capability	Capacity of a source object to become carrier-bearing only after adding a new incidence or attachment; it is not present candidate structure.

Term	Meaning in this paper
Source-malformed proposal	Locally typed data that violate a constitutive source equation and therefore fail to instantiate the asserted carrier object.
Law-only architecture	New occupancy-bearing law or process on existing carrier nodes without a new primitive node.
Existing-substrate branch	Candidate architectures whose primitive nodes lie in the fixed substrate.
Source-first independence	Invariance of the source theory, coherent generated universe, and candidate quotient under deletion or alteration of all downstream candidate outcomes.
Interface adequacy	the established interface theorem property tested later; not a candidate-generation predicate.
Survivor	A candidate remaining after simultaneous constraint analysis; never a field used to generate the candidate universe.
Role occupancy	A later assignment result under Role-Occupancy Closure; not inferred by candidate completeness.
Target update	A change in source/detection semantics, operator category, support ontology, spacetime regime, or other fixed target coordinate.
Source-theory reopening	Addition of a genuinely new same-target source sort, operation, incidence relation, or constitutive equation.
Comparison-class extension	Enlargement beyond finitely presented occupancy-bearing architectures over the current source carrier theory.
Semantic completeness	The theorem that every same-target carrier architecture in the comparison class has a source-coherent normal form up to carrier equivalence and that every generated universe element has a source-level carrier interpretation.
Finite-table coverage	Coverage of displayed representative families; not a semantic proof.

F Principal results and falsification conditions

F.1 Purpose

This appendix collects the theorem spine in a compact self-contained form. It is a navigation aid: the proofs remain in the main text and Appendix B. No theorem is strengthened by appearing in this package.

Principal theorem package

Fix the target fiber $\mathbb{T}_v^{(3)}$, comparison class $\mathfrak{C}_v^{(3)}$, upstream physical carrier ontology, and independently fixed source carrier theory $\mathbb{S}_{\text{car}}(\mathbb{T}_v^{(3)})$. Then the following results compose:

- T1. No-backflow.** Candidate generation factors through the source packet and not through interface values, simultaneous-constraint failures or survivor data, or later role occupancy data.
- T2. Canonical carrier core.** Every raw same-target world has an occupancy-bearing reduction given by the source-constitutive closure of its source-active, currently instantiated target-relative carrier paths.
- T3. Complete carrier trace.** Equality of χ_{car} is equivalent to equality under every carrier-semantic use.

- T4. Source-signature adequacy.** Every carrier node and relation in the comparison class is typed by support, persistence, derivation, constituent, attribute, parameter, incidence, and attachment data.
- T5. Source-constitutive coherence.** Typed source data form a candidate only when they satisfy the constitutive equations of the source objects and maps they instantiate.
- T6. Pointwise source activity.** Type-indexed activity support is evaluated at every source parameter point before candidate identity; source-null incidence is removed, non-null incidence is retained without a magnitude threshold, and dependent spectator structure is pruned source-first.
- T7. Source-coherence and activity soundness.** Every source-coherent participating point denotes a physically interpretable carrier proposal after activity reduction; merely typed but incoherent terms and empty-core points never enter the candidate universe.
- T8. Coherent constructor factorization.** Every finitely presented semantic carrier architecture factors into the source-coherent constructor algebra.
- T9. Hybrid closure.** Every finite lawful source-coherent composition of generated carrier components is generated without introducing an additional primitive family.
- T10. Spectator stripping.** Target-inert and merely latent carrier-capable sectors do not multiply candidate identity; actual source-active attachment or incidence activates the participating sector and changes the trace, while source-null degenerate incidence removes it pointwise when no independent active path remains.
- T11. Candidate quotient exactness.** Candidate equality is exactly carrier-trace equivalence and is finer than target-relative interface equality.
- T12. Normal-form existence and exactness.** Every semantic class has a source-coherent normal form, and two normal forms represent the same candidate exactly when their carrier traces agree.
- T13. Source-first independence.** The generated universe is invariant under deletion, randomization, or replacement of all downstream candidate outcomes.
- T14. Candidate completeness.**

$$\text{CarrierReal}_{\text{sem}}(\mathbb{T}_v^{(3)})/\equiv_{\text{car}} \cong \mathfrak{U}_v.$$

- T15. Omission disposition.** Every proposed omission is absorbed by a coherent normal form, stripped as spectator/presentation, rejected as a malformed source object, classified as target/comparison-class update, or retained as a genuine counterexample.
- T16. Downstream transfer immutability.** A subsequent simultaneous constraint analysis may append evaluation data but may not redefine the fixed source theory or candidate universe without reopening the relevant theorem.

F.2 Logical dependence

The load-bearing order is

$$T1 \rightarrow T2 \rightarrow T3 \rightarrow T4 \rightarrow T5 \rightarrow (T6, T7) \rightarrow (T8, T9, T10) \rightarrow T11 \rightarrow T13 \rightarrow (T12, T14, T15).$$

T12 is proved independently once the source theory is fixed, but it becomes scientifically useful only after the universe and quotient have been constructed.

F.3 Exact scope of completeness

The word *complete* means:

Every same-target, actually target-participating, physically interpretable carrier architecture in $\mathfrak{C}_v^{(3)}$, with a finite source-coherent occupancy-bearing presentation over the imported physical ontology, has a source-generated parametric normal form up to carrier-semantic equivalence; the source activity structure assigns the correct active carrier core at every parameter point before downstream evaluation; conversely, every generated universe element has a source-level physical carrier interpretation after that activity reduction.

It does not mean completeness over arbitrary UV ontologies, arbitrary quantum-gravity regimes, or architectures with no finite symbolic carrier presentation.

F.4 Direct falsifier

A direct refutation of the main theorem consists of an object X_* satisfying all of:

1. X_* has the unchanged target $\mathbb{T}_v^{(3)}$ and belongs to $\mathfrak{C}_v^{(3)}$;
2. X_* performs occupancy-bearing physical carrier work for the fixed role complex;
3. the support, persistence, derivation, constituent, attribute, parameter, incidence, constitutive, and target-attachment data of X_* are explicitly specified;
4. X_* is source coherent but has no source-generated normal form in $\mathfrak{U}_v$ up to $\equiv_{\text{car}}$;
5. the failure cannot be resolved by spectator stripping, lawful reparameterization, or a target/comparison-class update.

Such an object reopens the source theory and completeness theorem.

F.5 Near misses that do not refute the theorem

The following do not by themselves constitute counterexamples:

- a new named model lying in an existing open source family;
- a representation or multiplicity larger than those shown in representative construction tables;
- a finite source-coherent hybrid not previously displayed;
- a new parameter value or lawful reparameterization;
- a source-degenerate parameter point whose null incidence reduces to an already represented active carrier trace;
- an architecture that fails the target-relative interface theorem or the subsequent simultaneous constraints;
- a target-inert or merely latent-capable decoupled spectator sector;
- a typed associator, representation, dilation, derivation, or gluing map that violates the constitutive equations of the source object it claims to define;
- a richer target with new source/detection semantics, operator category, support ontology, or spacetime regime;
- an essentially infinitary architecture outside the finite-presentation comparison class.

F.6 Scientifically permitted endpoint branches

The theorem permits all of the following downstream outcomes:

1. $\mathcal{U}_v = \emptyset$ because the semantic carrier class is empty at the declared scope;
2. $\mathcal{U}_v \neq \emptyset$ but subsequent simultaneous-constraint analysis finds no simultaneous survivor;
3. multiple distinct survivor families remain;
4. one survivor family remains but with an open parameter domain;
5. a known-substrate or law-only candidate survives;
6. a nonlocal, topological, defect, composite, environmental, hidden, or local-matter candidate survives;
7. a genuine omitted carrier kind, source activity law, or constitutive law reopens the theorem.

The theorem does not presuppose a particle outcome. Its purpose is to ensure that any later particle claim is tested against an independently generated, source-coherent candidate universe.

F.7 Nonclaims adjacent to the main theorem

Nonclaim F.1. *Candidate completeness does not imply candidate viability, interface adequacy, role eligibility, joint compatibility, live-residue status, uniqueness, or realization.*

Nonclaim F.2. *The representative family table is not the proof of completeness; it is a readable projection of the extensional source-coherent family class.*

Nonclaim F.3. *No finite enumeration, computational consistency check, or representative construction suite substitutes for semantic factorization and source-coherence soundness.*

Nonclaim F.4. *Candidate equality is not inferred from equality of the established physical interface. Distinct microscopic carrier architectures may remain distinct candidates despite identical interface projections.*

Nonclaim F.5 (A dependency crosswalk is not a proof). *A dependency table can identify where a predeclared burden is addressed, but it does not establish the correctness of the local mathematical or physical argument. Correctness remains with the cited definitions and proofs.*

G Source-formation grammar

G.1 Role of the grammar

This appendix spells out the saturated source constructors and constitutive coherence guards used in the normal-form theorem. It is intentionally expressed in terms of physical formation roles rather than named neutrino mechanisms. New instances and parameter values do not require new rules. A genuinely new primitive source sort, incidence relation, or constitutive equation does.

G.2 Node constructors

Table 7: Source node constructors.

Constructor	Input	Output	Scientific guard
Existing	Fixed substrate carrier type and lawful restriction.	Existing carrier node.	No new primitive ontology.
LocalAtom	Bulk/local support, persistence mode, constituent kind, attributes, parameter domain.	New primitive local node.	Physical local carrier interpretation.

Constructor	Input	Output	Scientific guard
BoundaryAtom	Fixed boundary type, boundary support, attachment data.	Boundary/edge node.	Boundary semantics unchanged.
ExtendedAtom	Line/surface/defect support, fusion or action attributes.	Extended/defect node.	Operator category admitted by source signature.
TopologicalAtom	Global/topological support and response data.	Topological node.	Not merely a descriptive label.
SolitonAtom	Localized nonperturbative profile, stability/topological data.	Solitonic node.	Physical persistence and support.
EnvironmentAtom	Dynamical environment/medium state and carrier coupling type.	Environmental co-carrier node.	Environment is physical, not calibration skin.
HiddenAtom	Enlarged-state-space support and projection/dilation type.	Hidden-sector node.	Hidden structure is carrier-bearing.
LawNode	Existing carrier nodes and a new occupancy-bearing process/constitutive law.	Law/process object.	No hidden bearer omitted from trace.
Derived	Finite constituent diagram and lawful derivation/binding/emergence map.	Composite/collective/emergent node.	Constituent and derivation data retained.

G.3 Incidence constructors

Table 8: Source incidence constructors.

Constructor	Meaning	Examples of admissible source data	Deferred test
Couple	Physical coupling or shared carrier relation.	Interaction type, domain, coupling parameter space.	Coupling-map identities at source; gauge, anomaly, and viability downstream.
Bind	Constituents form a derived carrier.	Binding relation, order parameter, emergence map.	Constituent–derivation coherence at source; stability and interface adequacy downstream.
Project	Visible/hidden, bulk/boundary, or UV/IR projection.	Embedding, quotient, dilation, reduction map.	Defining projection/dilation identities at source; common-witness compatibility downstream.

Constructor	Meaning	Examples of admissible source data	Deferred test
Attach_∂	Carrier attaches to source/detection boundary type.	Boundary map, edge restriction, detector/source support.	Restriction and attachment functoriality at source; target-boundary preservation downstream.
Attach_T	Carrier core is typed to the open target role complex.	Unevaluated target attachment slot.	Attachment typing at source; interface adequacy and the subsequent simultaneous constraints downstream.
Fuse	Extended or defect objects form junction/fusion structure.	Fusion rule, junction space, associator and unit data.	Pentagon, triangle, and module coherence at source; target relevance and anomaly compatibility downstream.
Inflow	Topological/boundary support participates in obstruction or inflow.	Inflow map, topological support, global-form dependence.	Constitutive inflow/topological identities at source; full anomaly compatibility downstream.
CoCarry	Environment/medium shares occupancy-bearing work.	State-dependent coupling, memory, medium response.	Co-carriage map coherence at source; environment continuation and common record downstream.
ActLaw	A law/process object acts on carrier nodes.	Kernel, channel, constitutive map, memory functional.	Defining axioms of the claimed process class at source; physical-interface adequacy downstream.
Compose	Finite lawful hybrid of generated diagrams.	Pushout/gluing data, shared supports, interface maps.	Overlap and gluing commutativity at source; simultaneous physical realizability downstream.

G.4 Type-indexed source-coherence profiles

The source theory attaches constitutive equations to the source sorts actually used by a family. Representative profiles are:

Source structure	Constitutive coherence enforced before candidate formation
Representation-bearing node	Identity and composition laws; compatibility with declared source morphisms.
Derived/composite node	Constituent map, derivation map, and retained-incidence diagrams commute.
Projection/dilation	The equations defining the claimed embedding, projection, reduction, or dilation hold.
Fusion/defect/module data	Pentagon, triangle, unit, module, and declared junction coherence hold.
Boundary/attachment data	Restriction and attachment maps are functorial and preserve declared source identities.
Law/process node	The axioms constitutive of the claimed process class hold.
Finite hybrid	All overlap maps, shared supports, and source gluing diagrams commute.

Failure of one of these laws means that the typed symbols do not instantiate the asserted source object. It is not subsequent simultaneous-constraint analysis viability failure.

G.5 Type-indexed source activity profiles

Constitutive coherence and carrier activity are distinct. For every incidence or target-attachment sort, the source theory therefore records a lawful activity-support map and its null class. Representative cases are:

Source relation	Activity rule at an instantiated source point
Scalar or operator-valued coupling	Retain exactly when the instantiated physical interaction morphism has nonempty source activity support; no weak-coupling threshold.
Projection/dilation map	Retain the source-defined active support; a rank drop may shrink the participating subdiagram without invalidating the source object.
Boundary/target attachment	Count as current attachment only on its source-non-null support.
Fusion/junction/module action	Retain the instantiated non-null action and its constitutively required coherence cells.
Environmental/hidden co-carriage	Retain only the actually active carrier relation; an otherwise complete but disconnected sector is spectator structure.

A zero value of an unrelated intrinsic attribute does not by itself null a node. Activity is attached to the source relation that performs carrier work, not to the typography of a parameter or to a universal numerical zero rule. The pointwise reduction $\text{ActRed}_{\mathbb{T}_v^{(3)}}(H_u(\theta))$ is formed before candidate identity or subsequent simultaneous evaluation.

G.6 Parameter-domain constructors

A parameter domain is part of source identity and may be an ordinary set, manifold, representation category, moduli stack, discrete lattice, function space, operator family, or symbolic family. The grammar contains the following formation operations:

1. finite products and coproducts of source domains;

2. lawful subdomains defined by source typing rather than later viability;
3. quotient by source-level reparameterization groups;
4. symbolic multiplicity domains such as $n \in \mathbb{N}$;
5. functorial families over representation, support, or operator categories;
6. fibres over target/global-form variants explicitly declared in the source packet.

A parameter restriction obtained only after a subsequent simultaneous failure is not a source subdomain. It is a survivor restriction appended downstream. Source activity is different: the complete coherent domain is retained, while each source point is assigned its active carrier core before subsequent simultaneous-constraint analysis. A source-null incidence at a degenerate point can therefore change candidate identity without narrowing the source domain.

G.7 Family normal-form schema

A source family is represented by

$$u = (H_u, \omega_u, \Theta_u^{\text{coh}}, \Theta_u^{\text{part}}, A_u, \text{Prov}_u),$$

where H_u is a finite typed hypergraph template, ω_u its source-constitutive coherence structure, Θ_u^{coh} is the full source-coherent locus inside the raw primitive source domain, $\Theta_u^{\text{part}} \subseteq \Theta_u^{\text{coh}}$ is the source-defined locus on which pointwise activity reduction leaves a nonempty target-participating carrier core, and A_u records the active source-equational isomorphism class at each participating point. An instance is

$$X_u^{\text{act}}(\theta) = \text{Interpret}_{\text{src}}(\text{ActRed}_{\mathbb{T}_V^{(3)}}(H_u(\theta))), \quad \theta \in \Theta_u^{\text{part}}.$$

The source-coherent locus is not narrowed by downstream viability. Different points of one symbolic family may lie in different source-defined activity strata, and a degenerate point may be candidate-equivalent to a point in another family.

Definition G.1 (Lawful family equivalence). *Two families u, v are lawfully equivalent when there are lawful source reparameterizations carrying their participating loci onto one another such that*

$$X_u^{\text{act}}(\theta) \equiv_{\text{car}} X_v^{\text{act}}(f\theta)$$

for every participating point, with every candidate instance of either family represented in the image of the other.

This is stricter than overlap of interface images or equality at one parameter point and is evaluated after source activity reduction.

G.8 Formation rules by principal stratum

The principal family labels are obtained by classifying the complete carrier trace. They overlap when a carrier possesses several source features.

Table 11: Principal strata and formation routes.

Stratum	Formation route	Boundary against nearby strata
Existing substrate	Existing nodes plus lawful new incidence or collective organization.	New primitive node moves to another stratum; new law without node may also carry F_{law} .
Law/process	LawNode plus ActLaw.	Explicit hidden bearer moves to hidden/hybrid stratum.

Stratum	Formation route	Boundary against nearby strata
Local primitive	LocalAtom with local coupling/attachment edges.	Composite origin is recorded rather than erased by effective locality.
Composite/emergent	Derived from a finite constituent diagram.	Constituent-invisible descriptions are law-only only when no carrier use can detect the constituents.
Topological	TopologicalAtom or topological incidence on other nodes.	A factor with only latent capability and no current path is spectator; local topological excitations may also be solitonic.
Extended/defect	ExtendedAtom plus fusion/junction/attachment edges.	Point-local effective representatives do not erase the extended source trace.
Solitonic	SolitonAtom with profile/stability data.	Topological charge does not by itself make the whole carrier a TQFT.
Boundary/edge	BoundaryAtom plus boundary attachment.	A fixed experimental boundary is target data; only dynamical boundary structure is a carrier.
Environmental	EnvironmentAtom plus CoCarry.	External calibration or fixed matter profile is target/record data unless dynamically carrier-bearing.
Hidden/enlarged	HiddenAtom plus projection/dilation incidence.	A reduced law with no carrier-detectable dilation is law-only.
Hybrid	Compose on two or more generated diagrams.	Not an eleventh primitive kind; identity is the full composed trace.

G.9 Constructor completeness burden

The source theory is complete only if every finitely presented source-coherent carrier trace can be reconstructed by these formation roles and equations. A proposed omitted constructor must therefore identify either:

1. a new primitive carrier sort not represented by the node constructors;
2. a new physical incidence relation not represented by the incidence category;
3. a new type-indexed source activity or null-incidence law not captured by the activity-support structure;
4. a new source parameter-domain formation not captured by the domain grammar;
5. a new constitutive equation or higher-coherence law not represented in $\mathfrak{C}_{\text{src}}$;
6. an architecture whose occupancy-bearing part has no finite symbolic presentation.

The first five reopen the source theory; the sixth extends the comparison class. A new named mechanism or a new finite combination of existing roles does neither.

G.10 Why the formation table is not the completeness proof

No finite number of displayed rows is asserted to equal the candidate universe. Each row denotes a named open parametric source stratum, while the extensional class Fam_ν contains all source-coherent normal-form templates with their pointwise activity profiles modulo family equivalence. The saturated source theory generates arbitrarily many lawful finite hybrids. The semantic theorem is the factorization of carrier traces, not the enumeration of outputs.

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